Topicwise Solved Exam Answers

Compiled for Exam Preparation

June 2026

Contents

1 Topic 01: Eigenvalues and Eigenvectors

This file contains the organized questions and answers for **Eigenvalues and Eigenvectors**, priority ranked as **Priority 1** based on frequency and exam weight.

1.1 Q1 (04)

Determine the characteristic roots of

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -4 & 2 \\ 0 & 0 & 7 \end{pmatrix}$$

ID	PYQ-2017-5c
Source	2017 Q5(c) [04 marks]

Answer:

Let the matrix be:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -4 & 2 \\ 0 & 0 & 7 \end{pmatrix}$$

The characteristic equation is defined by $|A - \lambda I| = 0$:

$$\begin{vmatrix} 1-\lambda & 2 & 3 \\ 0 & -4-\lambda & 2 \\ 0 & 0 & 7-\lambda \end{vmatrix} = 0$$

Since this is an upper triangular matrix, its determinant is simply the product of its diagonal elements:

$$(1-\lambda)(-4-\lambda)(7-\lambda) = 0$$

This equation has three roots:

$$\lambda = 1, \quad \lambda = -4, \quad \lambda = 7$$

So the characteristic roots (or eigenvalues) are 1, -4, and 7.

1.2 Q2 (12)

What are the eigen-values and eigen-vectors? Find them for the matrix

$$\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

ID	PYQ-2017-7
Appeared in	2017 Q7 [12 marks], 2018 Q7(a) [06 marks]
Frequency	$\star\star$ (2 times)

Answer:

1.2.0.1 Definitions Let A be a square matrix of order n. If there exists a non-zero vector X and a scalar λ such that:

$$AX = \lambda X$$

then λ is called an **eigenvalue** of A, and X is called the corresponding **eigenvector**.

1.2.0.2 Finding eigenvalues

Let the matrix be:

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

We solve the characteristic equation $|A - \lambda I| = 0$:

$$\begin{vmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{vmatrix} = 0$$

Expanding the determinant:

$$(2-\lambda)\left[(3-\lambda)(2-\lambda)-2\right] - 2\left[(2-\lambda)-1\right] + 1\left[2-(3-\lambda)\right] = 0$$

$$(2-\lambda)\left[\lambda^2 - 5\lambda + 4\right] - 2[1-\lambda] + [\lambda - 1] = 0$$

$$-\lambda^3 + 7\lambda^2 - 11\lambda + 5 = 0 \implies \lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$$

For $\lambda = 1$, the equation holds $(1 - 7 + 11 - 5 = 0)$. Dividing by $\lambda - 1$ gives:

$$(\lambda - 1)(\lambda^2 - 6\lambda + 5) = 0 \implies (\lambda - 1)(\lambda - 1)(\lambda - 5) = 0$$

So the eigenvalues are:

$$\lambda = 5, \quad \lambda = 1, \quad \lambda = 1$$

1.2.0.3 Finding eigenvectors

We solve the system $(A - \lambda I)X = 0$ for each eigenvalue:

1.2.0.3.1 Case 1: For $\lambda = 5$

The system is:

$$\begin{bmatrix} -3 & 2 & 1 \\ 1 & -2 & 1 \\ 1 & 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We apply row operations: * Swap $R\_1 \leftrightarrow R\_2$:

$$\begin{bmatrix} 1 & -2 & 1 \\ -3 & 2 & 1 \\ 1 & 2 & -3 \end{bmatrix}$$

- $R\_2 \to R\_2 + 3R\_1$
- $R\_3 \to R\_3 - R\_1$

This gives:

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & -4 & 4 \\ 0 & 4 & -4 \end{bmatrix}$$

- $R\_3 \to R\_3 + R\_2$
- Divide row 2 by -4:

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

This gives the relationships:

$$y - z = 0 \implies y = z$$

$$x - 2y + z = 0 \implies x - 2z + z = 0 \implies x = z$$

Let $z = 1$. This gives the eigenvector:

$$X_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

1.2.0.3.2 Case 2: For $\lambda = 1$ The system is:

$$\begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This reduces to a single equation:

$$x + 2y + z = 0 \implies x = -2y - z$$

Here we have two free variables. Let $y = s$ and $z = t$. The eigenvectors are:

$$X = \begin{bmatrix} -2s - t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

So the two linearly independent eigenvectors for $\lambda = 1$ are:

$$X_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad X_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

1.3 Q3. Find the eigen vectors of the following matrix: (06)

ID	PYQ-2019-7a
Source	2019 Q7(a) [06 marks]

Answer:

$$\begin{pmatrix} 1 & 3 & 0 \\ 3 & -2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$$

Answer:

1.3.0.1 1. Find the Eigenvalues We solve the characteristic equation $|A - \lambda I| = 0$:

$$\begin{vmatrix} 1-\lambda & 3 & 0 \\ 3 & -2-\lambda & -1 \\ 0 & -1 & 1-\lambda \end{vmatrix} = 0$$

Expand the determinant along row 1:

$$(1-\lambda)\left[(-2-\lambda)(1-\lambda) - (-1)(-1)\right] - 3\left[3(1-\lambda) - 0\right] = 0$$

$$(1-\lambda)\left[(\lambda+2)(\lambda-1) - 1\right] - 9(1-\lambda) = 0$$

$$(1-\lambda)\left[\lambda^2 + \lambda - 2 - 1 - 9\right] = 0$$

$$(1-\lambda)\left[\lambda^2 + \lambda - 12\right] = 0$$

Factor the quadratic term:

$$(1-\lambda)(\lambda+4)(\lambda-3) = 0$$

So the eigenvalues are:

$$\lambda_1 = 1, \quad \lambda_2 = 3, \quad \lambda_3 = -4$$

1.3.0.2 2. Find the Eigenvectors We solve the system $(A - \lambda I)X = 0$ for each eigenvalue:

1.3.0.2.1 Case 1: For $\lambda\_1 = 1$

$$\begin{bmatrix} 0 & 3 & 0 \\ 3 & -3 & -1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

From row 1 and row 3, we get $y = 0$. Substitute $y = 0$ into row 2:

$$3x - 3(0) - z = 0 \implies z = 3x$$

Let $x = 1$. The first eigenvector is:

$$X_1 = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$$

1.3.0.2.2 Case 2: For $\lambda\_2 = 3$

$$\begin{bmatrix} -2 & 3 & 0 \\ 3 & -5 & -1 \\ 0 & -1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

From row 1:

$$-2x + 3y = 0 \implies x = \frac{3}{2}y$$

From row 3:

$$-y - 2z = 0 \implies z = -\frac{1}{2}y$$

Let $y = 2$. Then $x = 3$ and $z = -1$. The second eigenvector is:

$$X_2 = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$$

1.3.0.2.3 Case 3: For $\lambda\_3 = -4$

$$\begin{bmatrix} 5 & 3 & 0 \\ 3 & 2 & -1 \\ 0 & -1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

From row 1:

$$5x + 3y = 0 \implies y = -\frac{5}{3}x$$

From row 3:

$$-y + 5z = 0 \implies z = \frac{1}{5}y = \frac{1}{5}\left(-\frac{5}{3}x\right) = -\frac{1}{3}x$$

Let $x = 3$. Then $y = -5$ and $z = -1$. The third eigenvector is:

$$X_3 = \begin{bmatrix} 3 \\ -5 \\ -1 \end{bmatrix}$$

1.4 Q4 (06)

Find the eigen values and the corresponding eigen vectors for the matrix

$$A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

ID	PYQ-2020-7a
Source	2020 Q7(a) [06 marks]

Answer:

1.4.0.1 1. Find Eigenvalues

We solve the characteristic equation $|A - \lambda I| = 0$:

$$\begin{vmatrix} 1-\lambda & 1 & -2 \\ -1 & 2-\lambda & 1 \\ 0 & 1 & -1-\lambda \end{vmatrix} = 0$$

Expanding the determinant yields the equation:

$$\lambda^3 - 2\lambda^2 - \lambda + 2 = 0$$

Factor the expression:

$$(\lambda - 1)(\lambda + 1)(\lambda - 2) = 0$$

So the eigenvalues are:

$$\lambda = 1, \quad \lambda = -1, \quad \lambda = 2$$

1.4.0.2 2. Find Eigenvectors

1.4.0.2.1 Case 1: For $\lambda = 1$

We solve $(A - I)X = 0$:

$$\begin{bmatrix} 0 & 1 & -2 \\ -1 & 1 & 1 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

From row 1:

$$y - 2z = 0 \implies y = 2z$$

From row 2:

$$-x + y + z = 0 \implies x = y + z = 3z$$

Let $z = 1$. The eigenvector is:

$$X_1 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

1.4.0.2.2 Case 2: For $\lambda = -1$

We solve $(A + I)X = 0$:

$$\begin{bmatrix} 2 & 1 & -2 \\ -1 & 3 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

From row 3:

$$y = 0$$

From row 1:

$$2x - 2z = 0 \implies x = z$$

Let $z = 1$. The eigenvector is:

$$X_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

1.4.0.2.3 Case 3: For $\lambda = 2$

We solve $(A - 2I)X = 0$:

$$\begin{bmatrix} -1 & 1 & -2 \\ -1 & 0 & 1 \\ 0 & 1 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

From row 2:

$$-x + z = 0 \implies x = z$$

From row 3:

$$y - 3z = 0 \implies y = 3z$$

Let $z = 1$. The eigenvector is:

$$X_3 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

1.5 Q5 (06)

Find the eigenvalues and the corresponding eigen vectors of the matrix

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

ID	PYQ-2021-7a
Source	2021 Q7(a) [06 marks]

Answer:

1.5.0.1 1. Find Eigenvalues

We solve the characteristic equation $|A - \lambda I| = 0$:

$$\begin{vmatrix} 2-\lambda & 3 \\ 1 & 4-\lambda \end{vmatrix} = 0 \implies (2-\lambda)(4-\lambda) - 3 = 0$$

$$\lambda^2 - 6\lambda + 5 = 0 \implies (\lambda - 1)(\lambda - 5) = 0$$

So the eigenvalues are:

$$\lambda = 1, \quad \lambda = 5$$

1.5.0.2 2. Find Eigenvectors

1.5.0.2.1 Case 1: For $\lambda = 1$

We solve $(A - I)X = 0$:

$$\begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies x + 3y = 0 \implies x = -3y$$

Let $y = 1$. The eigenvector is:

$$X_1 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

1.5.0.2.2 Case 2: For $\lambda = 5$

We solve $(A - 5I)X = 0$:

$$\begin{bmatrix} -3 & 3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies x - y = 0 \implies x = y$$

Let $y = 1$. The eigenvector is:

$$X_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

1.6 Q6. Find the eigen values and eigen vectors of the matrix: (05)

ID	PYQ-2023-6a
Source	2023 Q6(a) [05 marks]

Answer:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

Answer:

1.6.0.1 1. Find Eigenvalues

We solve the characteristic equation $|A - \lambda I| = 0$:

$$\begin{vmatrix} 2-\lambda & 1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix} = 0$$

The expansion of this determinant yields:

$$(2-\lambda)(\lambda-1)(\lambda-3) = 0$$

So the eigenvalues are:

$$\lambda = 1, \quad \lambda = 2, \quad \lambda = 3$$

1.6.0.2 2. Find Eigenvectors

1.6.0.2.1 Case 1: For $\lambda = 1$

We solve $(A - I)X = 0$:

$$\begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Perform operations $R\_2 \to R\_2 + R\_1$, $R\_3 \to R\_3 - R\_1$:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & -2 & 0 \end{bmatrix} \implies y = 0 \quad \text{and} \quad x + z = 0 \implies x = -z$$

Let $z = 1$. The eigenvector is:

$$X_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

1.6.0.2.2 Case 2: For $\lambda = 2$

We solve $(A - 2I)X = 0$:

$$\begin{bmatrix} 0 & 1 & 1 \\ -1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

From row 1: $y + z = 0 \implies y = -z$. From row 2: $-x - z = 0 \implies x = -z$.
Let $z = -1$. The eigenvector is:

$$X_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

1.6.0.2.3 Case 3: For $\lambda = 3$

We solve $(A - 3I)X = 0$:

$$\begin{bmatrix} -1 & 1 & 1 \\ -1 & -1 & -1 \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Subtract row 1 from row 2 ($R\_2 \to R\_2 - R\_1$):

$$\begin{bmatrix} -1 & 1 & 1 \\ 0 & -2 & -2 \\ 1 & -1 & -1 \end{bmatrix} \implies y + z = 0 \implies y = -z \quad \text{and} \quad x = 0$$

Let $z = 1$. The eigenvector is:

$$X_3 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

1.7 Q7. Find the eigen vectors of the matrix: (05)

ID	PYQ-2024-6b
Source	2024 Q6(b) [05 marks]

Answer:

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

Answer:

1.7.0.1 1. Find the Eigenvalues

We solve the characteristic equation $|A - \lambda I| = 0$:

$$\begin{vmatrix} 3-\lambda & 1 & 1 \\ 1 & 5-\lambda & 1 \\ 1 & 1 & 3-\lambda \end{vmatrix} = 0$$

We expand the determinant:

$$(3-\lambda)\left[(5-\lambda)(3-\lambda) - 1\right] - 1\left[(3-\lambda) - 1\right] + 1\left[1 - (5-\lambda)\right] = 0$$

$$(3-\lambda)\left[\lambda^2 - 8\lambda + 14\right] - (2-\lambda) + (\lambda - 4) = 0$$

$$(3\lambda^2 - 24\lambda + 42 - \lambda^3 + 8\lambda^2 - 14\lambda) - 2 + \lambda + \lambda - 4 = 0$$

$$-\lambda^3 + 11\lambda^2 - 36\lambda + 36 = 0$$

We multiply by -1:

$$\lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0$$

We find the roots of this polynomial. We test $\lambda = 2$:

$$2^3 - 11(2^2) + 36(2) - 36 = 8 - 44 + 72 - 36 = 0$$

So, $\lambda = 2$ is a root. We factor it out:

$$(\lambda - 2)(\lambda^2 - 9\lambda + 18) = 0$$

We factor the quadratic term:

$$(\lambda - 2)(\lambda - 3)(\lambda - 6) = 0$$

The eigenvalues are:

$$\lambda = 2, \quad \lambda = 3, \quad \lambda = 6$$

1.7.0.2 2. Find the Eigenvectors

1.7.0.2.1 Case 1: For $\lambda = 2$

We solve $(A - 2I)X = 0$:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We apply row operations: * $R\_2 \to R\_2 - R\_1$ * $R\_3 \to R\_3 - R\_1$
This gives:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

From the second row, we get $2y = 0 \implies y = 0$. From the first row, we get $x + y + z = 0 \implies x + z = 0 \implies x = -z$.
Let $z = 1$. The eigenvector is:

$$X_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

1.7.0.2.2 Case 2: For $\lambda = 3$

We solve $(A - 3I)X = 0$:

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

From the first row, we get $y + z = 0 \implies y = -z$. From the third row, we get $x + y = 0 \implies x = -y = z$.
Let $z = 1$. The eigenvector is:

$$X_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

1.7.0.2.3 Case 3: For $\lambda = 6$

We solve $(A - 6I)X = 0$:

$$\begin{bmatrix} -3 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We swap rows $R\_1 \leftrightarrow R\_2$:

$$\begin{bmatrix} 1 & -1 & 1 \\ -3 & 1 & 1 \\ 1 & 1 & -3 \end{bmatrix}$$

We apply row operations: * $R\_2 \to R\_2 + 3R\_1$ * $R\_3 \to R\_3 - R\_1$
This gives:

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & -2 & 4 \\ 0 & 2 & -4 \end{bmatrix}$$

We scale the second row (

$$R\_2 \to -\frac{1}{2} R\_2$$

):

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 2 & -4 \end{bmatrix}$$

We perform $R\_3 \to R\_3 - 2R\_2$:

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

From the second row, we get $y - 2z = 0 \implies y = 2z$. From the first row, we get $x - y + z = 0 \implies x - 2z + z = 0 \implies x = z$.

Let $z = 1$. The eigenvector is:

$$X_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

2 Topic 02: System of Linear Equations

This file contains the organized questions and answers for **System of Linear Equations**, priority ranked as **Priority 2** based on frequency and exam weight.

2.1 Q1. Solve by elementary transformation: (06)

ID	PYQ-2017-6b
Source	2017 Q6(b) [06 marks]

Answer:

$$2x_1 + 2x_2 + x_3 = 2$$
$$3x_1 + x_2 - 2x_3 = 1$$
$$4x_1 - 3x_2 - x_3 = 3$$

Answer:
We write the augmented matrix of the system:

$$\begin{bmatrix} 2 & 2 & 1 & | & 2 \\ 3 & 1 & -2 & | & 1 \\ 4 & -3 & -1 & | & 3 \end{bmatrix}$$

We perform row operations to reduce it: * $R\_2 \to 2R\_2 - 3R\_1$ * $R\_3 \to R\_3 - 2R\_1$
This gives:

$$\begin{bmatrix} 2 & 2 & 1 & | & 2 \\ 0 & -4 & -7 & | & -4 \\ 0 & -7 & -3 & | & -1 \end{bmatrix}$$

Multiply the second and third rows by -1:

$$\begin{bmatrix} 2 & 2 & 1 & | & 2 \\ 0 & 4 & 7 & | & 4 \\ 0 & 7 & 3 & | & 1 \end{bmatrix}$$

Now perform row operation $R\_3 \to 4R\_3 - 7R\_2$:

$$\begin{bmatrix} 2 & 2 & 1 & | & 2 \\ 0 & 4 & 7 & | & 4 \\ 0 & 0 & -37 & | & -24 \end{bmatrix}$$

Now we solve the variables by back substitution. From the third row:

$$-37x_3 = -24 \implies x_3 = \frac{24}{37}$$

From the second row:

$$4x_2 + 7x_3 = 4 \implies 4x_2 + 7\left(\frac{24}{37}\right) = 4$$

$$4x_2 = 4 - \frac{168}{37} = \frac{148 - 168}{37} = -\frac{20}{37} \implies x_2 = -\frac{5}{37}$$

From the first row:

$$2x_1 + 2x_2 + x_3 = 2 \implies 2x_1 + 2\left(-\frac{5}{37}\right) + \frac{24}{37} = 2$$

$$2x_1 - \frac{10}{37} + \frac{24}{37} = 2 \implies 2x_1 + \frac{14}{37} = 2$$

$$2x_1 = 2 - \frac{14}{37} = \frac{74 - 14}{37} = \frac{60}{37} \implies x_1 = \frac{30}{37}$$

So the unique solution is:

$$x_1 = \frac{30}{37}, \quad x_2 = -\frac{5}{37}, \quad x_3 = \frac{24}{37}$$

2.2 Q2 (06)

Find the value of λ so that the following equations have a solution and solve completely in each case:

ID	PYQ-2018-6a
Source	2018 Q6(a) [06 marks]

Answer:

$$x + y + z = 1$$

$$x + 2y + 4z = \lambda$$

$$x + 4y + 10z = \lambda^2$$

Answer:
We write the augmented matrix of the system:

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 1 & 2 & 4 & | & \lambda \\ 1 & 4 & 10 & | & \lambda^2 \end{bmatrix}$$

Apply row operations: * $R\_2 \to R\_2 - R\_1$ * $R\_3 \to R\_3 - R\_1$
This gives:

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & 3 & | & \lambda - 1 \\ 0 & 3 & 9 & | & \lambda^2 - 1 \end{bmatrix}$$

Now eliminate row 3 using row 2 ($R\_3 \to R\_3 - 3R\_2$):

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & 3 & | & \lambda - 1 \\ 0 & 0 & 0 & | & \lambda^2 - 3\lambda + 2 \end{bmatrix}$$

A solution exists if the system is consistent:

$$\lambda^2 - 3\lambda + 2 = 0 \implies (\lambda - 1)(\lambda - 2) = 0 \implies \lambda = 1 \quad \text{or} \quad \lambda = 2$$

We solve the system for each case:

2.2.0.1 Case 1: For $\lambda = 1$

The augmented matrix is:

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & 3 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

From row 2:

$$y + 3z = 0 \implies y = -3z$$

From row 1:

$$x + y + z = 1 \implies x - 3z + z = 1 \implies x = 2z + 1$$

Let $z = t$ be a parameter. The general solution is:

$$x = 2t + 1, \quad y = -3t, \quad z = t$$

2.2.0.2 Case 2: For $\lambda = 2$

The augmented matrix is:

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & 3 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

From row 2:

$$y + 3z = 1 \implies y = 1 - 3z$$

From row 1:

$$x + y + z = 1 \implies x + (1 - 3z) + z = 1 \implies x = 2z$$

Let $z = t$ be a parameter. The general solution is:

$$x = 2t, \quad y = 1 - 3t, \quad z = t$$

2.3 Q3. Solve: (06)

ID	PYQ-2018-6b
Source	2018 Q6(b) [06 marks]

Answer:

$$2x_1 - x_2 + x_3 = 0$$
$$3x_1 + 2x_2 + x_3 = 0$$
$$x_1 - 3x_2 + 5x_3 = 0$$

Answer:
We write the augmented matrix of this homogeneous system:

$$\begin{bmatrix} 2 & -1 & 1 & | & 0 \\ 3 & 2 & 1 & | & 0 \\ 1 & -3 & 5 & | & 0 \end{bmatrix}$$

We swap row 1 and row 3 ($R\_1 \leftrightarrow R\_3$):

$$\begin{bmatrix} 1 & -3 & 5 & | & 0 \\ 3 & 2 & 1 & | & 0 \\ 2 & -1 & 1 & | & 0 \end{bmatrix}$$

Apply row operations: * $R\_2 \to R\_2 - 3R\_1$ * $R\_3 \to R\_3 - 2R\_1$
This gives:

$$\begin{bmatrix} 1 & -3 & 5 & | & 0 \\ 0 & 11 & -14 & | & 0 \\ 0 & 5 & -9 & | & 0 \end{bmatrix}$$

Perform the operation $R\_3 \to 11R\_3 - 5R\_2$:

$$\begin{bmatrix} 1 & -3 & 5 & | & 0 \\ 0 & 11 & -14 & | & 0 \\ 0 & 0 & -29 & | & 0 \end{bmatrix}$$

By back substitution, we solve the system: * From the third row:

$$-29x\_3 = 0 \implies x\_3 = 0.$$

- From the second row:

$$11x\_2 - 14(0) = 0 \implies x\_2 = 0.$$

- From the first row:

$$x\_1 - 3(0) + 5(0) = 0 \implies x\_1 = 0.$$

So the system only has the trivial solution:

$$x_1 = 0, \quad x_2 = 0, \quad x_3 = 0$$

2.4 Q4. Discuss about the consistency of a system and then find the solution set of the system: (06)

ID	PYQ-2019-6a
Source	2019 Q6(a) [06 marks]

Answer:

$$\begin{aligned} x + 2y + 2z &= 1 \\ 2x + y + z &= 2 \\ 3x + 2y + 2z &= 3 \\ y + z &= 0 \end{aligned}$$

Answer:

2.4.0.1 1. Discuss Consistency

A system of linear equations $AX = B$ is consistent if there exists at least one set of values for the unknowns that satisfies all equations simultaneously. In terms of rank: * The system is consistent if $\text{Rank}(A) = \text{Rank}([A|B])$, where A is the coefficient matrix and $[A|B]$ is the augmented matrix. * If $\text{Rank}(A) = \text{Rank}([A|B]) = n$ (number of variables), the system has a unique solution. * If $\text{Rank}(A) = \text{Rank}([A|B]) < n$, the system has infinitely many solutions (with $n - r$ free parameters). * If $\text{Rank}(A) \neq \text{Rank}([A|B])$, the system is inconsistent (no solution).

2.4.0.2 2. Solve the System

We write the augmented matrix $[A|B]$:

$$\begin{bmatrix} 1 & 2 & 2 & | & 1 \\ 2 & 1 & 1 & | & 2 \\ 3 & 2 & 2 & | & 3 \\ 0 & 1 & 1 & | & 0 \end{bmatrix}$$

Apply row operations to clear the first column: * $R\_2 \to R\_2 - 2R\_1$ * $R\_3 \to R\_3 - 3R\_1$

This gives:

$$\begin{bmatrix} 1 & 2 & 2 & | & 1 \\ 0 & -3 & -3 & | & 0 \\ 0 & -4 & -4 & | & 0 \\ 0 & 1 & 1 & | & 0 \end{bmatrix}$$

Scale row 2 and row 3: *

$$R\_2 \to -\frac{1}{3}R\_2$$

- $$R\_3 \to -\frac{1}{4}R\_3$$

This gives:

$$\begin{bmatrix} 1 & 2 & 2 & | & 1 \\ 0 & 1 & 1 & | & 0 \\ 0 & 1 & 1 & | & 0 \\ 0 & 1 & 1 & | & 0 \end{bmatrix}$$

Apply row operations: * $R\_3 \to R\_3 - R\_2$ * $R\_4 \to R\_4 - R\_2$

This yields the row echelon form:

$$\begin{bmatrix} 1 & 2 & 2 & | & 1 \\ 0 & 1 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

The rank of the coefficient matrix A is 2, and the rank of the augmented matrix $[A|B]$ is also 2. Since $\text{Rank}(A) = \text{Rank}([A|B]) = 2$, the system is consistent.

Since there are 3 variables (x, y, z) and the rank is 2, there is $3 - 2 = 1$ free variable.

From row 2:

$$y + z = 0 \implies y = -z$$

From row 1:

$$x + 2y + 2z = 1 \implies x + 2(-z) + 2z = 1 \implies x = 1$$

Let $z = k$ (where $k \in \mathbb{R}$ is a parameter). The solution set is:

$$x = 1, \quad y = -k, \quad z = k \quad (k \in \mathbb{R})$$

2.5 Q5. Find the trivial solution of: (06)

ID	PYQ-2019-6b
Source	2019 Q6(b) [06 marks]

Answer:

$$\begin{aligned} x + y + z + w &= 0 \\ x + 3y - 2z + 4w &= 0 \\ 2x + y - z - w &= 0 \end{aligned}$$

Answer:

For any homogeneous linear system $AX = 0$, the trivial solution is the solution where all variables are zero:

$$x = 0, \quad y = 0, \quad z = 0, \quad w = 0$$

To find if there are non-trivial solutions (which is the mathematical intent of the question), we write the coefficient matrix:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & -2 & 4 \\ 2 & 1 & -1 & -1 \end{bmatrix}$$

Apply row operations: * $R\_2 \to R\_2 - R\_1$ * $R\_3 \to R\_3 - 2R\_1$

This gives:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & -3 & 3 \\ 0 & -1 & -3 & -3 \end{bmatrix}$$

Swap row 2 and row 3 ($R\_2 \leftrightarrow R\_3$) and multiply the new row 2 by -1 ($R\_2 \to -R\_2$):

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 3 \\ 0 & 2 & -3 & 3 \end{bmatrix}$$

Apply row operation: * $R\_3 \to R\_3 - 2R\_2$

This gives:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & -9 & -3 \end{bmatrix}$$

Scale row 3 (

$$R\_3 \to -\frac{1}{3}R\_3$$

):

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 3 & 1 \end{bmatrix}$$

The rank of the matrix is 3. Since there are 4 variables (x, y, z, w) and the rank is 3, there is $4 - 3 = 1$ free variable.

From row 3:

$$3z + w = 0 \implies w = -3z$$

From row 2:

$$y + 3z + 3w = 0 \implies y + 3z + 3(-3z) = 0 \implies y - 6z = 0 \implies y = 6z$$

From row 1:

$$x + y + z + w = 0 \implies x + 6z + z - 3z = 0 \implies x + 4z = 0 \implies x = -4z$$

Let $z = k$ (where $k \in \mathbb{R}$ is a parameter). The complete solution set is:

$$x = -4k, \quad y = 6k, \quad z = k, \quad w = -3k \quad (k \in \mathbb{R})$$

- The **trivial solution** corresponds to $k = 0$, giving $(0, 0, 0, 0)$.
- For any $k \neq 0$, we obtain **non-trivial solutions**.

2.6 Q6 (07)

Investigate for what values of λ and μ the following system of equations have (i) no solution, (ii) unique solution, (iii) infinite number of solutions:

ID	PYQ-2020-6b
Appeared in	2020 Q6(b) [07 marks], 2024 Q5(b) [05 marks]
Frequency	⋆⋆ (2 times)

Answer:

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu$$

Answer:

We write the augmented matrix of the system:

$$\begin{bmatrix} 1 & 1 & 1 & | & 6 \\ 1 & 2 & 3 & | & 10 \\ 1 & 2 & \lambda & | & \mu \end{bmatrix}$$

Apply row operations: * $R\_2 \to R\_2 - R\_1$ * $R\_3 \to R\_3 - R\_1$

This gives:

$$\begin{bmatrix} 1 & 1 & 1 & | & 6 \\ 0 & 1 & 2 & | & 4 \\ 0 & 1 & \lambda - 1 & | & \mu - 6 \end{bmatrix}$$

Now eliminate row 3 using row 2 ($R\_3 \to R\_3 - R\_2$):

$$\begin{bmatrix} 1 & 1 & 1 & | & 6 \\ 0 & 1 & 2 & | & 4 \\ 0 & 0 & \lambda - 3 & | & \mu - 10 \end{bmatrix}$$

Now we investigate:

2.6.0.1 (i) No Solution

A contradiction occurs if the last row is of the form $[0, 0, 0|c]$ with $c \neq 0$:

$$\lambda - 3 = 0 \quad \text{and} \quad \mu - 10 \neq 0 \implies \lambda = 3 \quad \text{and} \quad \mu \neq 10$$

2.6.0.2 (ii) Unique Solution

A unique solution exists if the rank of the coefficient matrix is 3:

$$\lambda - 3 \neq 0 \implies \lambda \neq 3$$

for any value of μ.

2.6.0.3 (iii) Infinite Solutions

Infinite solutions exist if the last row is entirely zero:

$$\lambda - 3 = 0 \quad \text{and} \quad \mu - 10 = 0 \implies \lambda = 3 \quad \text{and} \quad \mu = 10$$

2.7 Q7 (06)

Discuss for what values of λ the following system of equations have (i) no solution (ii) unique solution and (iii) infinite number of solutions:

ID	PYQ-2021-6b
Source	2021 Q6(b) [06 marks]

Answer:

$$\lambda x + y + z = 1$$
$$x + \lambda y + z = \lambda$$
$$x + y + \lambda z = \lambda^2$$

Answer:
We write the determinant of the coefficient matrix:

$$D = \begin{vmatrix} \lambda & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix}$$

Add rows 2 and 3 to row 1:

$$D = \begin{vmatrix} \lambda+2 & \lambda+2 & \lambda+2 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix} = (\lambda+2)\begin{vmatrix} 1 & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix}$$

Apply column operations ($C\_2 \to C\_2 - C\_1$, $C\_3 \to C\_3 - C\_1$):

$$D = (\lambda+2)\begin{vmatrix} 1 & 0 & 0 \\ 1 & \lambda-1 & 0 \\ 1 & 0 & \lambda-1 \end{vmatrix} = (\lambda+2)(\lambda-1)^2$$

We investigate the cases:

2.7.0.1 (i) Unique Solution

A unique solution exists if the determinant is not zero:

$$D \neq 0 \implies \lambda \neq 1 \quad \text{and} \quad \lambda \neq -2$$

2.7.0.2 (ii) Infinite Solutions

If $\lambda = 1$, the system of equations becomes:

$$x + y + z = 1$$

$$x + y + z = 1$$

$$x + y + z = 1$$

All three equations are identical. The system is consistent and has an infinite number of solutions.

2.7.0.3 (iii) No Solution

If $\lambda = -2$, the system becomes:

$$-2x + y + z = 1 \quad \ldots(1)$$

$$x - 2y + z = -2 \quad \ldots(2)$$

$$x + y - 2z = 4 \quad \ldots(3)$$

Add these three equations:

$$(-2x + x + x) + (y - 2y + y) + (z + z - 2z) = 1 - 2 + 4 \implies 0 = 3$$

This is a contradiction. So there is no solution when $\lambda = -2$.

2.8 Q8. Solve the following system of linear equations: (05)

ID	PYQ-2023-6b
Source	2023 Q6(b) [05 marks]

Answer:

$$x + 2y - z = 2$$

$$2x + y + z = 1$$

$$x + 5y - 4z = 5$$

Answer:
We write the augmented matrix:

$$\begin{bmatrix} 1 & 2 & -1 & | & 2 \\ 2 & 1 & 1 & | & 1 \\ 1 & 5 & -4 & | & 5 \end{bmatrix}$$

Apply row operations: * $R\_2 \to R\_2 - 2R\_1$ * $R\_3 \to R\_3 - R\_1$
This gives:

$$\begin{bmatrix} 1 & 2 & -1 & | & 2 \\ 0 & -3 & 3 & | & -3 \\ 0 & 3 & -3 & | & 3 \end{bmatrix}$$

Perform the operation $R\_3 \to R\_3 + R\_2$:

$$\begin{bmatrix} 1 & 2 & -1 & | & 2 \\ 0 & -3 & 3 & | & -3 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Scale row 2 ($R\_2 \to -1/3R\_2$):

$$\begin{bmatrix} 1 & 2 & -1 & | & 2 \\ 0 & 1 & -1 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

This system is consistent. From row 2:

$$y - z = 1 \implies y = 1 + z$$

From row 1:

$$x + 2y - z = 2 \implies x + 2(1 + z) - z = 2 \implies x + 2 + z = 2 \implies x = -z$$

Let $z = t$ be a parameter. The general solution is:

$$x = -t, \quad y = 1 + t, \quad z = t$$

2.9 Q9 (07)

Consider a circuit with three mesh currents. The corresponding equations for the mesh currents i_1, i_2, and i_3 are:

ID	CT1M-1
Source	CT1M Q1 [07 marks]

Answer:

$$2i_1 + 3i_2 - i_3 = 5$$

$$i_1 - 2i_2 + 4i_3 = 8$$

$$3i_1 + i_2 + 2i_3 = -3$$

Solve this system of linear equations by reducing its augmented matrix to echelon form to find the mesh currents.

Answer:

First, we write the augmented matrix for this system:

$$\begin{bmatrix} 2 & 3 & -1 & | & 5 \\ 1 & -2 & 4 & | & 8 \\ 3 & 1 & 2 & | & -3 \end{bmatrix}$$

We swap the first and second rows to get a pivot of 1 in the top-left:

$$\begin{bmatrix} 1 & -2 & 4 & | & 8 \\ 2 & 3 & -1 & | & 5 \\ 3 & 1 & 2 & | & -3 \end{bmatrix} \quad (R_1 \leftrightarrow R_2)$$

Now we make the elements below the first pivot zero. We apply two row operations:

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - 3R_1$$

This gives the following matrix:

$$\begin{bmatrix} 1 & -2 & 4 & | & 8 \\ 0 & 7 & -9 & | & -11 \\ 0 & 7 & -10 & | & -27 \end{bmatrix}$$

Next, we make the element below the second pivot zero. We subtract the second row from the third row:

$$R_3 \to R_3 - R_2$$

This gives:

$$\begin{bmatrix} 1 & -2 & 4 & | & 8 \\ 0 & 7 & -9 & | & -11 \\ 0 & 0 & -1 & | & -16 \end{bmatrix}$$

We multiply the third row by -1 to make the pivot positive:

$$R_3 \to -R_3$$

$$\begin{bmatrix} 1 & -2 & 4 & | & 8 \\ 0 & 7 & -9 & | & -11 \\ 0 & 0 & 1 & | & 16 \end{bmatrix}$$

This is the echelon form.

Now we use back substitution to find the currents.

From the third row:

$$i_3 = 16$$

From the second row:

$$7i_2 - 9i_3 = -11$$

$$7i_2 - 9(16) = -11$$

$$7i_2 - 144 = -11$$

$$7i_2 = 133 \implies i_2 = 19$$

From the first row:

$$i_1 - 2i_2 + 4i_3 = 8$$

$$i_1 - 2(19) + 4(16) = 8$$

$$i_1 - 38 + 64 = 8$$

$$i_1 + 26 = 8 \implies i_1 = -18$$

So, the mesh currents are

$$i_1 = -18, \quad i_2 = 19, \quad i_3 = 16$$

2.10 Q10 (06)

Find the value(s) of k for which the following system of three linear equations in three variables has (i) a unique solution, (ii) no solution, and (iii) infinitely many solutions:

ID	CT1M-2
Source	CT1M Q2 [06 marks]

Answer:

$$x + y + z = 3$$

$$2x + 3y + z = 7$$

$$3x + 5y + kz = 10$$

Answer:

We write the augmented matrix for the system:

$$\begin{bmatrix} 1 & 1 & 1 & | & 3 \\ 2 & 3 & 1 & | & 7 \\ 3 & 5 & k & | & 10 \end{bmatrix}$$

We reduce the matrix to echelon form. First, we clear the first column below the pivot:

$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - 3R_1$$

This gives:

$$\begin{bmatrix} 1 & 1 & 1 & | & 3 \\ 0 & 1 & -1 & | & 1 \\ 0 & 2 & k-3 & | & 1 \end{bmatrix}$$

Next, we clear the second column below the pivot:

$$R_3 \to R_3 - 2R_2$$

This gives the echelon form:

$$\begin{bmatrix} 1 & 1 & 1 & | & 3 \\ 0 & 1 & -1 & | & 1 \\ 0 & 0 & k-1 & | & -1 \end{bmatrix}$$

Now we analyze the solutions:

2.10.0.1 (i) Unique Solution

A unique solution exists when the rank of the coefficient matrix equals the number of variables. This means the third pivot must not be zero:

$$k - 1 \neq 0 \implies k \neq 1$$

So, the system has a unique solution for any value $k \neq 1$.

2.10.0.2 (ii) No Solution

No solution exists when the rank of the coefficient matrix is less than the rank of the augmented matrix. This happens when the last row has a zero coefficient for z but a non-zero constant:

$$k - 1 = 0 \implies k = 1$$

When $k = 1$, the last row represents $0z = -1$, which is impossible. So, the system has no solution when $k = 1$.

2.10.0.3 (iii) Infinitely Many Solutions

Infinitely many solutions exist when the rank of the coefficient matrix equals the rank of the augmented matrix, and this rank is less than 3. This requires the entire last row to be all zeros. However, the right-hand side of the last row is -1, which is never zero. So, there is no value of k that gives infinitely many solutions.

3 Topic 03: Rank and Normal Form

This file contains the organized questions and answers for **Rank and Normal Form**, priority ranked as **Priority 3** based on frequency and exam weight.

3.1 Q1 (06)

What is the rank of a matrix? Find it for

$$\begin{bmatrix} 0 & 0 & 1 & -3 & -2 \\ 0 & 1 & 2 & 6 & 0 \\ 0 & 2 & 3 & 9 & 2 \\ 0 & 1 & 1 & 3 & 2 \end{bmatrix}$$

ID	PYQ-2017-6a
Source	2017 Q6(a) [06 marks]

Answer:

3.1.0.1 Rank of a Matrix The rank of a matrix is the maximum number of linearly independent row vectors (or column vectors) in it. It equals the number of non-zero rows when the matrix is reduced to its row-echelon form.

3.1.0.2 Finding the Rank Let the matrix be:

$$A = \begin{bmatrix} 0 & 0 & 1 & -3 & -2 \\ 0 & 1 & 2 & 6 & 0 \\ 0 & 2 & 3 & 9 & 2 \\ 0 & 1 & 1 & 3 & 2 \end{bmatrix}$$

We swap the first and second rows ($R\_1 \leftrightarrow R\_2$) to place a non-zero leading entry in the first row:

$$\begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -3 & -2 \\ 0 & 2 & 3 & 9 & 2 \\ 0 & 1 & 1 & 3 & 2 \end{bmatrix}$$

Now we perform row operations to make the entries below the leading entry of R_1 zero: * $R\_3 \to R\_3 - 2R\_1$ * $R\_4 \to R\_4 - R\_1$

This gives:

$$\begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -3 & -2 \\ 0 & 0 & -1 & -3 & 2 \\ 0 & 0 & -1 & -3 & 2 \end{bmatrix}$$

Next we make the entries below the leading entry of R_2 zero: * $R\_3 \to R\_3 + R\_2$ * $R\_4 \to R\_4 + R\_2$

This gives:

$$\begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & -6 & 0 \\ 0 & 0 & 0 & -6 & 0 \end{bmatrix}$$

Now eliminate the last entry in row 4 using row 3 ($R\_4 \to R\_4 - R\_3$):

$$\begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & -6 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Divide row 3 by -6:

$$\begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

This matrix is now in row-echelon form. The number of non-zero rows is 3.

So the rank of the matrix is 3.

3.2 Q2 (04)

What is rank of a matrix? Find the rank of

$$A = \begin{pmatrix} 1 & -2 & 1 & -1 \\ 1 & 1 & -2 & 3 \\ 4 & 1 & -5 & 8 \\ 5 & -7 & 2 & -1 \end{pmatrix}$$

ID	PYQ-2018-5c
Source	2018 Q5(c) [04 marks]

Answer:

3.2.0.1 Rank of a Matrix The rank of a matrix is the maximum number of linearly independent rows in it. It equals the number of non-zero rows in its row-echelon form.

3.2.0.2 Finding the Rank Let the matrix be:

$$A = \begin{pmatrix} 1 & -2 & 1 & -1 \\ 1 & 1 & -2 & 3 \\ 4 & 1 & -5 & 8 \\ 5 & -7 & 2 & -1 \end{pmatrix}$$

We apply row operations: * $R\_2 \to R\_2 - R\_1$ * $R\_3 \to R\_3 - 4R\_1$ * $R\_4 \to R\_4 - 5R\_1$

This gives:

$$\begin{pmatrix} 1 & -2 & 1 & -1 \\ 0 & 3 & -3 & 4 \\ 0 & 9 & -9 & 12 \\ 0 & 3 & -3 & 4 \end{pmatrix}$$

Now perform operations on rows 3 and 4: * $R\_3 \to R\_3 - 3R\_2$ * $R\_4 \to R\_4 - R\_2$

This gives:

$$\begin{pmatrix} 1 & -2 & 1 & -1 \\ 0 & 3 & -3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The matrix is now in echelon form. The number of non-zero rows is 2.

So the rank of the matrix is 2.

3.3 Q3. What is the normal form of a matrix? Find the rank of the following matrix reducing it to the normal form: (06)

ID	PYQ-2019-5b
Source	2019 Q5(b) [06 marks]

Answer:

$$\begin{pmatrix} 3 & -2 & 0 & -7 \\ 0 & 2 & 1 & -5 \\ 1 & -2 & -2 & 1 \\ 0 & 1 & 1 & -6 \end{pmatrix}$$

Answer:

3.3.0.1 1. Definition of Normal Form By applying a finite sequence of elementary row and column operations, any non-zero matrix A of rank r can be reduced to one of the following forms:

$$[I_r], \quad [I_r \quad | \quad 0], \quad \begin{bmatrix} I_r \\ -- \\ 0 \end{bmatrix}, \quad \begin{bmatrix} I_r & | & 0 \\ -- & - & -- \\ 0 & | & 0 \end{bmatrix}$$

where I_r is the identity matrix of order r. This form is called the **normal form** (or **canonical form**) of the matrix. The rank of the matrix is equal to the order of the identity matrix I_r in this form.

3.3.0.2 2. Reduction to Normal Form We write the matrix:

$$A = \begin{pmatrix} 3 & -2 & 0 & -7 \\ 0 & 2 & 1 & -5 \\ 1 & -2 & -2 & 1 \\ 0 & 1 & 1 & -6 \end{pmatrix}$$

Swap row 1 and row 3 ($R\_1 \leftrightarrow R\_3$):

$$\begin{pmatrix} 1 & -2 & -2 & 1 \\ 0 & 2 & 1 & -5 \\ 3 & -2 & 0 & -7 \\ 0 & 1 & 1 & -6 \end{pmatrix}$$

Apply row operation: * $R\_3 \to R\_3 - 3R\_1$
This gives:

$$\begin{pmatrix} 1 & -2 & -2 & 1 \\ 0 & 2 & 1 & -5 \\ 0 & 4 & 6 & -10 \\ 0 & 1 & 1 & -6 \end{pmatrix}$$

Swap row 2 and row 4 ($R\_2 \leftrightarrow R\_4$):

$$\begin{pmatrix} 1 & -2 & -2 & 1 \\ 0 & 1 & 1 & -6 \\ 0 & 4 & 6 & -10 \\ 0 & 2 & 1 & -5 \end{pmatrix}$$

Apply row operations: * $R\_3 \to R\_3 - 4R\_2$ * $R\_4 \to R\_4 - 2R\_2$
This yields:

$$\begin{pmatrix} 1 & -2 & -2 & 1 \\ 0 & 1 & 1 & -6 \\ 0 & 0 & 2 & 14 \\ 0 & 0 & -1 & 7 \end{pmatrix}$$

Scale row 3 (

$$R\_3 \to \frac{1}{2}R\_3$$

):

$$\begin{pmatrix} 1 & -2 & -2 & 1 \\ 0 & 1 & 1 & -6 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & -1 & 7 \end{pmatrix}$$

Add row 3 to row 4 ($R\_4 \to R\_4 + R\_3$):

$$\begin{pmatrix} 1 & -2 & -2 & 1 \\ 0 & 1 & 1 & -6 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 14 \end{pmatrix}$$

Scale row 4 (

$$R\_4 \to \frac{1}{14}R\_4$$

):

$$\begin{pmatrix} 1 & -2 & -2 & 1 \\ 0 & 1 & 1 & -6 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Now apply column operations to clear the off-diagonal elements. First, clear elements in row 1: * $C\_2 \to C\_2 + 2C\_1$ * $C\_3 \to C\_3 + 2C\_1$ * $C\_4 \to C\_4 - C\_1$

This gives:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -6 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Clear elements in row 2: * $C\_3 \to C\_3 - C\_2$ * $C\_4 \to C\_4 + 6C\_2$

This gives:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Clear elements in row 3: * $C\_4 \to C\_4 - 7C\_3$

This yields the identity matrix of order 4:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = I_4$$

So the normal form is I_4, and the rank of the matrix is:

$$\text{Rank} = 4$$

3.4 Q4 (05)

What is the rank of a matrix? Find the rank of the matrix,

$$A = \begin{bmatrix} 6 & 1 & 8 & 3 \\ 2 & 1 & 0 & 2 \\ 4 & -1 & -8 & -3 \end{bmatrix}$$

ID	PYQ-2020-6a
Source	2020 Q6(a) [05 marks]

Answer:

3.4.0.1 1. Definition of Rank The rank of a matrix is the maximum number of linearly independent row vectors in the matrix.

3.4.0.2 2. Find the Rank Let the matrix be:

$$A = \begin{bmatrix} 6 & 1 & 8 & 3 \\ 2 & 1 & 0 & 2 \\ 4 & -1 & -8 & -3 \end{bmatrix}$$

We swap row 1 and row 2 ($R\_1 \leftrightarrow R\_2$):

$$\begin{bmatrix} 2 & 1 & 0 & 2 \\ 6 & 1 & 8 & 3 \\ 4 & -1 & -8 & -3 \end{bmatrix}$$

Apply row operations: * $R\_2 \to R\_2 - 3R\_1$ * $R\_3 \to R\_3 - 2R\_1$
This gives:

$$\begin{bmatrix} 2 & 1 & 0 & 2 \\ 0 & -2 & 8 & -3 \\ 0 & -3 & -8 & -7 \end{bmatrix}$$

Perform the operation $R\_3 \to 2R\_3 - 3R\_2$:

$$\begin{bmatrix} 2 & 1 & 0 & 2 \\ 0 & -2 & 8 & -3 \\ 0 & 0 & -40 & -5 \end{bmatrix}$$

All three rows are non-zero. So the rank of the matrix is 3.

3.5 Q5. Find the rank of the following matrix and hence reduce it to the canonical form: (06)

ID	PYQ-2021-5b
Source	2021 Q5(b) [06 marks]

Answer:

$$A = \begin{pmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{pmatrix}$$

Answer:
We write the matrix:

$$A = \begin{pmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{pmatrix}$$

Swap row 1 and row 2 ($R\_1 \leftrightarrow R\_2$):

$$\begin{pmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{pmatrix}$$

Apply row operations: * $R\_2 \to R\_2 - 2R\_1$ * $R\_3 \to R\_3 - 3R\_1$ * $R\_4 \to R\_4 - 6R\_1$
This gives:

$$\begin{pmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{pmatrix}$$

Apply operations: * $R\_3 \to 5R\_3 - 4R\_2$ * $R\_4 \to 5R\_4 - 9R\_2$
This gives:

$$\begin{pmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 0 & 33 & 22 \\ 0 & 0 & 33 & 22 \end{pmatrix}$$

Simplify row 3 and row 4 (

$$R\_3 \to \frac{1}{11}R\_3,$$

$$R\_4 \to \frac{1}{11}R\_4$$

):

$$\begin{pmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 3 & 2 \end{pmatrix}$$

Subtract row 3 from row 4 ($R\_4 \to R\_4 - R\_3$):

$$\begin{pmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Since there are 3 non-zero rows in the echelon form, the rank is:

$$\text{Rank} = 3$$

3.5.0.1 Reduce to Canonical Form

We apply column operations to clear the first row: * $C\_2 \to C\_2 + C\_1$ * $C\_3 \to C\_3 + 2C\_1$ * $C\_4 \to C\_4 + 4C\_1$
This gives:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 3 & 7 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Scale row 2 (

$$R\_2 \to \frac{1}{5}R\_2$$

):

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 3/5 & 7/5 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Clear the second row using column operations: *

$$C\_3 \to C\_3 - \frac{3}{5}C\_2$$

•

$$C\_4 \to C\_4 - \frac{7}{5}C\_2$$

This gives:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Scale row 3 (

$$R\_3 \to \frac{1}{3}R\_3$$

):

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2/3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Clear the third column element (

$$C\_4 \to C\_4 - \frac{2}{3}C\_3$$

):

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

This is the canonical form $[I\_3|0]$:

$$\begin{pmatrix} I_3 & | & 0 \\ -- & - & -- \\ 0 & | & 0 \end{pmatrix}$$

3.6 Q6. Reduce the following matrix to the canonical form: (03)

ID	PYQ-2023-4c
Source	2023 Q4(c) [03 marks]

Answer:

$$A = \begin{bmatrix} 0 & 0 & 1 & 3 & -2 \\ 0 & 1 & 2 & 6 & 0 \\ 0 & 2 & 3 & 9 & 2 \\ 0 & 1 & 1 & 3 & 2 \end{bmatrix}$$

Answer:
We write the matrix:

$$A = \begin{bmatrix} 0 & 0 & 1 & 3 & -2 \\ 0 & 1 & 2 & 6 & 0 \\ 0 & 2 & 3 & 9 & 2 \\ 0 & 1 & 1 & 3 & 2 \end{bmatrix}$$

Swap row 1 and row 2 ($R\_1 \leftrightarrow R\_2$):

$$\begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 2 & 3 & 9 & 2 \\ 0 & 1 & 1 & 3 & 2 \end{bmatrix}$$

Apply row operations: * $R\_3 \to R\_3 - 2R\_1$ * $R\_4 \to R\_4 - R\_1$
This gives:

$$\begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & -1 & -3 & 2 \\ 0 & 0 & -1 & -3 & 2 \end{bmatrix}$$

Apply row operations: * $R\_3 \to R\_3 + R\_2$ * $R\_4 \to R\_4 + R\_2$
This yields:

$$\begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Clear columns using column operations: * $C\_3 \to C\_3 - 2C\_2$ * $C\_4 \to C\_4 - 6C\_2$
This gives:

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Clear column elements using column 3: * $C\_4 \to C\_4 - 3C\_3$ * $C\_5 \to C\_5 + 2C\_3$
This gives:

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Now swap columns to group the identity matrix blocks. Swap $C\_1 \leftrightarrow C\_2$:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Swap $C\_2 \leftrightarrow C\_3$:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

This is the canonical form $[I\_2|0]$:

$$\begin{pmatrix} I_2 & | & 0 \\ -- & - & -- \\ 0 & | & 0 \end{pmatrix}$$

3.7 Q7. Find the rank of the following matrix: (03)

ID	PYQ-2023-4d
Source	2023 Q4(d) [03 marks]

Answer:

$$A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$$

Answer:
We write the matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$$

Apply row operations: * $R\_2 \to R\_2 - 2R\_1$ * $R\_3 \to R\_3 - R\_1$
This gives:

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 1 & 1 & 3 \end{bmatrix}$$

Add row 2 to row 3 ($R\_3 \to R\_3 + R\_2$):

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

There are 2 non-zero rows in the echelon form. So the rank of the matrix is:

$$\text{Rank} = 2$$

3.8 SECTION - B

3.9 Q8. Define rank of a matrix. Find the rank of: (05)

ID	PYQ-2024-5a
Source	2024 Q5(a) [05 marks]

Answer:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & -2 & 1 \\ 2 & 0 & -3 & 2 \\ 3 & 3 & -3 & 3 \end{bmatrix}$$

Answer:

3.9.0.1 1. Definition of Rank The rank of a matrix is the maximum number of linearly independent row vectors in the matrix. This is also equal to the maximum number of linearly independent column vectors.

3.9.0.2 2. Find the Rank of Matrix A

We write the matrix:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & -2 & 1 \\ 2 & 0 & -3 & 2 \\ 3 & 3 & -3 & 3 \end{bmatrix}$$

We apply row operations to reduce it to echelon form: * $R\_2 \to R\_2 - R\_1$ * $R\_3 \to R\_3 - 2R\_1$ * $R\_4 \to R\_4 - 3R\_1$

This gives:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & -3 & 0 \\ 0 & -2 & -5 & 0 \\ 0 & 0 & -6 & 0 \end{bmatrix}$$

Next, we apply: * $R\_3 \to R\_3 + R\_2$

This gives:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & -3 & 0 \\ 0 & 0 & -8 & 0 \\ 0 & 0 & -6 & 0 \end{bmatrix}$$

We scale the third row (

$$R\_3 \to -\frac{1}{8}R\_3$$

):

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & -3 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -6 & 0 \end{bmatrix}$$

Now we apply: * $R\_4 \to R\_4 + 6R\_3$

This yields:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & -3 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

There are 3 non-zero rows in the echelon form. Thus, the rank of the matrix is 3.

3.10 Q9 (07)

What is meant by the rank of a matrix? Reduce the matrix

$$A = \begin{bmatrix} 6 & 3 & -4 \\ -4 & 1 & -6 \\ 1 & 2 & -5 \end{bmatrix}$$

ID	CT1M-3
Source	CT1M Q3 [07 marks]

Answer:

- **(i)** to echelon form
- **(ii)** to row canonical form, and
- **(iii)** determine the rank of the matrix.

Answer:

3.10.0.1 1. Definition of Rank

The rank of a matrix is the maximum number of linearly independent rows in the matrix. It also equals the number of non-zero rows in its row echelon form.

3.10.0.2 2. Reduce to Echelon Form

Let the matrix be:

$$A = \begin{bmatrix} 6 & 3 & -4 \\ -4 & 1 & -6 \\ 1 & 2 & -5 \end{bmatrix}$$

We swap the first and third rows to put a 1 in the top-left corner:

$$\begin{bmatrix} 1 & 2 & -5 \\ -4 & 1 & -6 \\ 6 & 3 & -4 \end{bmatrix} \quad (R_1 \leftrightarrow R_3)$$

Now we clear the first column:

$$R_2 \to R_2 + 4R_1$$

$$R_3 \to R_3 - 6R_1$$

$$\begin{bmatrix} 1 & 2 & -5 \\ 0 & 9 & -26 \\ 0 & -9 & 26 \end{bmatrix}$$

Next, we eliminate the second element in the third row:

$$R_3 \to R_3 + R_2$$

$$\begin{bmatrix} 1 & 2 & -5 \\ 0 & 9 & -26 \\ 0 & 0 & 0 \end{bmatrix}$$

We divide the second row by 9 to get a leading 1:

$$R_2 \to \frac{1}{9}R_2$$

$$\begin{bmatrix} 1 & 2 & -5 \\ 0 & 1 & -\frac{26}{9} \\ 0 & 0 & 0 \end{bmatrix}$$

This is the echelon form of the matrix.

3.10.0.3 3. Reduce to Row Canonical Form

Starting from the echelon form:

$$\begin{bmatrix} 1 & 2 & -5 \\ 0 & 1 & -\frac{26}{9} \\ 0 & 0 & 0 \end{bmatrix}$$

We eliminate the element above the leading 1 in the second row:

$$R_1 \to R_1 - 2R_2$$

$$R_1 \to \begin{bmatrix} 1 & 2-2(1) & -5-2\left(-\frac{26}{9}\right) \end{bmatrix} = \begin{bmatrix} 1 & 0 & -5+\frac{52}{9} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \frac{7}{9} \end{bmatrix}$$

This gives the row canonical form:

$$\begin{bmatrix} 1 & 0 & \frac{7}{9} \\ 0 & 1 & -\frac{26}{9} \\ 0 & 0 & 0 \end{bmatrix}$$

3.10.0.4 4. Determine the Rank

The row echelon form has two non-zero rows. So, the rank of the matrix A is 2.

4 Topic 08: Cayley-Hamilton Theorem

This file contains the organized questions and answers for **Cayley-Hamilton Theorem**, priority ranked as **Priority 8** based on frequency and exam weight.

4.1 Q1. State and prove Cayley-Hamilton theorem. (06)

ID	PYQ-2018-7b
Source	2018 Q7(b) [06 marks]

Answer:

4.1.0.1 Statement Every square matrix satisfies its own characteristic equation. If the characteristic polynomial of a matrix A of order n is:

$$p(\lambda) = (-1)^n \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_0 = 0$$

then we have:

$$p(A) = (-1)^n A^n + c_{n-1}A^{n-1} + \cdots + c_0 I = 0$$

4.1.0.2 Proof The characteristic matrix is $A - \lambda I$. The adjoint matrix $\text{adj}(A - \lambda I)$ contains polynomial elements in λ of degree at most $n-1$. We can write it as:

$$\text{adj}(A - \lambda I) = B_{n-1}\lambda^{n-1} + B_{n-2}\lambda^{n-2} + \cdots + B_0$$

where B_i are constant matrices of order n. We use the fundamental matrix relation:

$$(A - \lambda I)\text{adj}(A - \lambda I) = |A - \lambda I| I$$

Let the characteristic polynomial be

$$|A - \lambda I| = c\_0 + c\_1\lambda + \cdots + (-1)^n \lambda^n$$

. Substitute this and the adjoint expression:

$$(A - \lambda I)(B_{n-1}\lambda^{n-1} + B_{n-2}\lambda^{n-2} + \cdots + B_0) = (c_0 + c_1\lambda + \cdots + (-1)^n\lambda^n)I$$

Multiply the terms on the left side:

$$AB_{n-1}\lambda^{n-1} + \cdots + AB_0 - B_{n-1}\lambda^n - \cdots - B_0\lambda = c_0 I + c_1 I\lambda + \cdots + (-1)^n I\lambda^n$$

Equating the coefficients of like powers of λ:

$$AB_0 = c_0 I$$

$$AB_1 - B_0 = c_1 I$$

$$AB_2 - B_1 = c_2 I$$

$$\ldots$$

$$AB_{n-1} - B_{n-2} = c_{n-1} I$$

$$-B_{n-1} = (-1)^n I$$

Now we pre-multiply these equations by $I, A, A^2, \ldots, A^n$ respectively:

$$AB_0 = c_0 I$$

$$A^2B_1 - AB_0 = c_1A$$
$$A^3B_2 - A^2B_1 = c_2A^2$$
$$\dots$$
$$A^nB_{n-1} - A^{n-1}B_{n-2} = c_{n-1}A^{n-1}$$
$$-A^nB_{n-1} = (-1)^nA^n$$

Add all these equations together. The terms on the left side cancel out to zero:

$$0 = c_0I + c_1A + c_2A^2 + \cdots + (-1)^nA^n = p(A)$$

So $p(A) = 0$. The theorem is proven.

4.2 Q2 (06)

Verify Cayley-Hamilton's theorem for

$$A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$$

and hence find A^{-1}.

ID	PYQ-2019-7b
Source	2019 Q7(b) [06 marks]

Answer:

4.2.0.1 1. Find characteristic equation The characteristic equation is $|A - \lambda I| = 0$:

$$\begin{vmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix} = 0$$

Calculate the determinant:

$$(2-\lambda)\left[(2-\lambda)^2 - 1\right] - (-1)\left[-(2-\lambda) - (-1)\right] + 1\left[1 - (2-\lambda)\right] = 0$$

$$(2-\lambda)(\lambda^2 - 4\lambda + 3) + (\lambda - 1) + (\lambda - 1) = 0$$
$$-\lambda^3 + 6\lambda^2 - 9\lambda + 4 = 0 \implies \lambda^3 - 6\lambda^2 + 9\lambda - 4I = 0$$

4.2.0.2 2. Verification of Cayley-Hamilton Theorem According to the theorem, the matrix A satisfies its own characteristic equation:

$$A^3 - 6A^2 + 9A - 4I = 0$$

We calculate A^2 and A^3:

$$A^2 = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}\begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix} = \begin{pmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{pmatrix}$$

$$A^3 = A^2A = \begin{pmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{pmatrix}\begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix} = \begin{pmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{pmatrix}$$

Now substitute A^3, A^2, A, and I into the expression:

$$A^3 - 6A^2 + 9A - 4I = \begin{pmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{pmatrix} - \begin{pmatrix} 36 & -30 & 30 \\ -30 & 36 & -30 \\ 30 & -30 & 36 \end{pmatrix} + \begin{pmatrix} 18 & -9 & 9 \\ -9 & 18 & -9 \\ 9 & -9 & 18 \end{pmatrix} - \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

Calculate the matrix elements: * Diagonal elements: $22 - 36 + 18 - 4 = 0$ * Off-diagonal elements (e.g. Row 1 Col 2): $-21 - (-30) + (-9) - 0 = 0$

All entries are zero. Cayley-Hamilton's Theorem is verified.

4.2.0.3 3. Find A^{-1}

Multiply the equation by A^{-1}:

$$A^2 - 6A + 9I - 4A^{-1} = 0 \implies 4A^{-1} = A^2 - 6A + 9I$$

$$4A^{-1} = \begin{pmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{pmatrix} - \begin{pmatrix} 12 & -6 & 6 \\ -6 & 12 & -6 \\ 6 & -6 & 12 \end{pmatrix} + \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix}$$

$$4A^{-1} = \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{pmatrix} \implies A^{-1} = \frac{1}{4} \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{pmatrix}$$

4.3 Q3. Verify Cayley Hamilton's theorem for the matrix: (05)

ID	PYQ-2024-6a
Source	2024 Q6(a) [05 marks]

Answer:

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

Hence compute A^{-1}.

Answer:

4.3.0.1 1. Characteristic Equation

We find the characteristic equation $|A - \lambda I| = 0$:

$$\begin{vmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{vmatrix} = 0$$

We expand the determinant:

$$(2-\lambda)\left[(3-\lambda)(2-\lambda) - 2\right] - 2\left[1(2-\lambda) - 1\right] + 1\left[2 - (3-\lambda)\right] = 0$$

$$(2-\lambda)\left[\lambda^2 - 5\lambda + 4\right] - 2\left[1 - \lambda\right] + 1\left[\lambda - 1\right] = 0$$

$$(2\lambda^2 - 10\lambda + 8 - \lambda^3 + 5\lambda^2 - 4\lambda) - (2 - 2\lambda) + (\lambda - 1) = 0$$

$$-\lambda^3 + 7\lambda^2 - 14\lambda + 8 - 2 + 2\lambda + \lambda - 1 = 0$$

$$-\lambda^3 + 7\lambda^2 - 11\lambda + 5 = 0$$

We multiply by -1:

$$\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$$

4.3.0.2 2. Verification of Cayley-Hamilton Theorem

We show that matrix A satisfies this equation:

$$A^3 - 7A^2 + 11A - 5I = 0$$

First, we calculate A^2:

$$A^2 = A \cdot A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 12 & 6 \\ 6 & 13 & 6 \\ 6 & 12 & 7 \end{bmatrix}$$

Next, we calculate A^3:

$$A^3 = A^2 \cdot A = \begin{bmatrix} 7 & 12 & 6 \\ 6 & 13 & 6 \\ 6 & 12 & 7 \end{bmatrix} \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 32 & 62 & 31 \\ 31 & 63 & 31 \\ 31 & 62 & 32 \end{bmatrix}$$

Now we compute $A^3 - 7A^2 + 11A - 5I$:

$$\begin{bmatrix} 32 & 62 & 31 \\ 31 & 63 & 31 \\ 31 & 62 & 32 \end{bmatrix} - 7 \begin{bmatrix} 7 & 12 & 6 \\ 6 & 13 & 6 \\ 6 & 12 & 7 \end{bmatrix} + 11 \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 32 - 49 + 22 - 5 & 62 - 84 + 22 & 31 - 42 + 11 \\ 31 - 42 + 11 & 63 - 91 + 33 - 5 & 31 - 42 + 11 \\ 31 - 42 + 11 & 62 - 84 + 22 & 32 - 49 + 22 - 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

All entries are zero. Cayley-Hamilton's Theorem is verified.

4.3.0.3 3. Compute A^{-1}

We multiply the characteristic equation by A^{-1}:

$$A^{-1}(A^3 - 7A^2 + 11A - 5I) = 0$$

$$A^2 - 7A + 11I - 5A^{-1} = 0 \implies 5A^{-1} = A^2 - 7A + 11I$$

$$A^{-1} = \frac{1}{5}(A^2 - 7A + 11I)$$

We compute the right side:

$$A^2 - 7A + 11I = \begin{bmatrix} 7 & 12 & 6 \\ 6 & 13 & 6 \\ 6 & 12 & 7 \end{bmatrix} - \begin{bmatrix} 14 & 14 & 7 \\ 7 & 21 & 7 \\ 7 & 14 & 14 \end{bmatrix} + \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} 7 - 14 + 11 & 12 - 14 + 0 & 6 - 7 + 0 \\ 6 - 7 + 0 & 13 - 21 + 11 & 6 - 7 + 0 \\ 6 - 7 + 0 & 12 - 14 + 0 & 7 - 14 + 11 \end{bmatrix} = \begin{bmatrix} 4 & -2 & -1 \\ -1 & 3 & -1 \\ -1 & -2 & 4 \end{bmatrix}$$

So, the inverse is:

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 4 & -2 & -1 \\ -1 & 3 & -1 \\ -1 & -2 & 4 \end{bmatrix}$$

5 Topic 09: Adjoint and Inverse

This file contains the organized questions and answers for **Adjoint and Inverse**, priority ranked as **Priority 9** based on frequency and exam weight.

5.1 Q1 (05)

Find the adjoint and inverse of

$$\begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{pmatrix}$$

ID	PYQ-2017-5b
Source	2017 Q5(b) [05 marks]

Answer:
Let the matrix be:

$$A = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{pmatrix}$$

First we find the determinant of A:

$$|A| = 0(2-3) - 1(1-9) + 2(1-6) = 0 + 8 - 10 = -2$$

Since the determinant is not zero, the inverse exists.
Next we find the cofactor elements C_ij of the matrix:

$$C_{11} = +(2-3) = -1, \quad C_{12} = -(1-9) = 8, \quad C_{13} = +(1-6) = -5$$

$$C_{21} = -(1-2) = 1, \quad C_{22} = +(0-6) = -6, \quad C_{23} = -(0-3) = 3$$

$$C_{31} = +(3-4) = -1, \quad C_{32} = -(0-2) = 2, \quad C_{33} = +(0-1) = -1$$

This gives the cofactor matrix C:

$$C = \begin{pmatrix} -1 & 8 & -5 \\ 1 & -6 & 3 \\ -1 & 2 & -1 \end{pmatrix}$$

The adjoint of A is the transpose of the cofactor matrix:

$$\text{adj}(A) = C^T = \begin{pmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{pmatrix}$$

Now we find the inverse matrix A^{-1}:

$$A^{-1} = \frac{1}{|A|}\text{adj}(A) = -\frac{1}{2}\begin{pmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{pmatrix} = \begin{pmatrix} 1/2 & -1/2 & 1/2 \\ -4 & 3 & -1 \\ 5/2 & -3/2 & 1/2 \end{pmatrix}$$

5.2 Q2. What is an inverse of a matrix? Find the inverse of the following matrix: (06)

ID	PYQ-2019-5a
Source	2019 Q5(a) [06 marks]

Answer:

$$\begin{pmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{pmatrix}$$

Answer:

5.2.0.1 1. Definition of Inverse Matrix

Let A be a square matrix of order n. If there exists a square matrix B of order n such that:

$$AB = BA = I_n$$

(where I_n is the identity matrix of order n), then B is called the inverse of A. The inverse matrix is unique and is denoted by A^{-1}.

5.2.0.2 2. Find the Inverse

We set up the augmented matrix $[A|I]$:

$$\begin{bmatrix} 1 & 3 & 3 & | & 1 & 0 & 0 \\ 1 & 4 & 3 & | & 0 & 1 & 0 \\ 1 & 3 & 4 & | & 0 & 0 & 1 \end{bmatrix}$$

Apply row operations to clear the first column: * $R\_2 \to R\_2 - R\_1$ * $R\_3 \to R\_3 - R\_1$
This gives:

$$\begin{bmatrix} 1 & 3 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & 0 & | & -1 & 1 & 0 \\ 0 & 0 & 1 & | & -1 & 0 & 1 \end{bmatrix}$$

Apply row operation to clear the second column: * $R\_1 \to R\_1 - 3R\_2$
This gives:

$$\begin{bmatrix} 1 & 0 & 3 & | & 4 & -3 & 0 \\ 0 & 1 & 0 & | & -1 & 1 & 0 \\ 0 & 0 & 1 & | & -1 & 0 & 1 \end{bmatrix}$$

Apply row operation to clear the third column: * $R\_1 \to R\_1 - 3R\_3$
This yields:

$$\begin{bmatrix} 1 & 0 & 0 & | & 7 & -3 & -3 \\ 0 & 1 & 0 & | & -1 & 1 & 0 \\ 0 & 0 & 1 & | & -1 & 0 & 1 \end{bmatrix}$$

So the inverse matrix is:

$$A^{-1} = \begin{pmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

5.3 Q3 (05)

Define inverse of a matrix. Find the adjoint of matrix A and hence find A^{-1}, where:

ID	PYQ-2020-5c
Source	2020 Q5(c) [05 marks]

Answer:

$$A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Answer:

5.3.0.1 1. Definition

Let A be a square matrix. If there is a matrix B such that $AB = BA = I$, then B is the inverse of A (written as A^{-1}).

5.3.0.2 2. Find Adjoint and Inverse

We find the determinant of A:

$$|A| = 1(\cos^2\theta - (-\sin^2\theta)) = \cos^2\theta + \sin^2\theta = 1$$

Find the cofactors C_ij:

$$C_{11} = \cos\theta, \quad C_{12} = -\sin\theta, \quad C_{13} = 0$$
$$C_{21} = \sin\theta, \quad C_{22} = \cos\theta, \quad C_{23} = 0$$
$$C_{31} = 0, \quad C_{32} = 0, \quad C_{33} = 1$$

This gives the cofactor matrix C:

$$C = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The adjoint matrix is the transpose of the cofactor matrix:

$$\mathrm{adj}(A) = C^T = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Since the determinant is 1, the inverse is:

$$A^{-1} = \frac{1}{|A|}\mathrm{adj}(A) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

5.4 Q4. Find the inverse of the following matrix by using row operation: (06)

ID	PYQ-2021-6a
Source	2021 Q6(a) [06 marks]

Answer:

$$A = \begin{pmatrix} 2 & 1 & 2 \\ 2 & 2 & 1 \\ 1 & 2 & 2 \end{pmatrix}$$

Answer:

We write the augmented matrix $[A|I]$:

$$\begin{bmatrix} 2 & 1 & 2 & | & 1 & 0 & 0 \\ 2 & 2 & 1 & | & 0 & 1 & 0 \\ 1 & 2 & 2 & | & 0 & 0 & 1 \end{bmatrix}$$

Swap row 1 and row 3 ($R\_1 \leftrightarrow R\_3$):

$$\begin{bmatrix} 1 & 2 & 2 & | & 0 & 0 & 1 \\ 2 & 2 & 1 & | & 0 & 1 & 0 \\ 2 & 1 & 2 & | & 1 & 0 & 0 \end{bmatrix}$$

Apply row operations: * $R\_2 \to R\_2 - 2R\_1$ * $R\_3 \to R\_3 - 2R\_1$
This gives:

$$\begin{bmatrix} 1 & 2 & 2 & | & 0 & 0 & 1 \\ 0 & -2 & -3 & | & 0 & 1 & -2 \\ 0 & -3 & -2 & | & 1 & 0 & -2 \end{bmatrix}$$

Multiply row 2 by $-1/2$ ($R\_2 \to -1/2R\_2$):

$$\begin{bmatrix} 1 & 2 & 2 & | & 0 & 0 & 1 \\ 0 & 1 & 3/2 & | & 0 & -1/2 & 1 \\ 0 & -3 & -2 & | & 1 & 0 & -2 \end{bmatrix}$$

Eliminate the third row element ($R\_3 \to R\_3 + 3R\_2$):

$$\begin{bmatrix} 1 & 2 & 2 & | & 0 & 0 & 1 \\ 0 & 1 & 3/2 & | & 0 & -1/2 & 1 \\ 0 & 0 & 5/2 & | & 1 & -3/2 & 1 \end{bmatrix}$$

Multiply row 3 by $2/5$ ($R\_3 \to 2/5R\_3$):

$$\begin{bmatrix} 1 & 2 & 2 & | & 0 & 0 & 1 \\ 0 & 1 & 3/2 & | & 0 & -1/2 & 1 \\ 0 & 0 & 1 & | & 2/5 & -3/5 & 2/5 \end{bmatrix}$$

Perform the operation

$$R\_2 \to R\_2 - \frac{3}{2}R\_3:$$

$$\begin{bmatrix} 1 & 2 & 2 & | & 0 & 0 & 1 \\ 0 & 1 & 0 & | & -3/5 & 2/5 & 2/5 \\ 0 & 0 & 1 & | & 2/5 & -3/5 & 2/5 \end{bmatrix}$$

Perform the operation $R\_1 \to R\_1 - 2R\_3$:

$$\begin{bmatrix} 1 & 2 & 0 & | & -4/5 & 6/5 & 1/5 \\ 0 & 1 & 0 & | & -3/5 & 2/5 & 2/5 \\ 0 & 0 & 1 & | & 2/5 & -3/5 & 2/5 \end{bmatrix}$$

Perform the operation $R\_1 \to R\_1 - 2R\_2$:

$$\begin{bmatrix} 1 & 0 & 0 & | & 2/5 & 2/5 & -3/5 \\ 0 & 1 & 0 & | & -3/5 & 2/5 & 2/5 \\ 0 & 0 & 1 & | & 2/5 & -3/5 & 2/5 \end{bmatrix}$$

So the inverse is:

$$A^{-1} = \frac{1}{5}\begin{pmatrix} 2 & 2 & -3 \\ -3 & 2 & 2 \\ 2 & -3 & 2 \end{pmatrix}$$

5.5 Q5 (05)

Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 5 \\ 3 & 7 & 6 \end{bmatrix}$$

using row transformation.

ID	PYQ-2023-5b
Source	2023 Q5(b) [05 marks]

Answer:
We set up the augmented matrix $[A|I]$:

$$\begin{bmatrix} 1 & 3 & 5 & | & 1 & 0 & 0 \\ 2 & 4 & 5 & | & 0 & 1 & 0 \\ 3 & 7 & 6 & | & 0 & 0 & 1 \end{bmatrix}$$

Apply row operations: * $R\_2 \to R\_2 - 2R\_1$ * $R\_3 \to R\_3 - 3R\_1$
This gives:

$$\begin{bmatrix} 1 & 3 & 5 & | & 1 & 0 & 0 \\ 0 & -2 & -5 & | & -2 & 1 & 0 \\ 0 & -2 & -9 & | & -3 & 0 & 1 \end{bmatrix}$$

Subtract row 2 from row 3 ($R\_3 \to R\_3 - R\_2$):

$$\begin{bmatrix} 1 & 3 & 5 & | & 1 & 0 & 0 \\ 0 & -2 & -5 & | & -2 & 1 & 0 \\ 0 & 0 & -4 & | & -1 & -1 & 1 \end{bmatrix}$$

Multiply row 2 by $-1/2$ and row 3 by $-1/4$: * $R\_2 \to -1/2R\_2$ * $R\_3 \to -1/4R\_3$
This gives:

$$\begin{bmatrix} 1 & 3 & 5 & | & 1 & 0 & 0 \\ 0 & 1 & 5/2 & | & 1 & -1/2 & 0 \\ 0 & 0 & 1 & | & 1/4 & 1/4 & -1/4 \end{bmatrix}$$

Perform the operation

$$R\_2 \to R\_2 - \frac{5}{2}R\_3 :$$

$$\begin{bmatrix} 1 & 3 & 5 & | & 1 & 0 & 0 \\ 0 & 1 & 0 & | & 3/8 & -9/8 & 5/8 \\ 0 & 0 & 1 & | & 1/4 & 1/4 & -1/4 \end{bmatrix}$$

Perform the operation $R\_1 \to R\_1 - 5R\_3$:

$$\begin{bmatrix} 1 & 3 & 0 & | & -1/4 & -5/4 & 5/4 \\ 0 & 1 & 0 & | & 3/8 & -9/8 & 5/8 \\ 0 & 0 & 1 & | & 1/4 & 1/4 & -1/4 \end{bmatrix}$$

Perform the operation $R\_1 \to R\_1 - 3R\_2$:

$$\begin{bmatrix} 1 & 0 & 0 & | & -11/8 & 17/8 & -5/8 \\ 0 & 1 & 0 & | & 3/8 & -9/8 & 5/8 \\ 0 & 0 & 1 & | & 1/4 & 1/4 & -1/4 \end{bmatrix}$$

So the inverse is:

$$A^{-1} = \frac{1}{8} \begin{bmatrix} -11 & 17 & -5 \\ 3 & -9 & 5 \\ 2 & 2 & -2 \end{bmatrix}$$

6 Topic 10: Matrix Types and Properties

This file contains the organized questions and answers for **Matrix Types and Properties**, priority ranked as **Priority 10** based on frequency and exam weight.

6.1 Q1 (03)

If

$$A_\alpha = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix}$$

Then show that

$$A\_\alpha \cdot A\_\beta = A\_\alpha + \beta = A\_\beta \cdot A\_\alpha.$$

ID	PYQ-2017-5a
Source	2017 Q5(a) [03 marks]

Answer:

Let us multiply the two matrices A_α and A_β:

$$A_\alpha \cdot A_\beta = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} \cos\beta & \sin\beta \\ -\sin\beta & \cos\beta \end{pmatrix}$$

Multiply rows by columns:

$$A_\alpha \cdot A_\beta = \begin{pmatrix} \cos\alpha\cos\beta - \sin\alpha\sin\beta & \cos\alpha\sin\beta + \sin\alpha\cos\beta \\ -\sin\alpha\cos\beta - \cos\alpha\sin\beta & -\sin\alpha\sin\beta + \cos\alpha\cos\beta \end{pmatrix}$$

We apply standard trigonometric addition formulas:

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Substitute these identities back into the product matrix:

$$A_\alpha \cdot A_\beta = \begin{pmatrix} \cos(\alpha+\beta) & \sin(\alpha+\beta) \\ -\sin(\alpha+\beta) & \cos(\alpha+\beta) \end{pmatrix} = A_{\alpha+\beta}$$

Since addition of angles is commutative ($\alpha + \beta = \beta + \alpha$), we also have:

$$A_{\alpha+\beta} = A_{\beta+\alpha} = A_\beta \cdot A_\alpha$$

So we have shown that

$$A\_\alpha \cdot A\_\beta = A\_\alpha + \beta = A\_\beta \cdot A\_\alpha.$$

6.2 Q2. Define symmetric, skew-symmetric, and Hermitian matrices with examples. (03)

ID	PYQ-2018-5a
Source	2018 Q5(a) [03 marks]

Answer:

6.2.0.1 Symmetric Matrix

A square matrix A is symmetric if it equals its transpose:

$$A^T = A$$

Example:

$$\begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$$

6.2.0.2 Skew-Symmetric Matrix

A square matrix A is skew-symmetric if its transpose equals its negative:

$$A^T = -A$$

The diagonal elements of a skew-symmetric matrix are always zero.

Example:

$$\begin{pmatrix} 0 & -3 \\ 3 & 0 \end{pmatrix}$$

6.2.0.3 Hermitian Matrix

A square matrix A with complex entries is Hermitian if it equals its conjugate transpose:

$$A^\dagger = (\bar{A})^T = A$$

Example:

$$\begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix}$$

6.3 Q3 (05)

Define singular matrix. If A and B are non-singular matrices of the same order, then show that

$$(AB)^{-1} = B^{-1}A^{-1}$$

. Hence prove that

$$(A^{-1})^n = (A^n)^{-1}$$

for any positive integer.

ID	PYQ-2018-5b
Appeared in	2018 Q5(b) [05 marks], 2023 Q4(b) [02 marks]
Frequency	⋆⋆ (2 times)

Answer:

6.3.0.1 1. Singular Matrix

A square matrix is singular if its determinant is zero. It has no inverse.

6.3.0.2 2. Prove $(AB)^{-1} = B^{-1}A^{-1}$

We multiply AB by $B^{-1}A^{-1}$:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I$$

Since the product in both directions is the identity matrix, we get:

$$(AB)^{-1} = B^{-1}A^{-1}$$

6.3.0.3 3. Prove $(A^{-1})^n = (A^n)^{-1}$

We use mathematical induction on n:

- **Base Case ($n = 1$):**

$$(A^{-1})^1 = (A^1)^{-1}$$

, which is true. * **Inductive Step:** Assume the statement is true for $n = k$:

$$(A^{-1})^k = (A^k)^{-1}$$

We test for $n = k + 1$:

$$(A^{k+1})^{-1} = (A^k \cdot A)^{-1}$$

Using the product inverse property shown above:

$$(A^k \cdot A)^{-1} = A^{-1} \cdot (A^k)^{-1}$$

Substitute the induction assumption:

$$A^{-1} \cdot (A^k)^{-1} = A^{-1} \cdot (A^{-1})^k = (A^{-1})^{k+1}$$

So the statement is true for $n = k + 1$. The proof is complete.

6.4 Q4 (03)

For two matrices A and B, prove that $(AB)' = B'A'$.

ID	PYQ-2020-5a
Source	2020 Q5(a) [03 marks]

Answer:
Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Let the product matrix be $C = AB$. The (i, j)-th element of C is:

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

The transpose element $(C')\_ij$ is defined as the (j, i)-th element of C:

$$(C')_{ij} = c_{ji} = \sum_{k=1}^{n} a_{jk} b_{ki}$$

Now look at the product of the transposes $B'A'$. The (i, j)-th element of this product is:

$$(B'A')_{ij} = \sum_{k=1}^{n} (B')_{ik} (A')_{kj}$$

Using the transpose definitions

$$(B')\_ik = b\_ki$$

and

$$(A')\_kj = a\_jk$$

, we substitute:

$$(B'A')_{ij} = \sum_{k=1}^{n} b_{ki} a_{jk} = \sum_{k=1}^{n} a_{jk} b_{ki}$$

Since

$$(AB)'\_ij = (B'A')\_ij$$

for all indices, we get:

$$(AB)' = B'A'$$

The proof is complete.

6.5 Q5. Show that every square matrix can be uniquely expressed as the sum of a symmetric and a skew symmetric matrices. (04)

ID	PYQ-2020-5b
Appeared in	2020 Q5(b) [04 marks], 2021 Q5(a) [06 marks]
Frequency	$\star\star$ (2 times)

Answer:
Let A be a square matrix. We can decompose A as:

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

Let

$$P = \frac{1}{2}(A + A^T)$$

and

$$Q = \frac{1}{2}(A - A^T).$$

6.5.0.1 1. Show P is symmetric

Calculate the transpose of P:

$$P^T = \left[\frac{1}{2}(A + A^T)\right]^T = \frac{1}{2}(A^T + (A^T)^T) = \frac{1}{2}(A^T + A) = P$$

So P is symmetric.

6.5.0.2 2. Show Q is skew-symmetric

Calculate the transpose of Q:

$$Q^T = \left[\frac{1}{2}(A - A^T)\right]^T = \frac{1}{2}(A^T - (A^T)^T) = \frac{1}{2}(A^T - A) = -Q$$

So Q is skew-symmetric.

6.5.0.3 3. Prove Uniqueness

Suppose there is another decomposition:

$$A = P' + Q'$$

where P' is symmetric and Q' is skew-symmetric. Taking the transpose:

$$A^T = (P' + Q')^T = P'^T + Q'^T = P' - Q'$$

We solve the equations: 1. $P' + Q' = A$ 2.

$$P' - Q' = A^T$$

Adding them gives:

$$2P' = A + A^T \implies P' = \frac{1}{2}(A + A^T) = P$$

Subtracting them gives:

$$2Q' = A - A^T \implies Q' = \frac{1}{2}(A - A^T) = Q$$

The decomposition is unique.

6.6 Q6. Define commutative, idempotent and involutory matrix. (02)

ID	PYQ-2023-4a
Source	2023 Q4(a) [02 marks]

Answer:

- **Commutative Matrices:** Two square matrices A and B are commutative if $AB = BA$.
- **Idempotent Matrix:** A square matrix A is idempotent if its square equals itself:
$$A^2 = A.$$
- **Involutory Matrix:** A square matrix A is involutory if its square equals the identity matrix:
$$A^2 = I.$$

6.7 Q7. Show that every square matrix can be expressed as the sum of a Hermitian matrix and a Skew-Hermitian matrix. (05)

ID	PYQ-2023-5a
Source	2023 Q5(a) [05 marks]

Answer:

Let A be a square matrix. We write A as:

$$A = \frac{1}{2}(A + A^\dagger) + \frac{1}{2}(A - A^\dagger)$$

where

$$A^\dagger = (\bar{A})^T$$

is the conjugate transpose of A. Let:

$$P = \frac{1}{2}(A + A^\dagger) \quad \text{and} \quad Q = \frac{1}{2}(A - A^\dagger)$$

6.7.0.1 1. Show P is Hermitian

$$P^\dagger = \left[\frac{1}{2}(A + A^\dagger)\right]^\dagger = \frac{1}{2}(A^\dagger + (A^\dagger)^\dagger) = \frac{1}{2}(A^\dagger + A) = P$$

So P is a Hermitian matrix.

6.7.0.2 2. Show Q is Skew-Hermitian

$$Q^\dagger = \left[\frac{1}{2}(A - A^\dagger)\right]^\dagger = \frac{1}{2}(A^\dagger - (A^\dagger)^\dagger) = \frac{1}{2}(A^\dagger - A) = -Q$$

So Q is a Skew-Hermitian matrix.

So $A = P + Q$ is the sum of a Hermitian matrix and a Skew-Hermitian matrix.

7 Topic 18: Matrix ODE and Diagonalization

This file contains the organized questions and answers for **Matrix ODE and Diagonalization**, priority ranked as **Priority 18** based on frequency and exam weight.

7.1 Q1 (12)

Diagonalize the matrix

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Hence evaluate e^A, and then solve

$$\frac{dx\_1}{dt} = x\_2,$$
$$\frac{dx\_2}{dt} = x\_1.$$

ID	PYQ-2017-8
Source	2017 Q8 [12 marks]

Answer:

7.1.0.1 1. Diagonalization check

Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Its eigenvalues are the diagonal entries:

$$\lambda_1 = 1, \quad \lambda_2 = 1$$

To check if A is diagonalizable, we find the eigenvectors by solving $(A - I)X = 0$:

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies y = 0$$

The eigenvectors are of the form

$$\begin{bmatrix} x \\ 0 \end{bmatrix}$$

So there is only one linearly independent eigenvector:

$$X_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Since the geometric multiplicity (1) is less than the algebraic multiplicity (2), **the matrix A is not diagonalizable**.

7.1.0.2 2. Evaluation of e^A

Even though A is not diagonalizable, we can evaluate e^A using the power series definition. We decompose A into:

$$A = I + B, \quad \text{where } B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Since the identity matrix I commutes with any matrix ($IB = BI$), we can write:

$$e^A = e^{I+B} = e^I \cdot e^B = e \cdot e^B$$

Note that

$$B^2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

So

$$B^n = 0$$

for all $n \geq 2$.
Using the series definition for e^B:

$$e^B = I + B + \frac{B^2}{2!} + \cdots = I + B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Therefore:

$$e^A = e \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} e & e \\ 0 & e \end{bmatrix}$$

7.1.0.3 3. Solving the differential equations We solve the systems in two different interpretations:

7.1.0.3.1 Interpretation 1: Solving the system as printed The printed system is:

$$\frac{dx_1}{dt} = x_2, \quad \frac{dx_2}{dt} = x_1$$

We differentiate the first equation:

$$\frac{d^2x_1}{dt^2} = \frac{dx_2}{dt} = x_1 \implies \frac{d^2x_1}{dt^2} - x_1 = 0$$

The characteristic equation is

$$r^2 - 1 = 0$$

, which gives $r = \pm 1$. So the general solution for $x\_1(t)$ is:

$$x_1(t) = C_1 e^t + C_2 e^{-t}$$

Then we find $x\_2(t)$ using the first relation:

$$x_2(t) = \frac{dx_1}{dt} = C_1 e^t - C_2 e^{-t}$$

7.1.0.3.2 Interpretation 2: Solving the intended system $\frac{dX}{dt} = AX$ The intended system associated with matrix A is:

$$\frac{dx_1}{dt} = x_1 + x_2, \quad \frac{dx_2}{dt} = x_2$$

The solution to a system of the form

$$\frac{dX}{dt} = AX$$

is:

$$X(t) = e^{At} X(0)$$

We calculate the matrix exponential e^{At} by decomposing $At = It + Bt$:

$$e^{At} = e^t e^{Bt} = e^t(I + Bt) = e^t \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} e^t & te^t \\ 0 & e^t \end{bmatrix}$$

So the solution is:

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} e^t & te^t \\ 0 & e^t \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$$

This gives the general solution:

$$x_1(t) = (C_1 + C_2 t)e^t$$
$$x_2(t) = C_2 e^t$$

where

$$C\_1 = x\_1(0)$$

and

$$C\_2 = x\_2(0).$$

7.2 Q2. Solve the following system of differential equations by matrix method: (06)

ID	PYQ-2020-7b
Appeared in	2020 Q7(b) [06 marks], 2021 Q7(b) [06 marks]
Frequency	⋆⋆ (2 times)

Answer:

$$\frac{dx}{dt} = 6x - 3y$$
$$\frac{dy}{dt} = 2x + y$$

Answer:
We write the system in matrix form:

$$\frac{dX}{dt} = MX, \quad \text{where } M = \begin{bmatrix} 6 & -3 \\ 2 & 1 \end{bmatrix}$$

First we find the eigenvalues of M by solving $|M - \lambda I| = 0$:

$$\begin{vmatrix} 6-\lambda & -3 \\ 2 & 1-\lambda \end{vmatrix} = 0 \implies (6-\lambda)(1-\lambda) + 6 = \lambda^2 - 7\lambda + 12 = 0$$

$$(\lambda - 3)(\lambda - 4) = 0 \implies \lambda = 3 \quad \text{or} \quad \lambda = 4$$

Now find the eigenvectors: * **For $\lambda = 3$:**

$$M - 3I = \begin{bmatrix} 3 & -3 \\ 2 & -2 \end{bmatrix} \implies 3x - 3y = 0 \implies x = y$$

The eigenvector is:

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- **For $\lambda = 4$:**

$$M - 4I = \begin{bmatrix} 2 & -3 \\ 2 & -3 \end{bmatrix} \implies 2x - 3y = 0 \implies x = \frac{3}{2}y$$

The eigenvector (for $y = 2$) is:

$$v_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

The general solution is a linear combination of the fundamental solutions:

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = C_1 e^{3t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + C_2 e^{4t} \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

So the system solution is:

$$x(t) = C_1 e^{3t} + 3C_2 e^{4t}$$
$$y(t) = C_1 e^{3t} + 2C_2 e^{4t}$$

8 Topic 04: Green's Theorem

This file contains the organized questions and answers for **Green's Theorem**, priority ranked as **Priority 4** based on frequency and exam weight.

8.1 Q1 (06)

If

$$\vec{F} = (2x + y^2)\hat{i} + (3y - 4x)\hat{j}$$

Then evaluate

$$\oint_C \vec{F} \cdot d\vec{r}$$

around the curve C of the following figure:

ID	PYQ-2017-2a
Source	2017 Q2(a) [06 marks]

Answer:

> [!NOTE] **Figure Description:** The curve C is a closed loop in the first quadrant of the xy-plane. It is bounded by the parabolas $y = x^2$ and $y^2 = x$. The curves intersect at the points $(0, 0)$ and $(1, 1)$. The boundary is traversed in the counterclockwise direction.

Answer:

We can solve this problem in two ways: using Green's Theorem or using direct line integration.

8.1.0.1 Method 1: Using Green's Theorem

Green's Theorem states:

$$\oint_C (Pdx + Qdy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

Here we identify the functions:

$$P = 2x + y^2 \implies \frac{\partial P}{\partial y} = 2y$$

$$Q = 3y - 4x \implies \frac{\partial Q}{\partial x} = -4$$

Substitute these into the integrand:

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = -4 - 2y$$

The region R lies between

$$y = x^2$$

and $y = \sqrt{x}$ for x from 0 to 1. We set up the double integral:

$$\oint_C \vec{F} \cdot d\vec{r} = \int_{x=0}^{1} \int_{y=x^2}^{\sqrt{x}} (-4 - 2y) dy dx$$

Integrate with respect to y:

$$\oint_C \vec{F} \cdot d\vec{r} = \int_0^1 \left[-4y - y^2\right]_{x^2}^{\sqrt{x}} dx$$

$$\oint_C \vec{F} \cdot d\vec{r} = \int_0^1 \left[(-4\sqrt{x} - x) - (-4x^2 - x^4)\right] dx = \int_0^1 \left(-4x^{1/2} - x + 4x^2 + x^4\right) dx$$

Now integrate with respect to x:

$$\oint_C \vec{F} \cdot d\vec{r} = \left[-\frac{8}{3}x^{3/2} - \frac{1}{2}x^2 + \frac{4}{3}x^3 + \frac{1}{5}x^5 \right]_0^1$$

$$\oint_C \vec{F} \cdot d\vec{r} = -\frac{8}{3} - \frac{1}{2} + \frac{4}{3} + \frac{1}{5} = -\frac{4}{3} - \frac{1}{2} + \frac{1}{5}$$

Find a common denominator of 30:

$$\oint_C \vec{F} \cdot d\vec{r} = \frac{-40 - 15 + 6}{30} = -\frac{49}{30}$$

8.1.0.2 Method 2: Direct Line Integration The boundary curve C consists of two paths: 1. Path C_1: Along the parabola

$$y = x^2$$

from $(0, 0)$ to $(1, 1)$. Here $dy = 2xdx$. 2. Path C_2: Along the parabola

$$y^2 = x$$

from $(1, 1)$ to $(0, 0)$. We can write this as

$$x = y^2$$

from $y = 1$ to $y = 0$. Here $dx = 2ydy$.
Evaluate the integral along Path C_1:

$$\int_{C_1} \vec{F} \cdot d\vec{r} = \int_0^1 \left[(2x + x^4)dx + (3x^2 - 4x)(2xdx) \right] = \int_0^1 \left(2x + x^4 + 6x^3 - 8x^2 \right) dx$$

$$\int_{C_1} \vec{F} \cdot d\vec{r} = \left[x^2 + \frac{1}{5}x^5 + \frac{3}{2}x^4 - \frac{8}{3}x^3 \right]_0^1 = 1 + \frac{1}{5} + \frac{3}{2} - \frac{8}{3}$$

$$\int_{C_1} \vec{F} \cdot d\vec{r} = \frac{30 + 6 + 45 - 80}{30} = \frac{1}{30}$$

Evaluate the integral along Path C_2:

$$\int_{C_2} \vec{F} \cdot d\vec{r} = \int_1^0 \left[(2y^2 + y^2)(2ydy) + (3y - 4y^2)dy \right] = \int_1^0 \left(6y^3 + 3y - 4y^2 \right) dy$$

$$\int_{C_2} \vec{F} \cdot d\vec{r} = \left[\frac{3}{2}y^4 + \frac{3}{2}y^2 - \frac{4}{3}y^3 \right]_1^0 = 0 - \left(\frac{3}{2} + \frac{3}{2} - \frac{4}{3} \right) = -\left(3 - \frac{4}{3} \right) = -\frac{5}{3} = -\frac{50}{30}$$

Add the two line integrals together:

$$\oint_C \vec{F} \cdot d\vec{r} = \int_{C_1} + \int_{C_2} = \frac{1}{30} - \frac{50}{30} = -\frac{49}{30}$$

Both methods yield the same result of $-\frac{49}{30}$.

8.2 Q2. State and prove Green's theorem in the plane. (06)

ID	PYQ-2017-3a
Appeared in	2017 Q3(a) [06 marks], 2021 Q4(a) [06 marks], 2024 Q4(a) [05 marks]
Frequency	⋆⋆⋆ (3 times)

Answer:

8.2.0.1 Statement

Let R be a closed, flat region bounded by a simple, closed curve C in the xy-plane. If $P(x,y)$ and $Q(x,y)$ are continuous functions with continuous first-order partial derivatives in R, then:

$$\oint_C (Pdx + Qdy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dxdy$$

Here the curve C is traversed in the counterclockwise (positive) direction.

8.2.0.2 Proof

We will prove the theorem for a simple region R that is bounded by curves. We split the theorem into two parts:

$$\text{Part 1: } \oint_C Pdx = -\iint_R \frac{\partial P}{\partial y} dxdy$$
$$\text{Part 2: } \oint_C Qdy = \iint_R \frac{\partial Q}{\partial x} dxdy$$

8.2.0.2.1 Proof of Part 1

Let the region R be bounded by the curves

$$y = y\_1(x)$$

(lower boundary) and

$$y = y\_2(x)$$

(upper boundary) for x from a to b.

Evaluate the double integral on the right-hand side:

$$\iint_R \frac{\partial P}{\partial y} dxdy = \int_a^b \left[\int_{y_1(x)}^{y_2(x)} \frac{\partial P}{\partial y} dy \right] dx = \int_a^b \left[P(x, y_2(x)) - P(x, y_1(x)) \right] dx \quad \ldots(1)$$

Now evaluate the line integral

$$\oint_C Pdx$$

The closed boundary curve C consists of two parts: * Path C_1: Along the curve

$$y = y\_1(x)$$

from $x = a$ to $x = b$. * Path C_2: Along the curve

$$y = y\_2(x)$$

from $x = b$ to $x = a$.

We calculate the line integral over each path:

$$\oint_C Pdx = \int_{C_1} Pdx + \int_{C_2} Pdx = \int_a^b P(x, y_1(x))dx + \int_b^a P(x, y_2(x))dx$$

Reverse the limits on the second integral:

$$\oint_C Pdx = \int_a^b P(x, y_1(x))dx - \int_a^b P(x, y_2(x))dx = -\int_a^b \left[P(x, y_2(x)) - P(x, y_1(x)) \right] dx \quad \ldots(2)$$

Comparing equations (1) and (2), we get:

$$\oint_C Pdx = -\iint_R \frac{\partial P}{\partial y} dxdy$$

8.2.0.2.2 Proof of Part 2

Let the region R be bounded by the curves

$$x = x\_1(y)$$

(left boundary) and

$$x = x\_2(y)$$

(right boundary) for y from c to d. By using the same steps, we get:

$$\oint_C Qdy = \iint_R \frac{\partial Q}{\partial x} dxdy$$

Adding the equations from Part 1 and Part 2 gives:

$$\oint_C (Pdx + Qdy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dxdy$$

The proof is complete.

8.3 Q3 (06)

Evaluate

$$\oint_C (y - \sin x)dx + \cos y dy$$

where C is the triangle of the adjoining figure:

ID	PYQ-2017-3b
Source	2017 Q3(b) [06 marks]

Answer:

> [!NOTE] **Figure Description:** The curve C is a triangle in the xy-plane with vertices at $(0,0)$, $(\pi/2, 0)$, and $(\pi/2, 1)$. The boundary is traversed in the counterclockwise direction.

Answer:
We can solve this problem in two ways: using Green's Theorem or using direct line integration.

8.3.0.1 Method 1: Using Green's Theorem

We identify the functions:

$$P = y - \sin x \implies \frac{\partial P}{\partial y} = 1$$

$$Q = \cos y \implies \frac{\partial Q}{\partial x} = 0$$

Substitute these into Green's Theorem formula:

$$\oint_C (Pdx + Qdy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \iint_R (0 - 1)dA = -\iint_R dA$$

The integral

$$\iint_R dA$$

is simply the area of the triangular region R:

$$\text{Area}(R) = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times \frac{\pi}{2} \times 1 = \frac{\pi}{4}$$

So we get:

$$\oint_C (y - \sin x)dx + \cos y dy = -\text{Area}(R) = -\frac{\pi}{4}$$

8.3.0.2 Method 2: Direct Line Integration The boundary curve C consists of three straight paths: 1. Path C_1: Along the x-axis ($y = 0$, $dy = 0$) from $(0, 0)$ to $(\pi/2, 0)$. 2. Path C_2: Along the vertical line $x = \pi/2$ ($dx = 0$) from $(\pi/2, 0)$ to $(\pi/2, 1)$. 3. Path C_3: Along the line

$$y = \frac{2}{\pi}x$$

(or

$$x = \frac{\pi}{2}y,$$

$$dx = \frac{\pi}{2}dy$$

) from $(\pi/2, 1)$ to $(0, 0)$.

Evaluate the integral along Path C_1:

$$\int_{C_1} (y - \sin x)dx + \cos y dy = \int_0^{\pi/2} (0 - \sin x)dx = [\cos x]_0^{\pi/2} = \cos\left(\frac{\pi}{2}\right) - \cos(0) = 0 - 1 = -1$$

Evaluate the integral along Path C_2:

$$\int_{C_2} (y - \sin x)dx + \cos y dy = \int_0^1 \cos y dy = [\sin y]_0^1 = \sin(1)$$

Evaluate the integral along Path C_3:

Here we integrate with respect to y from $y = 1$ to $y = 0$:

$$\int_{C_3} (y - \sin x)dx + \cos y dy = \int_1^0 \left[\left(\frac{\pi}{2}y - \sin\left(\frac{\pi}{2}y\right)\right)\frac{\pi}{2} + \cos y\right] dy$$

$$\int_{C_3} (y - \sin x)dx + \cos y dy = \left[\frac{\pi}{4}y^2 + \cos\left(\frac{\pi}{2}y\right) + \sin y\right]_1^0$$

$$\int_{C_3} (y - \sin x)dx + \cos y dy = (0 + 1 + 0) - \left(\frac{\pi}{4} + 0 + \sin(1)\right) = 1 - \frac{\pi}{4} - \sin(1)$$

Add the three line integrals together:

$$\oint_C = \int_{C_1} + \int_{C_2} + \int_{C_3} = -1 + \sin(1) + 1 - \frac{\pi}{4} - \sin(1) = -\frac{\pi}{4}$$

Both methods yield the same result of $-\frac{\pi}{4}$.

8.4 Q4 (06)

Verify Green's theorem in the plane for

$$\oint_C (2x - y^2)dx - xydy$$

where C is the boundary of the region enclosed by the circles

$$x^2 + y^2 = 1$$

and

$$x^2 + y^2 = 9.$$

ID	PYQ-2019-4b
Source	2019 Q4(b) [06 marks]

Answer:
Green's Theorem states:

$$\oint_C (Pdx + Qdy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Here we identify the functions:

$$P = 2x - y^2 \implies \frac{\partial P}{\partial y} = -2y$$

$$Q = -xy \implies \frac{\partial Q}{\partial x} = -y$$

Substitute these:

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = -y - (-2y) = y$$

8.4.0.1 1. Evaluate Double Integral

The region R is the annulus between the circles $r = 1$ and $r = 3$. We convert to polar coordinates

$$x = r\cos\theta, \quad y = r\sin\theta, \quad dA = r\,dr\,d\theta :$$

$$\iint_R y\,dA = \int_{\theta=0}^{2\pi} \int_{r=1}^{3} (r\sin\theta) r\,dr\,d\theta = \left(\int_1^3 r^2\,dr \right) \left(\int_0^{2\pi} \sin\theta\,d\theta \right)$$

Calculate the angular integral:

$$\int_0^{2\pi} \sin\theta\,d\theta = [-\cos\theta]_0^{2\pi} = -1 - (-1) = 0$$

So the double integral is:

$$\iint_R y\,dA = 0$$

8.4.0.2 2. Evaluate Line Integral

The closed boundary C consists of two circular paths: the outer circle C_1 ($r = 3$, traversed counterclockwise) and the inner circle C_2 ($r = 1$, traversed clockwise):

- **On Outer Circle C_1:** $x = 3\cos\theta, y = 3\sin\theta \implies dx = -3\sin\theta\,d\theta, dy = 3\cos\theta\,d\theta$ with θ from 0 to 2π:

$$\int_{C_1} (2x - y^2)dx - xydy = \int_0^{2\pi} \left[(6\cos\theta - 9\sin^2\theta)(-3\sin\theta) - 9\cos\theta\sin\theta(3\cos\theta) \right] d\theta$$

$$= \int_0^{2\pi} \left[-18\cos\theta\sin\theta + 27\sin^3\theta - 27\cos^2\theta\sin\theta \right] d\theta = 0$$

(since each component integrates to 0 over the full period $[0, 2\pi]$).

- **On Inner Circle C_2:** $x = \cos\theta, y = \sin\theta \implies dx = -\sin\theta\,d\theta, dy = \cos\theta\,d\theta$ with θ from 2π to 0:

$$\int_{C_2} (2x - y^2)dx - xydy = \int_{2\pi}^{0} \left[(2\cos\theta - \sin^2\theta)(-\sin\theta) - \cos\theta\sin\theta(\cos\theta) \right] d\theta = 0$$

The total line integral is:

$$\oint_C = \int_{C_1} + \int_{C_2} = 0 + 0 = 0$$

Both the double integral and the line integral equal 0, so Green's theorem is verified.

8.4.1 SECTION - B

8.5 Q5 (06)

Verify Green's theorem in the plane for

$$\oint_C (xy + y^2)dx + x^2 dy$$

where C is a closed region bounded by $y = x$ and

$$y = x^2.$$

ID	PYQ-2020-4b
Appeared in	2020 Q4(b) [06 marks], 2023 Q3(b) [05 marks]
Frequency	⋆⋆ (2 times)

Answer:
Green's Theorem states:

$$\oint_C (Pdx + Qdy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

We identify the functions:

$$P = xy + y^2 \implies \frac{\partial P}{\partial y} = x + 2y$$

$$Q = x^2 \implies \frac{\partial Q}{\partial x} = 2x$$

Substitute these:

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2x - (x + 2y) = x - 2y$$

8.5.0.1 1. Evaluate Double Integral

The boundaries intersect at $(0, 0)$ and $(1, 1)$. We integrate:

$$\iint_R (x - 2y)dA = \int_{x=0}^{1} \int_{y=x^2}^{x} (x - 2y)dydx$$

Integrate with respect to y:

$$\iint_R (x - 2y)dA = \int_0^1 \left[xy - y^2\right]_{x^2}^{x} dx = \int_0^1 \left[0 - (x^3 - x^4)\right] dx = \int_0^1 (x^4 - x^3)dx$$

Integrate with respect to x:

$$\iint_R (x - 2y)dA = \left[\frac{x^5}{5} - \frac{x^4}{4}\right]_0^1 = \frac{1}{5} - \frac{1}{4} = -\frac{1}{20}$$

8.5.0.2 2. Evaluate Line Integral

The closed path C has two parts: 1. Path C_1 (

$$y = x^2$$

, $dy = 2xdx$) from $(0, 0)$ to $(1, 1)$. 2. Path C_2 ($y = x$, $dy = dx$) from $(1, 1)$ to $(0, 0)$.
Evaluate along Path C_1:

$$\int_{C_1} (xy + y^2)dx + x^2 dy = \int_0^1 \left[(x^3 + x^4)dx + x^2(2xdx)\right] = \int_0^1 (3x^3 + x^4)dx$$

$$\int_{C_1} = \left[\frac{3}{4}x^4 + \frac{x^5}{5}\right]_0^1 = \frac{3}{4} + \frac{1}{5} = \frac{19}{20}$$

Evaluate along Path C_2:

$$\int_{C_2} (xy + y^2)dx + x^2dy = \int_1^0 \left[(x^2 + x^2)dx + x^2dx\right] = \int_1^0 3x^2dx = \left[x^3\right]_1^0 = -1$$

Add the line integrals together:

$$\oint_C = \frac{19}{20} - 1 = -\frac{1}{20}$$

Both integrals give $-\frac{1}{20}$. So Green's theorem is verified.

8.6 SECTION - B

9 Topic 05: Gradient and Directional Derivative

This file contains the organized questions and answers for **Gradient and Directional Derivative**, priority ranked as **Priority 5** based on frequency and exam weight.

9.1 Q1 (04)

Find the angle between the surfaces

$$x^2 + y^2 + z^2 = 9$$

and

$$z = x^2 + y^2 - 3$$

at the point $(2, -1, 2)$.

ID	PYQ-2017-1a
Appeared in	2017 Q1(a) [04 marks], 2019 Q3(a) [04 marks], 2021 Q3(a) [04 marks]
Frequency	⋆⋆⋆ (3 times)

Answer:

Let the first surface be defined by the function:

$$f_1(x, y, z) = x^2 + y^2 + z^2 - 9 = 0$$

The gradient of f_1 gives the normal vector to this surface:

$$\vec{\nabla} f_1 = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

At the point $P(2, -1, 2)$, the normal vector $\vec{n}$_1 is:

$$\vec{n}_1 = 2(2)\hat{i} + 2(-1)\hat{j} + 2(2)\hat{k} = 4\hat{i} - 2\hat{j} + 4\hat{k}$$

Let the second surface be defined by the function:

$$f_2(x, y, z) = x^2 + y^2 - z - 3 = 0$$

The gradient of f_2 gives the normal vector to this surface:

$$\vec{\nabla} f_2 = 2x\hat{i} + 2y\hat{j} - \hat{k}$$

At the point $P(2, -1, 2)$, the normal vector $\vec{n}$_2 is:

$$\vec{n}_2 = 2(2)\hat{i} + 2(-1)\hat{j} - \hat{k} = 4\hat{i} - 2\hat{j} - \hat{k}$$

Let θ be the angle between these two surfaces. We use the dot product of their normal vectors:

$$\cos\theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|}$$

Calculate the dot product:

$$\vec{n}_1 \cdot \vec{n}_2 = (4)(4) + (-2)(-2) + (4)(-1) = 16 + 4 - 4 = 16$$

Calculate the magnitudes of the normal vectors:

$$|\vec{n}_1| = \sqrt{4^2 + (-2)^2 + 4^2} = \sqrt{16 + 4 + 16} = \sqrt{36} = 6$$

$$|\vec{n}_2| = \sqrt{4^2 + (-2)^2 + (-1)^2} = \sqrt{16 + 4 + 1} = \sqrt{21}$$

Substitute these values back into the cosine equation:

$$\cos\theta = \frac{16}{6\sqrt{21}} = \frac{8}{3\sqrt{21}}$$

So the angle θ is:

$$\theta = \cos^{-1}\left(\frac{8}{3\sqrt{21}}\right)$$

9.2 Q2 (04)

Find the directional derivative of

$$\phi = 4e^{2x-y+z}$$

at $(1,1,-1)$ in a direction toward the point $(-3,5,6)$.

ID	PYQ-2017-1c
Source	2017 Q1(c) [04 marks]

Answer:
Let P be the point $(1,1,-1)$ and Q be the point $(-3,5,6)$.
The direction vector from P to Q is:

$$\vec{d} = \vec{PQ} = (-3-1)\hat{i} + (5-1)\hat{j} + (6-(-1))\hat{k} = -4\hat{i} + 4\hat{j} + 7\hat{k}$$

We find the unit vector $\hat{u}$ in this direction:

$$\hat{u} = \frac{\vec{d}}{|\vec{d}|} = \frac{-4\hat{i}+4\hat{j}+7\hat{k}}{\sqrt{(-4)^2+4^2+7^2}} = \frac{-4\hat{i}+4\hat{j}+7\hat{k}}{\sqrt{16+16+49}} = \frac{-4\hat{i}+4\hat{j}+7\hat{k}}{\sqrt{81}} = \frac{-4\hat{i}+4\hat{j}+7\hat{k}}{9}$$

Next we find the gradient of ϕ:

$$\vec{\nabla}\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$$

For

$$\phi = 4e^{2x-y+z}$$

, we calculate the partial derivatives:

$$\frac{\partial\phi}{\partial x} = 8e^{2x-y+z}$$

$$\frac{\partial\phi}{\partial y} = -4e^{2x-y+z}$$

$$\frac{\partial\phi}{\partial z} = 4e^{2x-y+z}$$

So the gradient vector is:

$$\vec{\nabla}\phi = 4e^{2x-y+z}(2\hat{i} - \hat{j} + \hat{k})$$

At the point $P(1,1,-1)$, the exponent is $2(1) - 1 + (-1) = 0$. So we get:

$$\vec{\nabla}\phi|_P = 4e^0(2\hat{i} - \hat{j} + \hat{k}) = 8\hat{i} - 4\hat{j} + 4\hat{k}$$

The directional derivative is the dot product of the gradient and the unit vector:

$$D_{\hat{u}}\phi = \vec{\nabla}\phi|_P \cdot \hat{u} = (8\hat{i} - 4\hat{j} + 4\hat{k}) \cdot \left(\frac{-4\hat{i}+4\hat{j}+7\hat{k}}{9}\right)$$

$$D_{\hat{u}}\phi = \frac{(8)(-4)+(-4)(4)+(4)(7)}{9} = \frac{-32-16+28}{9} = \frac{-20}{9}$$

So the directional derivative is $-\frac{20}{9}$.

9.3 Q3 (03)

Find an equation for the tangent plane to the surface

$$2xz^2 - 3xy - 4x = 7$$

at the point $(1, -1, 2)$.

ID	PYQ-2018-4b
Source	2018 Q4(b) [03 marks]

Answer:
Let the surface be defined by the function:

$$F(x, y, z) = 2xz^2 - 3xy - 4x - 7 = 0$$

The gradient of F gives the normal vector to the surface:

$$\vec{\nabla} F = \frac{\partial F}{\partial x}\hat{i} + \frac{\partial F}{\partial y}\hat{j} + \frac{\partial F}{\partial z}\hat{k}$$

Calculate the partial derivatives:

$$\frac{\partial F}{\partial x} = 2z^2 - 3y - 4$$
$$\frac{\partial F}{\partial y} = -3x$$
$$\frac{\partial F}{\partial z} = 4xz$$

At the point $(1, -1, 2)$, the gradient vector is:

$$\vec{\nabla} F = (2(4) - 3(-1) - 4)\hat{i} - 3(1)\hat{j} + 4(1)(2)\hat{k} = 7\hat{i} - 3\hat{j} + 8\hat{k}$$

The equation of the tangent plane at

$$(x\_0, y\_0, z\_0) = (1, -1, 2)$$

is:

$$7(x - 1) - 3(y - (-1)) + 8(z - 2) = 0$$
$$7x - 7 - 3y - 3 + 8z - 16 = 0 \implies 7x - 3y + 8z = 26$$

So the equation of the tangent plane is $7x - 3y + 8z = 26$.

9.4 Q4 (04)

What is directional derivative? Find the directional derivative of

$$\phi = x^2yz + 4xz^2$$

at $(1, -2, -1)$ in the direction

$$2\hat{i} - \hat{j} - 2\hat{k}$$

ID	PYQ-2018-4c
Source	2018 Q4(c) [04 marks]

Answer:

9.4.0.1 1. Definition The directional derivative is the rate of change of a scalar function at a given point in a specific direction. It equals the dot product of the function's gradient and the unit direction vector.

9.4.0.2 2. Calculation Let the direction vector be

$$\vec{d} = 2\hat{i} - \hat{j} - 2\hat{k}$$

The unit direction vector $\hat{u}$ is:

$$\hat{u} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{\sqrt{2^2 + (-1)^2 + (-2)^2}} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{\sqrt{9}} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{3}$$

Next we find the gradient of

$$\phi = x^2yz + 4xz^2 :$$

$$\vec{\nabla}\phi = (2xyz + 4z^2)\hat{i} + x^2z\hat{j} + (x^2y + 8xz)\hat{k}$$

At the point $(1, -2, -1)$, the gradient vector is:

$$\vec{\nabla}\phi = (2(1)(-2)(-1) + 4(1))\hat{i} + (1)(-1)\hat{j} + ((1)(-2) + 8(1)(-1))\hat{k}$$

$$\vec{\nabla}\phi = (4 + 4)\hat{i} - \hat{j} + (-2 - 8)\hat{k} = 8\hat{i} - \hat{j} - 10\hat{k}$$

The directional derivative is:

$$D_{\hat{u}}\phi = \vec{\nabla}\phi \cdot \hat{u} = (8\hat{i} - \hat{j} - 10\hat{k}) \cdot \left(\frac{2\hat{i} - \hat{j} - 2\hat{k}}{3}\right)$$

$$D_{\hat{u}}\phi = \frac{8(2) + (-1)(-1) + (-10)(-2)}{3} = \frac{16 + 1 + 20}{3} = \frac{37}{3}$$

So the directional derivative is $\frac{37}{3}$.

9.5 SECTION - B

9.6 Q5 (04)

Find the directional derivative of

$$\phi = x^2yz + 4xz^2$$

at $(1, -2, 1)$ in the direction

$$2\hat{i} - \hat{j} - 2\hat{k}$$

ID	PYQ-2019-2c
Source	2019 Q2(c) [04 marks]

Answer:

The gradient of ϕ is:

$$\vec{\nabla}\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$$

$$\vec{\nabla}\phi = (2xyz + 4z^2)\hat{i} + x^2z\hat{j} + (x^2y + 8xz)\hat{k}$$

Evaluate this gradient at the point $(1, -2, 1)$:

- $$\frac{\partial\phi}{\partial x} = 2(1)(-2)(1) + 4(1)^2 = -4 + 4 = 0$$

-

$$\frac{\partial \phi}{\partial y} = (1)^2(1) = 1$$

- $$\frac{\partial \phi}{\partial z} = (1)^2(-2) + 8(1)(1) = -2 + 8 = 6$$

So the gradient vector is:

$$\vec{\nabla}\phi(1, -2, 1) = \hat{j} + 6\hat{k}$$

The direction vector is

$$\vec{d} = 2\hat{i} - \hat{j} - 2\hat{k}$$

The unit vector $\hat{u}$ is:

$$\hat{u} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{\sqrt{2^2 + (-1)^2 + (-2)^2}} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{3}$$

The directional derivative is the dot product of the gradient and the unit vector:

$$D_{\hat{u}}\phi = \vec{\nabla}\phi \cdot \hat{u} = (\hat{j} + 6\hat{k}) \cdot \left(\frac{2\hat{i} - \hat{j} - 2\hat{k}}{3}\right) = \frac{0(2) + 1(-1) + 6(-2)}{3} = -\frac{13}{3}$$

9.7 Q6 (03)

Show that, $\nabla\Phi$ is a vector perpendicular to the surface $\Phi(x, y, z) = C$, where C is a constant.

ID	PYQ-2020-2c
Source	2020 Q2(c) [03 marks]

Answer:

Let P be a point on the surface $\Phi(x, y, z) = C$. Let the vector function:

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

define any curve that lies entirely on the surface and passes through P.

Since all points of the curve lie on the surface, they satisfy the surface equation:

$$\Phi(x(t), y(t), z(t)) = C$$

Differentiate this equation with respect to t using the chain rule:

$$\frac{\partial \Phi}{\partial x}\frac{dx}{dt} + \frac{\partial \Phi}{\partial y}\frac{dy}{dt} + \frac{\partial \Phi}{\partial z}\frac{dz}{dt} = 0$$

We can write this as a dot product:

$$\left(\frac{\partial \Phi}{\partial x}\hat{i} + \frac{\partial \Phi}{\partial y}\hat{j} + \frac{\partial \Phi}{\partial z}\hat{k}\right) \cdot \left(\frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}\right) = 0$$

$$\vec{\nabla}\Phi \cdot \frac{d\vec{r}}{dt} = 0$$

Here the vector $\frac{d\vec{r}}{dt}$ is a tangent vector to the curve. Because the dot product is zero, $\vec{\nabla}\Phi$ is perpendicular to this tangent vector.

Since this relation holds for any curve on the surface passing through P, the gradient vector $\vec{\nabla}\Phi$ is perpendicular to the surface.

9.8 Q7 (03)

Find the directional derivative of the scalar function

$$f(x, y, z) = x^2 + xy + z^2$$

at the point $A(1, -1, -1)$ in the direction of the line AB where B has co-ordinates $(3, 2, 1)$.

ID	PYQ-2021-2a
Source	2021 Q2(a) [03 marks]

Answer:
We construct the direction vector $\vec{AB}$:

$$\vec{AB} = (3-1)\hat{i} + (2-(-1))\hat{j} + (1-(-1))\hat{k} = 2\hat{i} + 3\hat{j} + 2\hat{k}$$

Find the unit vector $\hat{u}$ in this direction:

$$\hat{u} = \frac{2\hat{i} + 3\hat{j} + 2\hat{k}}{\sqrt{2^2 + 3^2 + 2^2}} = \frac{2\hat{i} + 3\hat{j} + 2\hat{k}}{\sqrt{17}}$$

Calculate the gradient of the function

$$f = x^2 + xy + z^2 :$$

$$\vec{\nabla} f = (2x + y)\hat{i} + x\hat{j} + 2z\hat{k}$$

Evaluate the gradient at the point $A(1, -1, -1)$:

$$\vec{\nabla} f(A) = (2(1) - 1)\hat{i} + (1)\hat{j} + 2(-1)\hat{k} = \hat{i} + \hat{j} - 2\hat{k}$$

The directional derivative is the dot product of the gradient and the unit vector:

$$D_{\hat{u}} f = \vec{\nabla} f \cdot \hat{u} = (\hat{i} + \hat{j} - 2\hat{k}) \cdot \left(\frac{2\hat{i} + 3\hat{j} + 2\hat{k}}{\sqrt{17}} \right) = \frac{2 + 3 - 4}{\sqrt{17}} = \frac{1}{\sqrt{17}}$$

So the directional derivative is $\frac{1}{\sqrt{17}}$.

9.9 Q8 (02)

If

$$Q = 3x^2 y - y^3 z^2$$

find ∇Q at the point $(1, -2, -1)$.

ID	PYQ-2024-2c
Source	2024 Q2(c) [02 marks]

Answer:
The gradient of Q is:

$$\nabla Q = \frac{\partial Q}{\partial x}\hat{i} + \frac{\partial Q}{\partial y}\hat{j} + \frac{\partial Q}{\partial z}\hat{k}$$

We calculate the partial derivatives:

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(3x^2 y - y^3 z^2) = 6xy$$

$$\frac{\partial Q}{\partial y} = \frac{\partial}{\partial y}(3x^2y - y^3z^2) = 3x^2 - 3y^2z^2$$

$$\frac{\partial Q}{\partial z} = \frac{\partial}{\partial z}(3x^2y - y^3z^2) = -2y^3z$$

We substitute

$$x = 1, \quad y = -2, \quad z = -1 :$$

$$\frac{\partial Q}{\partial x} = 6(1)(-2) = -12$$

$$\frac{\partial Q}{\partial y} = 3(1)^2 - 3(-2)^2(-1)^2 = 3 - 3(4)(1) = 3 - 12 = -9$$

$$\frac{\partial Q}{\partial z} = -2(-2)^3(-1) = -2(-8)(-1) = -16$$

So, the gradient at $(1, -2, -1)$ is:

$$\nabla Q = -12\hat{i} - 9\hat{j} - 16\hat{k}$$

9.10 Q9 (05)

Find the unit outward drawn normal to the surface

$$(x-1)^2 + y^2 + (z+2)^2 = 9$$

at the point $(3, 1, -4)$.

ID	CT2V-2
Source	CT2V Q2 [05 marks]

Answer:

Let the surface function be:

$$f(x, y, z) = (x-1)^2 + y^2 + (z+2)^2 - 9$$

The gradient of f gives a vector normal to the surface. We calculate this gradient:

$$\vec{\nabla} f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}$$

$$\vec{\nabla} f = 2(x-1)\hat{i} + 2y\hat{j} + 2(z+2)\hat{k}$$

We evaluate the gradient at $(3, 1, -4)$:

$$\vec{\nabla} f|_{(3,1,-4)} = 2(3-1)\hat{i} + 2(1)\hat{j} + 2(-4+2)\hat{k}$$

$$\vec{\nabla} f|_{(3,1,-4)} = 4\hat{i} + 2\hat{j} - 4\hat{k}$$

Now we find the magnitude of this normal vector:

$$|\vec{\nabla} f| = \sqrt{4^2 + 2^2 + (-4)^2} = \sqrt{16 + 4 + 16} = \sqrt{36} = 6$$

We divide the normal vector by its magnitude. This gives the unit outward normal vector $\hat{n}$:

$$\hat{n} = \frac{4\hat{i} + 2\hat{j} - 4\hat{k}}{6} = \frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k}$$

10 Topic 06: Line Integrals

This file contains the organized questions and answers for **Line Integrals**, priority ranked as **Priority 6** based on frequency and exam weight.

10.1 Q1 (04)

If

$$\vec{F} = 3xy\hat{i} - y^2\hat{j}$$

Then evaluate

$$\int_C \vec{F} \cdot d\vec{r}$$

When C is the curve in the xy plane,

$$y = 2x^2$$

from $(0, 0)$ to $(1, 2)$.

ID	PYQ-2017-1b
Source	2017 Q1(b) [04 marks]

Answer:

The position vector in the xy-plane is

$$\vec{r} = x\hat{i} + y\hat{j}$$

So we have:

$$d\vec{r} = dx\hat{i} + dy\hat{j}$$

We can write the line integral as:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C (3xy\hat{i} - y^2\hat{j}) \cdot (dx\hat{i} + dy\hat{j}) = \int_C \left(3xydx - y^2dy\right)$$

The path C is defined by

$$y = 2x^2$$

. This gives:

$$dy = 4xdx$$

We substitute these terms into our integral. The limit for x goes from 0 to 1:

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 \left[3x(2x^2) - (2x^2)^2(4x)\right] dx$$

Simplify the integrand:

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 \left(6x^3 - 16x^5\right) dx$$

Now integrate term by term:

$$\int_C \vec{F} \cdot d\vec{r} = \left[\frac{6x^4}{4} - \frac{16x^6}{6}\right]_0^1 = \left[\frac{3}{2}x^4 - \frac{8}{3}x^6\right]_0^1$$

Evaluate at the limits:

$$\int_C \vec{F} \cdot d\vec{r} = \left(\frac{3}{2} - \frac{8}{3}\right) - 0 = \frac{9 - 16}{6} = -\frac{7}{6}$$

So the value of the line integral is $-\frac{7}{6}$.

10.2 Q2 (06)

If

$$\bar{A} = (3x^2 + 6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k}$$

Then evaluate

$$\int_C \bar{A} \cdot d\bar{r}$$

from $(0,0,0)$ to $(1,1,1)$ along the following paths C:

ID	PYQ-2018-2b
Appeared in	2018 Q2(b) [06 marks], 2020 Q3(b) [06 marks]
Frequency	⋆⋆ (2 times)

Answer:

- **(i)**

$$x = t, y = t^2, z = t^3$$

- **(ii)** the straight line joining $(0,0,0)$ to $(1,1,1)$.

Answer:

We write the line integral as:

$$\int_C \bar{A} \cdot d\bar{r} = \int_C \left[(3x^2 + 6y)dx - 14yzdy + 20xz^2dz\right]$$

10.2.0.1 Path (i): $x = t, y = t^2, z = t^3$

Here we calculate the differentials:

$$dx = dt, \quad dy = 2tdt, \quad dz = 3t^2dt$$

As the path goes from $(0,0,0)$ to $(1,1,1)$, the parameter t goes from 0 to 1. Substitute these into the integral:

$$\int_C \bar{A} \cdot d\bar{r} = \int_0^1 \left[(3t^2 + 6t^2)dt - 14(t^2)(t^3)(2tdt) + 20(t)(t^3)^2(3t^2dt)\right]$$

$$\int_C \bar{A} \cdot d\bar{r} = \int_0^1 \left(9t^2 - 28t^6 + 60t^9\right) dt$$

Integrate the terms:

$$\int_C \bar{A} \cdot d\bar{r} = \left[3t^3 - 4t^7 + 6t^{10}\right]_0^1 = 3 - 4 + 6 = 5$$

So the integral along this path is 5.

10.2.0.2 Path (ii): The straight line

The straight line from $(0,0,0)$ to $(1,1,1)$ is defined by:

$$x = t, \quad y = t, \quad z = t$$

with t from 0 to 1. The differentials are:

$$dx = dt, \quad dy = dt, \quad dz = dt$$

Substitute these into the integral:

$$\int_C \bar{A} \cdot d\bar{r} = \int_0^1 \left[(3t^2 + 6t)dt - 14(t)(t)dt + 20(t)(t)^2dt\right]$$

$$\int_C \bar{A} \cdot d\bar{r} = \int_0^1 \left(20t^3 - 11t^2 + 6t\right) dt$$

Integrate the terms:

$$\int_C \bar{A} \cdot d\bar{r} = \left[5t^4 - \frac{11}{3}t^3 + 3t^2\right]_0^1 = 5 - \frac{11}{3} + 3 = 8 - \frac{11}{3} = \frac{13}{3}$$

So the integral along this path is $\frac{13}{3}$.

10.3 Q3 (04)

If

$$\vec{A} = (3x^2 + 6y)\hat{\imath} - 14yz\hat{\jmath} + 20xz^2\hat{k}$$

Evaluate

$$\int_C \vec{A} \cdot d\vec{r}$$

along the straight lines from $(0,0,0)$ to $(1,0,0)$ then to $(1,1,0)$, and then to $(1,1,1)$ along the paths C.

ID	PYQ-2021-3b
Source	2021 Q3(b) [04 marks]

Answer:
We write the line integral as:

$$\int_C \vec{A} \cdot d\vec{r} = \int_C \left[(3x^2 + 6y)dx - 14yzdy + 20xz^2dz\right]$$

We divide the path C into three line segments:

10.3.0.1 1. Segment C__1: From $(0,0,0)$ to $(1,0,0)$

On this segment, $y = 0$ and $z = 0$, which means $dy = 0$ and $dz = 0$. The variable x goes from 0 to 1:

$$\int_{C_1} \vec{A} \cdot d\vec{r} = \int_{x=0}^{1} (3x^2 + 0)dx = [x^3]_0^1 = 1$$

10.3.0.2 2. Segment C__2: From $(1,0,0)$ to $(1,1,0)$

On this segment, $x = 1$ and $z = 0$, which means $dx = 0$ and $dz = 0$. The variable y goes from 0 to 1:

$$\int_{C_2} \vec{A} \cdot d\vec{r} = \int_{y=0}^{1} -14y(0)dy = 0$$

10.3.0.3 3. Segment C__3: From $(1,1,0)$ to $(1,1,1)$

On this segment, $x = 1$ and $y = 1$, which means $dx = 0$ and $dy = 0$. The variable z goes from 0 to 1:

$$\int_{C_3} \vec{A} \cdot d\vec{r} = \int_{z=0}^{1} 20(1)z^2dz = \left[\frac{20}{3}z^3\right]_0^1 = \frac{20}{3}$$

10.3.0.4 Total Line Integral

$$\int_C \vec{A} \cdot d\vec{r} = \int_{C_1} + \int_{C_2} + \int_{C_3} = 1 + 0 + \frac{20}{3} = \frac{23}{3}$$

So the value of the line integral is $\frac{23}{3}$.

10.4 Q4 (03)

Find the work done in moving a particle in a force field given by

$$\vec{F} = 3xy\hat{i} - 5z\hat{j} + 10x\hat{k}$$

along the curve

$$x = t^2 + 1, y = 2t^2, z = t^3$$

from $t = 1$ to $t = 2$.

ID	PYQ-2024-3b
Source	2024 Q3(b) [03 marks]

Answer:

The work done is:

$$W = \int_C \vec{F} \cdot d\vec{r} = \int_C (3xydx - 5zdy + 10xdz)$$

We express all terms using the parameter t: *

$$x = t^2 + 1 \implies dx = 2tdt$$

- $$y = 2t^2 \implies dy = 4tdt$$

- $$z = t^3 \implies dz = 3t^2dt$$

We substitute these into the integrand: 1.

$$3xydx = 3(t^2 + 1)(2t^2)(2tdt) = 12t^3(t^2 + 1)dt = (12t^5 + 12t^3)dt$$

2.

$$-5zdy = -5(t^3)(4tdt) = -20t^4dt$$

3.

$$10xdz = 10(t^2 + 1)(3t^2dt) = 30t^2(t^2 + 1)dt = (30t^4 + 30t^2)dt$$

We sum these three parts:

$$\vec{F} \cdot d\vec{r} = (12t^5 + 12t^3 - 20t^4 + 30t^4 + 30t^2)dt = (12t^5 + 10t^4 + 12t^3 + 30t^2)dt$$

We integrate from $t = 1$ to $t = 2$:

$$W = \int_1^2 (12t^5 + 10t^4 + 12t^3 + 30t^2)dt$$

We find the antiderivative:

$$\int (12t^5 + 10t^4 + 12t^3 + 30t^2)dt = 2t^6 + 2t^5 + 3t^4 + 10t^3$$

We evaluate at the limits: * At $t = 2$:

$$2(2^6) + 2(2^5) + 3(2^4) + 10(2^3) = 2(64) + 2(32) + 3(16) + 10(8) = 128 + 64 + 48 + 80 = 320$$

- At $t = 1$:

$$2(1^6) + 2(1^5) + 3(1^4) + 10(1^3) = 2 + 2 + 3 + 10 = 17$$

We subtract the values:

$$W = 320 - 17 = 303$$

So, the work done is 303.

10.5 Q5 (06)

If

$$\vec{F} = (5xy - 6x^2)\hat{\imath} + (2y - 4x)\hat{\jmath}$$

Evaluate

$$\int_C \vec{F} \cdot d\vec{r}$$

along the curve C in the xy plane,

$$y = x^3$$

from the point $(1, 1)$ to $(2, 8)$.

ID	CT2V-3
Source	CT2V Q3 [06 marks]

Answer:

We write the line integral:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C [(5xy - 6x^2)dx + (2y - 4x)dy]$$

We are given the curve relation:

$$y = x^3$$

We find the differential dy:

$$dy = 3x^2 dx$$

We substitute

$$y = x^3$$

and

$$dy = 3x^2 dx$$

into the line integral. The variable x goes from 1 to 2:

$$\int_C \vec{F} \cdot d\vec{r} = \int_1^2 [(5x(x^3) - 6x^2)dx + (2(x^3) - 4x)(3x^2 dx)]$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_1^2 [(5x^4 - 6x^2) + (6x^5 - 12x^3)]dx$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_1^2 (6x^5 + 5x^4 - 12x^3 - 6x^2)dx$$

We integrate each term with respect to x:

$$\int_C \vec{F} \cdot d\vec{r} = \left[x^6 + x^5 - 3x^4 - 2x^3\right]_1^2$$

We evaluate at the upper limit $x = 2$:

$$2^6 + 2^5 - 3(2^4) - 2(2^3) = 64 + 32 - 48 - 16 = 32$$

We evaluate at the lower limit $x = 1$:

$$1^6 + 1^5 - 3(1^4) - 2(1^3) = 1 + 1 - 3 - 2 = -3$$

We subtract the lower limit value from the upper limit value:

$$\int_C \vec{F} \cdot d\vec{r} = 32 - (-3) = 35$$

So, the value of the line integral is 35.

11 Topic 07: Irrotational and Conservative Fields

This file contains the organized questions and answers for **Irrotational and Conservative Fields**, priority ranked as **Priority 7** based on frequency and exam weight.

11.1 Q1 (03)

Find the constants a, b, c so that

$$\bar{V} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$$

is irrotational.

ID	PYQ-2018-1b
Source	2018 Q1(b) [03 marks]

Answer:
A vector field $\bar{V}$ is irrotational if its curl is zero:

$$\bar{\nabla} \times \bar{V} = 0$$

We set up the curl calculation:

$$\bar{\nabla} \times \bar{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x + 2y + az & bx - 3y - z & 4x + cy + 2z \end{vmatrix} = 0$$

Expand the determinant:

$$\bar{\nabla} \times \bar{V} = \hat{i}\left(\frac{\partial}{\partial y}(4x + cy + 2z) - \frac{\partial}{\partial z}(bx - 3y - z)\right) - \hat{j}\left(\frac{\partial}{\partial x}(4x + cy + 2z) - \frac{\partial}{\partial z}(x + 2y + az)\right) + \hat{k}\left(\frac{\partial}{\partial x}(bx - 3y - z) - \frac{\partial}{\partial y}(x +$$

Calculate the derivatives:

$$\bar{\nabla} \times \bar{V} = \hat{i}(c - (-1)) - \hat{j}(4 - a) + \hat{k}(b - 2) = 0$$

This gives the system:

$$c + 1 = 0 \implies c = -1$$

$$4 - a = 0 \implies a = 4$$

$$b - 2 = 0 \implies b = 2$$

So the constants are

$$a = 4, \quad b = 2, \quad c = -1$$

11.2 Q2 (06)

Prove that

$$\bar{F} = (y^2 \cos x + z^3)\hat{i} + (2y \sin x - 4)\hat{j} + (3xz^2 + 2)\hat{k}$$

represents a conservative force field. Also find the scalar potential for $\bar{F}$.

ID	PYQ-2018-2a
Source	2018 Q2(a) [06 marks]

Answer:

11.2.0.1 1. Prove Conservative

The force field is conservative if its curl is zero:

$$\bar{\nabla} \times \bar{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 \cos x + z^3 & 2y \sin x - 4 & 3xz^2 + 2 \end{vmatrix}$$

Calculate the components: * $\hat{i}$-component:

$$\frac{\partial}{\partial y}(3xz^2 + 2) - \frac{\partial}{\partial z}(2y \sin x - 4) = 0 - 0 = 0$$

- $\hat{j}$-component:

$$\frac{\partial}{\partial z}(y^2 \cos x + z^3) - \frac{\partial}{\partial x}(3xz^2 + 2) = 3z^2 - 3z^2 = 0$$

- $\hat{k}$-component:

$$\frac{\partial}{\partial x}(2y \sin x - 4) - \frac{\partial}{\partial y}(y^2 \cos x + z^3) = 2y \cos x - 2y \cos x = 0$$

Since the curl is the zero vector, the force field is conservative.

11.2.0.2 2. Find Scalar Potential

We solve the relation

$$\bar{F} = \bar{\nabla}\phi :$$

$$\frac{\partial \phi}{\partial x} = y^2 \cos x + z^3 \quad \ldots(1)$$

$$\frac{\partial \phi}{\partial y} = 2y \sin x - 4 \quad \ldots(2)$$

$$\frac{\partial \phi}{\partial z} = 3xz^2 + 2 \quad \ldots(3)$$

Integrate equation (1) with respect to x:

$$\phi = y^2 \sin x + xz^3 + g(y, z) \quad \ldots(4)$$

Differentiate equation (4) with respect to y and equate it to equation (2):

$$\frac{\partial \phi}{\partial y} = 2y \sin x + \frac{\partial g}{\partial y} = 2y \sin x - 4 \implies \frac{\partial g}{\partial y} = -4$$

Integrate with respect to y:

$$g(y, z) = -4y + h(z)$$

Substitute this back into equation (4):

$$\phi = y^2 \sin x + xz^3 - 4y + h(z) \quad \ldots(5)$$

Now differentiate equation (5) with respect to z and equate it to equation (3):

$$\frac{\partial \phi}{\partial z} = 3xz^2 + h'(z) = 3xz^2 + 2 \implies h'(z) = 2$$

Integrate with respect to z:

$$h(z) = 2z + C$$

Substitute this back into equation (5) to get the final potential function:

$$\phi(x, y, z) = y^2 \sin x + xz^3 - 4y + 2z + C$$

11.3 Q3 (04)

If $\bar{F}$ represents a conservative force field then show that

$$\nabla \times \bar{F} = 0.$$

ID	PYQ-2019-3c
Source	2019 Q3(c) [04 marks]

Answer:
A force field $\vec{F}$ is conservative if the work done in moving a particle between two points is independent of the path. This implies that there exists a scalar potential function ϕ such that:

$$\vec{F} = \vec{\nabla}\phi$$

We calculate the curl of the field:

$$\vec{\nabla} \times \vec{F} = \vec{\nabla} \times (\vec{\nabla}\phi)$$

Writing this in determinant form:

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix}$$

$$= \hat{i}\left(\frac{\partial^2 \phi}{\partial y \partial z} - \frac{\partial^2 \phi}{\partial z \partial y}\right) - \hat{j}\left(\frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \phi}{\partial z \partial x}\right) + \hat{k}\left(\frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial y \partial x}\right)$$

Assuming the second-order partial derivatives of ϕ are continuous, mixed partial derivatives are equal:

$$\frac{\partial^2 \phi}{\partial y \partial z} = \frac{\partial^2 \phi}{\partial z \partial y}, \quad \frac{\partial^2 \phi}{\partial x \partial z} = \frac{\partial^2 \phi}{\partial z \partial x}, \quad \frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}$$

So we get:

$$\vec{\nabla} \times \vec{F} = 0\hat{i} - 0\hat{j} + 0\hat{k} = \vec{0}$$

This completes the proof.

11.4 Q4 (06)

Show that,

$$\bar{A} = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k}$$

is irrotational. Find Φ such that

$$\bar{A} = \nabla\Phi.$$

ID	PYQ-2020-3a
Source	2020 Q3(a) [06 marks]

Answer:

11.4.0.1 1. Prove Irrotational

The field is irrotational if its curl is zero:

$$\bar{\nabla} \times \bar{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xy + z^3 & 3x^2 - z & 3xz^2 - y \end{vmatrix}$$

Calculate the components: * $\hat{i}$-component:

$$\frac{\partial}{\partial y}(3xz^2 - y) - \frac{\partial}{\partial z}(3x^2 - z) = -1 - (-1) = 0$$

- $\hat{j}$-component:

$$\frac{\partial}{\partial z}(6xy + z^3) - \frac{\partial}{\partial x}(3xz^2 - y) = 3z^2 - 3z^2 = 0$$

- $\hat{k}$-component:

$$\frac{\partial}{\partial x}(3x^2 - z) - \frac{\partial}{\partial y}(6xy + z^3) = 6x - 6x = 0$$

The curl is the zero vector, so the field is irrotational.

11.4.0.2 2. Find Potential Function

We solve the relation

$$\bar{A} = \bar{\nabla}\Phi :$$

$$\frac{\partial \Phi}{\partial x} = 6xy + z^3 \implies \Phi = 3x^2y + xz^3 + g(y, z) \quad \ldots (1)$$

Differentiate equation (1) with respect to y:

$$\frac{\partial \Phi}{\partial y} = 3x^2 + \frac{\partial g}{\partial y} = 3x^2 - z \implies \frac{\partial g}{\partial y} = -z$$

Integrate with respect to y:

$$g(y, z) = -yz + h(z)$$

Substitute this back into equation (1):

$$\Phi = 3x^2y + xz^3 - yz + h(z) \quad \ldots (2)$$

Differentiate equation (2) with respect to z:

$$\frac{\partial \Phi}{\partial z} = 3xz^2 - y + h'(z) = 3xz^2 - y \implies h'(z) = 0 \implies h(z) = C$$

So the scalar potential is:

$$\Phi = 3x^2y + xz^3 - yz + C$$

11.5 Q5 (05)

Show that

$$\vec{F} = (y^2 + 2xz^2)\hat{\imath} + (2xy - z)\hat{\jmath} + (2x^2z - y + 2z)\hat{k}$$

is irrotational vector and hence find corresponding scalar potential.

ID	PYQ-2021-2b
Source	2021 Q2(b) [05 marks]

Answer:

11.5.0.1 1. Show Irrotational

A vector field $\vec{F}$ is irrotational if its curl is zero:

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 + 2xz^2 & 2xy - z & 2x^2z - y + 2z \end{vmatrix}$$

Calculate the components of the curl: * $\hat{\imath}$-component:

$$\frac{\partial}{\partial y}(2x^2z - y + 2z) - \frac{\partial}{\partial z}(2xy - z) = -1 - (-1) = 0$$

- $\hat{\jmath}$-component:

$$\frac{\partial}{\partial z}(y^2 + 2xz^2) - \frac{\partial}{\partial x}(2x^2z - y + 2z) = 4xz - 4xz = 0$$

- $\hat{k}$-component:

$$\frac{\partial}{\partial x}(2xy - z) - \frac{\partial}{\partial y}(y^2 + 2xz^2) = 2y - 2y = 0$$

The curl is the zero vector, so the field is irrotational.

11.5.0.2 2. Find Potential Function

We solve the relation

$$\vec{\nabla}\Phi = \vec{F}:$$

$$\frac{\partial \Phi}{\partial x} = y^2 + 2xz^2 \implies \Phi = xy^2 + x^2z^2 + g(y,z) \quad \ldots(1)$$

Differentiate equation (1) with respect to y:

$$\frac{\partial \Phi}{\partial y} = 2xy + \frac{\partial g}{\partial y} = 2xy - z \implies \frac{\partial g}{\partial y} = -z$$

Integrate with respect to y:

$$g(y,z) = -yz + h(z)$$

Substitute this back into equation (1):

$$\Phi = xy^2 + x^2z^2 - yz + h(z) \quad \ldots(2)$$

Differentiate equation (2) with respect to z:

$$\frac{\partial \Phi}{\partial z} = 2x^2z - y + h'(z) = 2x^2z - y + 2z \implies h'(z) = 2z \implies h(z) = z^2 + C$$

So the scalar potential is:

$$\Phi = xy^2 + x^2z^2 - yz + z^2 + C$$

11.6 Q6 (03+06)

Find constants a, b, c so that

$$\vec{F} = (x + 2y + az)\hat{\imath} + (bx - 3y - z)\hat{\jmath} + (4x + cy + 2z)\hat{k}$$

is a conservative force field. Also obtain the scalar potential.

ID	CT2V-1
Source	CT2V Q1 [03+06 marks]

Answer:

11.6.0.1 1. Find Constants a, b, c

A vector field $\vec{F}$ is conservative if its curl is zero:

$$\vec{\nabla} \times \vec{F} = 0$$

We calculate the curl of $\vec{F}$:

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x + 2y + az & bx - 3y - z & 4x + cy + 2z \end{vmatrix} = 0$$

Let us expand this determinant:

$$\vec{\nabla}\times\vec{F} = \hat{i}\left[\frac{\partial}{\partial y}(4x + cy + 2z) - \frac{\partial}{\partial z}(bx - 3y - z)\right] - \hat{j}\left[\frac{\partial}{\partial x}(4x + cy + 2z) - \frac{\partial}{\partial z}(x + 2y + az)\right] + \hat{k}\left[\frac{\partial}{\partial x}(bx - 3y - z) - \frac{\partial}{\partial y}(x + 2$$

We calculate the partial derivatives:

$$\vec{\nabla} \times \vec{F} = \hat{i}(c - (-1)) - \hat{j}(4 - a) + \hat{k}(b - 2) = 0$$

$$\vec{\nabla} \times \vec{F} = \hat{i}(c + 1) + \hat{j}(a - 4) + \hat{k}(b - 2) = 0$$

We equate the components to zero:

$$c + 1 = 0 \implies c = -1$$

$$a - 4 = 0 \implies a = 4$$

$$b - 2 = 0 \implies b = 2$$

So, the constants are

$$a = 4, \quad b = 2, \quad c = -1$$

11.6.0.2 2. Find Scalar Potential

We substitute a, b, and c back into $\vec{F}$:

$$\vec{F} = (x + 2y + 4z)\hat{i} + (2x - 3y - z)\hat{j} + (4x - y + 2z)\hat{k}$$

Let ϕ be the scalar potential such that

$$\vec{F} = \vec{\nabla}\phi$$

. This gives three partial differential equations:

$$\frac{\partial\phi}{\partial x} = x + 2y + 4z \quad \ldots(1)$$

$$\frac{\partial\phi}{\partial y} = 2x - 3y - z \quad \ldots(2)$$

$$\frac{\partial\phi}{\partial z} = 4x - y + 2z \quad \ldots(3)$$

We integrate equation (1) with respect to x:

$$\phi = \frac{1}{2}x^2 + 2xy + 4xz + g(y, z) \quad \ldots(4)$$

Here, $g(y, z)$ is an arbitrary integration function.
Now we differentiate equation (4) with respect to y. We equate it to equation (2):

$$\frac{\partial\phi}{\partial y} = 2x + \frac{\partial g}{\partial y} = 2x - 3y - z$$

$$\frac{\partial g}{\partial y} = -3y - z$$

We integrate this with respect to y:

$$g(y,z) = -\frac{3}{2}y^2 - yz + h(z)$$

Here, $h(z)$ is an arbitrary integration function.
We substitute $g(y,z)$ back into equation (4):

$$\phi = \frac{1}{2}x^2 + 2xy + 4xz - \frac{3}{2}y^2 - yz + h(z) \quad \ldots(5)$$

Now we differentiate equation (5) with respect to z. We equate it to equation (3):

$$\frac{\partial \phi}{\partial z} = 4x - y + h'(z) = 4x - y + 2z$$

$$h'(z) = 2z$$

We integrate this with respect to z:

$$h(z) = z^2 + C$$

Here, C is a constant.
We substitute $h(z)$ back into equation (5) to get the final potential:

$$\phi = \frac{1}{2}x^2 - \frac{3}{2}y^2 + z^2 + 2xy + 4xz - yz + C$$

12 Topic 15: Vector Basics

This file contains the organized questions and answers for **Vector Basics**, priority ranked as **Priority 15** based on frequency and exam weight.

12.1 Q1 (04)

Find an equation for the plane perpendicular to the vector

$$\bar{A} = 2\hat{i} + 3\hat{j} + 6\hat{k}$$

and passing through the terminal point of the vector

$$\bar{B} = \hat{i} + 5\hat{j} + 3\hat{k}$$

ID	PYQ-2019-1a
Appeared in	2019 Q1(a) [04 marks], 2024 Q2(a) [04 marks]
Frequency	⋆⋆ (2 times)

Answer:
We assume the vector $\vec{B}$ starts at the origin. Its terminal point is $P(1, 5, 3)$.
The equation of a plane passing through a point $P(x\_0, y\_0, z\_0)$ and perpendicular to a normal vector

$$\vec{A} = a\hat{i} + b\hat{j} + c\hat{k}$$

is:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

Substitute the point $(1, 5, 3)$ and the components of

$$\vec{A} = 2\hat{i} + 3\hat{j} + 6\hat{k} :$$

$$2(x - 1) + 3(y - 5) + 6(z - 3) = 0$$

Expand the equation:

$$2x - 2 + 3y - 15 + 6z - 18 = 0$$

$$2x + 3y + 6z - 35 = 0 \implies 2x + 3y + 6z = 35$$

So the equation of the plane is $2x + 3y + 6z = 35$.

12.2 Q2 (04)

Find the acute angles which the line joining the points $(1, -3, 2)$ and $(3, -5, 1)$ makes with the coordinate axes.

ID	PYQ-2019-1c
Source	2019 Q1(c) [04 marks]

Answer:
Let the points be $P(1, -3, 2)$ and $Q(3, -5, 1)$. The vector $\vec{PQ}$ representing the line segment is:

$$\vec{PQ} = (3 - 1)\hat{i} + (-5 - (-3))\hat{j} + (1 - 2)\hat{k} = 2\hat{i} - 2\hat{j} - \hat{k}$$

Find the magnitude of the vector $\vec{PQ}$:

$$|\vec{PQ}| = \sqrt{2^2 + (-2)^2 + (-1)^2} = \sqrt{4 + 4 + 1} = \sqrt{9} = 3$$

The direction cosines of this vector are:

$$\cos\alpha = \frac{2}{3}, \quad \cos\beta = -\frac{2}{3}, \quad \cos\gamma = -\frac{1}{3}$$

where α, β, γ are the angles made with the positive directions of the x, y, and z axes respectively. For the acute angles, the cosines must be positive:

$$\cos\alpha = \frac{2}{3} \implies \alpha = \cos^{-1}\left(\frac{2}{3}\right) \approx 48.19^\circ$$

$$\cos\beta = \frac{2}{3} \implies \beta = \cos^{-1}\left(\frac{2}{3}\right) \approx 48.19^\circ$$

$$\cos\gamma = \frac{1}{3} \implies \gamma = \cos^{-1}\left(\frac{1}{3}\right) \approx 70.53^\circ$$

12.3 Q3 (04)

What is the vector? Find a unit vector parallel to the resultant of vectors

$$\vec{r}_1 = 2\hat{\imath} + 4\hat{\jmath} - 5\hat{k}$$

$$\vec{r}_2 = \hat{\imath} + 2\hat{\jmath} + \hat{k}$$

ID	PYQ-2020-1a
Source	2020 Q1(a) [04 marks]

Answer:

12.3.0.1 1. Definition of a Vector A vector is a mathematical quantity that has both magnitude and direction. It also satisfies the parallelogram law of vector addition.

12.3.0.2 2. Find the Unit Vector We calculate the resultant vector $\vec{R}$ by adding the two given vectors:

$$\vec{R} = \vec{r}_1 + \vec{r}_2 = (2\hat{\imath} + 4\hat{\jmath} - 5\hat{k}) + (\hat{\imath} + 2\hat{\jmath} + \hat{k}) = 3\hat{\imath} + 6\hat{\jmath} - 4\hat{k}$$

Find the magnitude of the resultant vector:

$$|\vec{R}| = \sqrt{3^2 + 6^2 + (-4)^2} = \sqrt{9 + 36 + 16} = \sqrt{61}$$

The unit vector parallel to $\vec{R}$ is:

$$\hat{u} = \frac{\vec{R}}{|\vec{R}|} = \frac{3\hat{\imath} + 6\hat{\jmath} - 4\hat{k}}{\sqrt{61}}$$

12.4 Q4 (04)

If

$$\bar{A} = 2\hat{\imath} + \hat{\jmath} - 3\hat{k}$$

and

$$\bar{B} = \hat{\imath} - 2\hat{\jmath} + \hat{k}$$

Find a vector of magnitude 5 perpendicular to both $\bar{A}$ and $\bar{B}$.

ID	PYQ-2020-1c
Source	2020 Q1(c) [04 marks]

Answer:

We calculate the cross product of the two vectors to find a perpendicular direction:

$$\bar{A} \times \bar{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -3 \\ 1 & -2 & 1 \end{vmatrix} = \hat{i}(1-6) - \hat{j}(2-(-3)) + \hat{k}(-4-1) = -5\hat{i} - 5\hat{j} - 5\hat{k}$$

We simplify the perpendicular direction to:

$$\vec{v} = \hat{i} + \hat{j} + \hat{k}$$

The unit normal vector $\hat{n}$ is:

$$\hat{n} = \pm \frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{1^2 + 1^2 + 1^2}} = \pm \frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$$

The vector with a magnitude of 5 is:

$$\vec{P} = \pm 5\hat{n} = \pm \frac{5}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$$

12.5 Q5 (04)

What are the vector and scalar quantity? Give some examples. If

$$\vec{A} = 3\hat{i} - \hat{j} + 4\hat{k}$$

$$\vec{B} = -2\hat{i} + 4\hat{j} - 3\hat{k}$$

and

$$\vec{C} = \hat{i} + 2\hat{j} - \hat{k}$$

Find the magnitude of $2\vec{A} - 3\vec{B} + 2\vec{C}$.

ID	PYQ-2021-1a
Source	2021 Q1(a) [04 marks]

Answer:

12.5.0.1 1. Definitions and Examples

- **Scalar Quantity:** A physical quantity that has only magnitude but no direction. Examples: Mass, Temperature, Time.
- **Vector Quantity:** A physical quantity that has both magnitude and direction, and obeys vector addition rules. Examples: Velocity, Force, Acceleration.

12.5.0.2 2. Find the Magnitude We calculate the vector expression:

$$\vec{V} = 2\vec{A} - 3\vec{B} + 2\vec{C}$$

Substitute the given vectors:

$$2\vec{A} = 6\hat{i} - 2\hat{j} + 8\hat{k}$$

$$-3\vec{B} = 6\hat{i} - 12\hat{j} + 9\hat{k}$$

$$2\vec{C} = 2\hat{\imath} + 4\hat{\jmath} - 2\hat{k}$$

Sum the components together:

$$\vec{V} = (6+6+2)\hat{\imath} + (-2-12+4)\hat{\jmath} + (8+9-2)\hat{k} = 14\hat{\imath} - 10\hat{\jmath} + 15\hat{k}$$

Find the magnitude:

$$|\vec{V}| = \sqrt{14^2 + (-10)^2 + 15^2} = \sqrt{196+100+225} = \sqrt{521}$$

So the magnitude is $\sqrt{521}$.

12.6 Q6 (04)

What is meant by dot and cross product in vector analysis? Determine a unit vector perpendicular to the plane of

$$\vec{A} = 2\hat{\imath} - 6\hat{\jmath} - 3\hat{k}$$

and

$$\vec{B} = 4\hat{\imath} + 3\hat{\jmath} - \hat{k}$$

ID	PYQ-2021-1b
Appeared in	2021 Q1(b) [04 marks], 2023 Q1(a) [03 marks]
Frequency	⋆⋆ (2 times)

Answer:

12.6.0.1 1. Definitions

- **Dot Product:** The scalar product of two vectors $\vec{A}$ and $\vec{B}$ is defined as

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}|\cos\theta$$

, where θ is the angle between them. * **Cross Product:** The vector product of two vectors $\vec{A}$ and $\vec{B}$ is defined as

$$\vec{A} \times \vec{B} = |\vec{A}||\vec{B}|\sin\theta\hat{n}$$

Where $\hat{n}$ is a unit vector perpendicular to the plane of the two vectors.

12.6.0.2 2. Find the Perpendicular Unit Vector

We calculate the cross product of $\vec{A}$ and $\vec{B}$ to get a perpendicular vector:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & -6 & -3 \\ 4 & 3 & -1 \end{vmatrix} = \hat{\imath}(6-(-9)) - \hat{\jmath}(-2-(-12)) + \hat{k}(6-(-24)) = 15\hat{\imath} - 10\hat{\jmath} + 30\hat{k}$$

Find the magnitude of this cross product:

$$|\vec{A} \times \vec{B}| = \sqrt{15^2 + (-10)^2 + 30^2} = \sqrt{225+100+900} = \sqrt{1225} = 35$$

The unit vector perpendicular to the plane is:

$$\hat{n} = \pm\frac{15\hat{\imath} - 10\hat{\jmath} + 30\hat{k}}{35} = \pm\left(\frac{3}{7}\hat{\imath} - \frac{2}{7}\hat{\jmath} + \frac{6}{7}\hat{k}\right)$$

12.7 Q7 (03)

Determine the angles α, β, and γ which the vector

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

makes with the positive directions of the coordinate axes and show that

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1.$$

ID	PYQ-2023-2a
Source	2023 Q2(a) [03 marks]

Answer:
Let the magnitude of the position vector $\vec{r}$ be:

$$r = |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

The cosine of the angles made with the coordinate axes are given by:

$$\cos\alpha = \frac{\vec{r}\cdot\hat{i}}{r} = \frac{x}{r}$$

$$\cos\beta = \frac{\vec{r}\cdot\hat{j}}{r} = \frac{y}{r}$$

$$\cos\gamma = \frac{\vec{r}\cdot\hat{k}}{r} = \frac{z}{r}$$

Sum the squares of these cosines:

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{x^2}{r^2} + \frac{y^2}{r^2} + \frac{z^2}{r^2} = \frac{x^2 + y^2 + z^2}{r^2} = \frac{r^2}{r^2} = 1$$

This completes the proof.

12.8 Q8 (03)

Find the constant 'a' such that the vectors

$$2\hat{i} - \hat{j} + \hat{k}$$

$$\hat{i} + 2\hat{j} - 3\hat{k}$$

and

$$3\hat{i} + a\hat{j} + 5\hat{k}$$

are coplanar.

ID	PYQ-2023-2b
Source	2023 Q2(b) [03 marks]

Answer:
Three vectors are coplanar if and only if their scalar triple product is zero. We set up the determinant:

$$\begin{vmatrix} 2 & -1 & 1 \\ 1 & 2 & -3 \\ 3 & a & 5 \end{vmatrix} = 0$$

Expand this determinant along the first row:

$$2(10-(-3a))-(-1)(5-(-9))+1(a-6)=0$$

$$2(10+3a)+14+a-6=0$$

$$20+6a+14+a-6=0$$

$$7a+28=0 \implies 7a=-28 \implies a=-4$$

So the constant is $a=-4$.

12.9 Q9. What is vector field? Graph the vector fields defined by: (05)

ID	PYQ-2024-1a
Source	2024 Q1(a) [05 marks]

Answer:

- (i)

$$\vec{V}(x,y)=x\hat{i}+y\hat{j}$$

- (ii)

$$\vec{V}(x,y)=-x\hat{i}-y\hat{j}$$

Answer:

12.9.0.1 1. Definition of a Vector Field A vector field is a function. It assigns a vector to every point in a given region. In a two-dimensional space, we write it as:

$$\vec{V}(x,y)=P(x,y)\hat{i}+Q(x,y)\hat{j}$$

Here, $P(x,y)$ and $Q(x,y)$ are scalar functions.

12.9.0.2 2. Graphing the Vector Fields

12.9.0.2.1 (i)

$$\vec{V}(x,y)=x\hat{i}+y\hat{j}$$

For any point (x,y), the vector points directly away from the origin $(0,0)$. The length of the vector equals the distance of the point from the origin,

$$r=\sqrt{x^2+y^2}.$$

We choose a few points to show the field: * At $(1,0)$, the vector is $\hat{i}$. It points right with a length of 1. * At $(0,1)$, the vector is $\hat{j}$. It points up with a length of 1. * At $(-1,0)$, the vector is

$$-\hat{i}$$

It points left with a length of 1. * At $(0,-1)$, the vector is

$$-\hat{j}$$

It points down with a length of 1. * At $(2,2)$, the vector is

$$2\hat{i}+2\hat{j}$$

It points diagonally outward with a length of $2\sqrt{2}$.

This field looks like a source. All vectors point radially outward from the center.

12.9.0.2.2 (ii)

$$\vec{V}(x,y) = -x\hat{i} - y\hat{j}$$

For any point (x, y), the vector points directly toward the origin $(0, 0)$. Its length equals the distance from the origin.

We choose a few points to show the field: * At $(1, 0)$, the vector is

$$-\hat{i}$$

It points left with a length of 1. * At $(0, 1)$, the vector is

$$-\hat{j}$$

It points down with a length of 1. * At $(-1, 0)$, the vector is $\hat{i}$. It points right with a length of 1. * At $(0, -1)$, the vector is $\hat{j}$. It points up with a length of 1. * At $(2, 2)$, the vector is

$$-2\hat{i} - 2\hat{j}$$

It points diagonally inward with a length of $2\sqrt{2}$.

This field looks like a sink. All vectors point radially inward toward the center.

12.10 Q10 (2+2+6)

Explain the physical significance of dot and cross product. Show that

$$\vec{A} = \frac{1}{3}(2\hat{i} - 2\hat{j} + \hat{k})$$

$$\vec{B} = \frac{1}{3}(\hat{i} + 2\hat{j} + 2\hat{k})$$

and

$$\vec{C} = \frac{1}{3}(2\hat{i} + \hat{j} - 2\hat{k})$$

are mutually orthogonal unit vectors.

ID	CT1V-1
Source	CT1V Q1 [2+2+6 marks]

Answer:

12.10.0.1 1. Physical Significance of Dot Product The dot product represents the projection of one vector onto another. It shows how much the two vectors point in the same direction. It results in a scalar quantity. In physics, we use it to calculate work done by a force:

$$W = \vec{F} \cdot \vec{d}$$

Here, W is work. $\vec{F}$ is force. $\vec{d}$ is displacement.

12.10.0.2 2. Physical Significance of Cross Product The cross product represents a rotational effect. It results in a new vector. This vector is perpendicular to the plane of the two original vectors. The magnitude of the cross product equals the area of the parallelogram formed by the two vectors. In physics, we use it to calculate torque:

$$\vec{\tau} = \vec{r} \times \vec{F}$$

Here, $\vec{\tau}$ is torque. $\vec{r}$ is the position vector. $\vec{F}$ is force.

12.10.0.3 3. Orthogonal Unit Vectors Verification First, we check the magnitude of each vector.
For vector $\vec{A}$:

$$|\vec{A}| = \sqrt{\left(\frac{2}{3}\right)^2 + \left(-\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2} = \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{1}{9}} = \sqrt{\frac{9}{9}} = 1$$

For vector $\vec{B}$:

$$|\vec{B}| = \sqrt{\left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2} = \sqrt{\frac{1}{9} + \frac{4}{9} + \frac{4}{9}} = \sqrt{\frac{9}{9}} = 1$$

For vector $\vec{C}$:

$$|\vec{C}| = \sqrt{\left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(-\frac{2}{3}\right)^2} = \sqrt{\frac{4}{9} + \frac{1}{9} + \frac{4}{9}} = \sqrt{\frac{9}{9}} = 1$$

So, all three vectors are unit vectors.

Next, we check if they are mutually orthogonal. We calculate their dot products.

Dot product of $\vec{A}$ and $\vec{B}$:

$$\vec{A} \cdot \vec{B} = \left[\frac{1}{3}(2\hat{i} - 2\hat{j} + \hat{k})\right] \cdot \left[\frac{1}{3}(\hat{i} + 2\hat{j} + 2\hat{k})\right]$$

$$\vec{A} \cdot \vec{B} = \frac{1}{9}[2(1) + (-2)(2) + 1(2)] = \frac{1}{9}[2 - 4 + 2] = 0$$

Dot product of $\vec{B}$ and $\vec{C}$:

$$\vec{B} \cdot \vec{C} = \left[\frac{1}{3}(\hat{i} + 2\hat{j} + 2\hat{k})\right] \cdot \left[\frac{1}{3}(2\hat{i} + \hat{j} - 2\hat{k})\right]$$

$$\vec{B} \cdot \vec{C} = \frac{1}{9}[1(2) + 2(1) + 2(-2)] = \frac{1}{9}[2 + 2 - 4] = 0$$

Dot product of $\vec{C}$ and $\vec{A}$:

$$\vec{C} \cdot \vec{A} = \left[\frac{1}{3}(2\hat{i} + \hat{j} - 2\hat{k})\right] \cdot \left[\frac{1}{3}(2\hat{i} - 2\hat{j} + \hat{k})\right]$$

$$\vec{C} \cdot \vec{A} = \frac{1}{9}[2(2) + 1(-2) + (-2)(1)] = \frac{1}{9}[4 - 2 - 2] = 0$$

The dot product of each pair is zero. So, the vectors are mutually orthogonal.

13 Topic 16: Area, Volume, and Triple Products

This file contains the organized questions and answers for **Area, Volume, and Triple Products**, priority ranked as **Priority 16** based on frequency and exam weight.

13.1 Q1 (04)

Find the volume of the parallelepiped whose edges are represented by the vector

$$\bar{A} = 2\hat{\imath} - 3\hat{\jmath} + 4\hat{k}$$

$$\bar{B} = \hat{\imath} + 2\hat{\jmath} - \hat{k}$$

and

$$\bar{C} = 3\hat{\imath} - \hat{\jmath} + 2\hat{k}$$

ID	PYQ-2019-1b
Source	2019 Q1(b) [04 marks]

Answer:
The volume V of a parallelepiped with adjacent edges represented by vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$ is given by the absolute value of their scalar triple product:

$$V = |\vec{A} \cdot (\vec{B} \times \vec{C})|$$

We calculate the scalar triple product using the determinant:

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \begin{vmatrix} 2 & -3 & 4 \\ 1 & 2 & -1 \\ 3 & -1 & 2 \end{vmatrix}$$

Expand along the first row:

$$= 2\begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} - (-3)\begin{vmatrix} 1 & -1 \\ 3 & 2 \end{vmatrix} + 4\begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix}$$

$$= 2[4 - 1] + 3[2 - (-3)] + 4[-1 - 6]$$

$$= 2(3) + 3(5) + 4(-7) = 6 + 15 - 28 = -7$$

Taking the absolute value:

$$V = |-7| = 7 \text{ cubic units}$$

13.2 Q2 (04)

Define triple product of vectors. Show that, $\bar{A} \cdot (\bar{B} \times \bar{C})$ is in absolute value equal to the volume of a parallelepiped with sides $\bar{A}$, $\bar{B}$ and $\bar{C}$.

ID	PYQ-2020-1b
Source	2020 Q1(b) [04 marks]

Answer:

13.2.0.1 1. Definition A triple product is a product involving three vectors. There are two kinds: * **Scalar Triple Product:** The expression $\bar{A} \cdot (\bar{B} \times \bar{C})$, which results in a scalar. * **Vector Triple Product:** The expression $\bar{A} \times (\bar{B} \times \bar{C})$, which results in a vector.

13.2.0.2 2. Geometric Proof

Let the three vectors $\bar{A}$, $\bar{B}$, and $\bar{C}$ represent the three adjacent edges of a parallelepiped.

The cross product vector $\bar{B} \times \bar{C}$ is perpendicular to the base face of the parallelepiped. The magnitude of this cross product is the area of the base parallelogram:

$$\text{Area of Base} = |\bar{B} \times \bar{C}|$$

Let θ be the angle between the vector $\bar{A}$ and the normal vector $\bar{B} \times \bar{C}$. The height h of the parallelepiped is the projection of $\bar{A}$ along this normal vector:

$$h = |\bar{A}| \cos\theta$$

Now calculate the scalar triple product:

$$\bar{A} \cdot (\bar{B} \times \bar{C}) = |\bar{A}||\bar{B} \times \bar{C}| \cos\theta = h \times (\text{Area of Base}) = \text{Volume of Parallelepiped}$$

Since a volume is always positive, we take the absolute value:

$$\text{Volume} = |\bar{A} \cdot (\bar{B} \times \bar{C})|$$

The proof is complete.

13.3 Q3 (04)

Find the area of a triangle with vertices at $(3, -1, 2)$, $(1, -1, -3)$ and $(4, -3, 1)$.

ID	PYQ-2021-1c
Appeared in	2021 Q1(c) [04 marks], 2023 Q1(b) [03 marks]
Frequency	⋆⋆ (2 times)

Answer:

Let the vertices of the triangle be $P(3, -1, 2)$, $Q(1, -1, -3)$, and $R(4, -3, 1)$.

We construct two side vectors starting from point P:

$$\vec{PQ} = (1-3)\hat{\imath} + (-1-(-1))\hat{\jmath} + (-3-2)\hat{k} = -2\hat{\imath} - 5\hat{k}$$

$$\vec{PR} = (4-3)\hat{\imath} + (-3-(-1))\hat{\jmath} + (1-2)\hat{k} = \hat{\imath} - 2\hat{\jmath} - \hat{k}$$

Calculate the cross product of these side vectors:

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ -2 & 0 & -5 \\ 1 & -2 & -1 \end{vmatrix} = \hat{\imath}(0-10) - \hat{\jmath}(2-(-5)) + \hat{k}(4-0) = -10\hat{\imath} - 7\hat{\jmath} + 4\hat{k}$$

Find the magnitude of the cross product:

$$|\vec{PQ} \times \vec{PR}| = \sqrt{(-10)^2 + (-7)^2 + 4^2} = \sqrt{100 + 49 + 16} = \sqrt{165}$$

The area of the triangle is:

$$\text{Area} = \frac{1}{2}|\vec{PQ} \times \vec{PR}| = \frac{1}{2}\sqrt{165}$$

13.4 Q4. Evaluate: (04)

ID	PYQ-2024-3c
Source	2024 Q3(c) [04 marks]

Answer:

$$\iint_R \sqrt{x^2 + y^2}dxdy$$

over the region R in the xy plane bounded by

$$x^2 + y^2 = 36.$$

Answer:
We switch to polar coordinates:

$$x = r\cos\theta, \quad y = r\sin\theta, \quad dxdy = rdrd\theta$$

The region R is a disk of radius 6 centered at the origin, because

$$x^2 + y^2 = 36 = 6^2.$$

The integration limits are: * r goes from 0 to 6 * θ goes from 0 to 2π
The integrand is:

$$\sqrt{x^2 + y^2} = r$$

We rewrite the double integral:

$$\iint_R \sqrt{x^2 + y^2}dxdy = \int_0^{2\pi}\int_0^6 r \cdot rdrd\theta = \int_0^{2\pi}\int_0^6 r^2 drd\theta$$

We integrate with respect to r:

$$\int_0^6 r^2 dr = \left[\frac{r^3}{3}\right]_0^6 = \frac{216}{3} = 72$$

We integrate with respect to θ:

$$\int_0^{2\pi} 72d\theta = 72 \cdot 2\pi = 144\pi$$

The value of the integral is 144π.

14 Topic 17: Velocity, Acceleration, and Particle Motion

This file contains the organized questions and answers for **Velocity, Acceleration, and Particle Motion**, priority ranked as **Priority 17** based on frequency and exam weight.

14.1 Q1 (04)

A particle moves along the curve

$$x = 2t^2, y = t^2 - 4t, z = 3t - 5$$

, where t is the time. Find the components of its velocity and acceleration at time $t = 1$ in the direction

$$\hat{i} - 3\hat{j} + 2\hat{k}$$

ID	PYQ-2019-2a
Appeared in	2019 Q2(a) [04 marks], 2023 Q2(c) [04 marks]
Frequency	⋆⋆ (2 times)

Answer:

Let the direction vector be

$$\vec{d} = \hat{i} - 3\hat{j} + 2\hat{k}$$

The unit direction vector $\hat{u}$ is:

$$\hat{u} = \frac{\hat{i} - 3\hat{j} + 2\hat{k}}{\sqrt{1^2 + (-3)^2 + 2^2}} = \frac{\hat{i} - 3\hat{j} + 2\hat{k}}{\sqrt{14}}$$

The position vector of the particle is:

$$\vec{r}(t) = 2t^2\hat{i} + (t^2 - 4t)\hat{j} + (3t - 5)\hat{k}$$

14.1.0.1 1. Velocity Component Differentiate the position vector to get the velocity vector:

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = 4t\hat{i} + (2t - 4)\hat{j} + 3\hat{k}$$

Evaluate velocity at $t = 1$:

$$\vec{v}(1) = 4\hat{i} - 2\hat{j} + 3\hat{k}$$

The component of velocity in the direction of $\hat{u}$ is:

$$v_d = \vec{v}(1) \cdot \hat{u} = (4\hat{i} - 2\hat{j} + 3\hat{k}) \cdot \left(\frac{\hat{i} - 3\hat{j} + 2\hat{k}}{\sqrt{14}}\right) = \frac{4 + 6 + 6}{\sqrt{14}} = \frac{16}{\sqrt{14}}$$

14.1.0.2 2. Acceleration Component Differentiate the velocity vector to get the acceleration vector:

$$\vec{a}(t) = \frac{d\vec{v}}{dt} = 4\hat{i} + 2\hat{j}$$

Evaluate acceleration at $t = 1$:

$$\vec{a}(1) = 4\hat{i} + 2\hat{j}$$

The component of acceleration in the direction of $\hat{u}$ is:

$$a_d = \vec{a}(1) \cdot \hat{u} = (4\hat{i} + 2\hat{j}) \cdot \left(\frac{\hat{i} - 3\hat{j} + 2\hat{k}}{\sqrt{14}}\right) = \frac{4 - 6 + 0}{\sqrt{14}} = -\frac{2}{\sqrt{14}}$$

14.2 Q2 (04)

If $\bar{A}$ has a constant magnitude show that $\bar{A}$ and $\frac{d\bar{A}}{dt}$ are perpendicular provided

$$\left|\frac{d\bar{A}}{dt}\right| \neq 0.$$

ID	PYQ-2019-2b
Source	2019 Q2(b) [04 marks]

Answer:
Let the constant magnitude of $\vec{A}$ be

$$|\vec{A}| = c$$

(where c is a constant).
Then we can write:

$$\vec{A} \cdot \vec{A} = |\vec{A}|^2 = c^2$$

Differentiate both sides with respect to t:

$$\frac{d}{dt}(\vec{A} \cdot \vec{A}) = \frac{d}{dt}(c^2)$$

Using the product rule for vector dot products:

$$\vec{A} \cdot \frac{d\vec{A}}{dt} + \frac{d\vec{A}}{dt} \cdot \vec{A} = 0$$

Since the dot product is commutative (

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

):

$$2\left(\vec{A} \cdot \frac{d\vec{A}}{dt}\right) = 0 \implies \vec{A} \cdot \frac{d\vec{A}}{dt} = 0$$

Since the dot product of $\vec{A}$ and $\frac{d\vec{A}}{dt}$ is zero, and neither vector is zero (given

$$\left|\frac{d\vec{A}}{dt}\right| \neq 0$$

and magnitude of $\vec{A}$ is a non-zero constant), the vectors must be perpendicular.

14.3 Q3 (04)

A particle moves along a curve whose parametric equations are

$$x = e^{-t}, y = 2\cos 3t, z = 2\sin 3t$$

, where t is the time. Find the magnitudes of the velocity and acceleration at $t = 0$.

ID	PYQ-2020-2a
Source	2020 Q2(a) [04 marks]

Answer:
The position vector of the particle is:

$$\vec{r}(t) = e^{-t}\hat{i} + 2\cos 3t\hat{j} + 2\sin 3t\hat{k}$$

14.3.0.1 1. Magnitude of Velocity

Differentiate the position vector to find the velocity vector:

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = -e^{-t}\hat{\imath} - 6\sin 3t\hat{\jmath} + 6\cos 3t\hat{k}$$

At $t = 0$:

$$\vec{v}(0) = -\hat{\imath} - 0\hat{\jmath} + 6\hat{k} = -\hat{\imath} + 6\hat{k}$$

The magnitude is:

$$|\vec{v}(0)| = \sqrt{(-1)^2 + 6^2} = \sqrt{1+36} = \sqrt{37}$$

14.3.0.2 2. Magnitude of Acceleration

Differentiate the velocity vector to find the acceleration vector:

$$\vec{a}(t) = \frac{d\vec{v}}{dt} = e^{-t}\hat{\imath} - 18\cos 3t\hat{\jmath} - 18\sin 3t\hat{k}$$

At $t = 0$:

$$\vec{a}(0) = \hat{\imath} - 18\hat{\jmath} - 0\hat{k} = \hat{\imath} - 18\hat{\jmath}$$

The magnitude is:

$$|\vec{a}(0)| = \sqrt{1^2 + (-18)^2} = \sqrt{1+324} = \sqrt{325} = 5\sqrt{13}$$

14.4 Q4 (04)

A particle moves so that its position vector is given by

$$\vec{r} = \cos\omega t\hat{\imath} + \sin\omega t\hat{\jmath}$$

where ω is constant. Show that the velocity $\vec{v}$ of the particle is perpendicular to $\vec{r}$, also show that

$$\vec{r} \times \vec{v} = \vec{a}$$

constant vector.

ID	PYQ-2024-2b
Source	2024 Q2(b) [04 marks]

Answer:

14.4.0.1 1. Show that Velocity is Perpendicular to Position

We find the velocity vector $\vec{v}$ by taking the derivative of $\vec{r}$ with respect to t:

$$\vec{v} = \frac{d\vec{r}}{dt} = -\omega\sin\omega t\hat{\imath} + \omega\cos\omega t\hat{\jmath}$$

We calculate the dot product of $\vec{r}$ and $\vec{v}$:

$$\vec{r}\cdot\vec{v} = (\cos\omega t)(-\omega\sin\omega t) + (\sin\omega t)(\omega\cos\omega t) = -\omega\sin\omega t\cos\omega t + \omega\sin\omega t\cos\omega t = 0$$

The dot product is zero. So, the velocity vector $\vec{v}$ is perpendicular to $\vec{r}$.

14.4.0.2 2. Show that $\vec{r} \times \vec{v}$ is a Constant Vector

We calculate the cross product:

$$\vec{r} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\omega t & \sin\omega t & 0 \\ -\omega\sin\omega t & \omega\cos\omega t & 0 \end{vmatrix}$$

$$\vec{r} \times \vec{v} = \hat{i}(0) - \hat{j}(0) + \hat{k}\left[(\cos\omega t)(\omega\cos\omega t) - (\sin\omega t)(-\omega\sin\omega t)\right]$$

$$\vec{r} \times \vec{v} = \hat{k}\left(\omega\cos^2\omega t + \omega\sin^2\omega t\right) = \omega(\cos^2\omega t + \sin^2\omega t)\hat{k} = \omega\hat{k}$$

The term ω is a constant. So, the vector $\omega\hat{k}$ is constant. Let

$$\vec{a} = \omega\hat{k}$$

We get:

$$\vec{r} \times \vec{v} = \vec{a}$$

This is a constant vector.

14.5 Q5 (2+8)

State Frenet-Serret formulas. Show that

$$\vec{r} = e^{-t}(\vec{A}\cos 2t + \vec{B}\sin 2t)$$

, where $\vec{A}$ and $\vec{B}$ are constant vectors, is a solution of the differential equation

$$\frac{d^2\vec{r}}{dt^2} + 2\frac{d\vec{r}}{dt} + 5\vec{r} = 0.$$

ID	CT1V-2
Source	CT1V Q2 [2+8 marks]

Answer:

14.5.0.1 1. Frenet-Serret Formulas

Let s be the arc length along a space curve. Let $\vec{T}$ be the unit tangent vector. Let $\vec{N}$ be the unit principal normal vector. Let $\vec{B}\_b$ be the unit binormal vector. Let κ be curvature. Let τ be torsion. The Frenet-Serret formulas are: 1.

$$\frac{d\vec{T}}{ds} = \kappa\vec{N}$$

2.

$$\frac{d\vec{N}}{ds} = -\kappa\vec{T} + \tau\vec{B}\_b$$

3.

$$\frac{d\vec{B}\_b}{ds} = -\tau\vec{N}$$

14.5.0.2 2. Verification of Differential Equation

We are given:

$$\vec{r} = e^{-t}(\vec{A}\cos 2t + \vec{B}\sin 2t)$$

We find the first derivative of $\vec{r}$ with respect to t:

$$\frac{d\vec{r}}{dt} = -e^{-t}(\vec{A}\cos 2t + \vec{B}\sin 2t) + e^{-t}(-2\vec{A}\sin 2t + 2\vec{B}\cos 2t)$$

$$\frac{d\vec{r}}{dt} = e^{-t}[(-\vec{A} + 2\vec{B})\cos 2t + (-2\vec{A} - \vec{B})\sin 2t]$$

Now we find the second derivative:

$$\frac{d^2\vec{r}}{dt^2} = -e^{-t}[(-\vec{A} + 2\vec{B})\cos 2t + (-2\vec{A} - \vec{B})\sin 2t] + e^{-t}[-2(-\vec{A} + 2\vec{B})\sin 2t + 2(-2\vec{A} - \vec{B})\cos 2t]$$

$$\frac{d^2\vec{r}}{dt^2} = e^{-t}[(\vec{A} - 2\vec{B} - 4\vec{A} - 2\vec{B})\cos 2t + (2\vec{A} + \vec{B} + 2\vec{A} - 4\vec{B})\sin 2t]$$

$$\frac{d^2\vec{r}}{dt^2} = e^{-t}[(-3\vec{A} - 4\vec{B})\cos 2t + (4\vec{A} - 3\vec{B})\sin 2t]$$

We substitute these derivatives into the left-hand side of the differential equation:

$$\text{LHS} = \frac{d^2\vec{r}}{dt^2} + 2\frac{d\vec{r}}{dt} + 5\vec{r}$$

$$\text{LHS} = e^{-t}[(-3\vec{A}-4\vec{B})\cos 2t+(4\vec{A}-3\vec{B})\sin 2t]+2e^{-t}[(-\vec{A}+2\vec{B})\cos 2t+(-2\vec{A}-\vec{B})\sin 2t]+5e^{-t}[\vec{A}\cos 2t+\vec{B}\sin 2t]$$

Combine the coefficient terms for $\cos 2t$:

$$(-3\vec{A} - 4\vec{B}) + 2(-\vec{A} + 2\vec{B}) + 5\vec{A} = -3\vec{A} - 4\vec{B} - 2\vec{A} + 4\vec{B} + 5\vec{A} = 0$$

Combine the coefficient terms for $\sin 2t$:

$$(4\vec{A} - 3\vec{B}) + 2(-2\vec{A} - \vec{B}) + 5\vec{B} = 4\vec{A} - 3\vec{B} - 4\vec{A} - 2\vec{B} + 5\vec{B} = 0$$

So, the equation becomes:

$$\text{LHS} = e^{-t}[0\cos 2t + 0\sin 2t] = 0 = \text{RHS}$$

This confirms that $\vec{r}$ is a solution.

15 Topic 19: Vector Identities

This file contains the organized questions and answers for **Vector Identities**, priority ranked as **Priority 19** based on frequency and exam weight.

15.1 Q1 (03)

If $\bar{A}$ is a constant vector, then prove that

$$\nabla(\bar{r} \cdot \bar{A}) = \bar{A}.$$

ID	PYQ-2018-1c
Source	2018 Q1(c) [03 marks]

Answer:
Let the constant vector be:

$$\bar{A} = A_1\hat{i} + A_2\hat{j} + A_3\hat{k}$$

Let the position vector be:

$$\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

Calculate the dot product:

$$\bar{r} \cdot \bar{A} = A_1x + A_2y + A_3z$$

Now calculate the gradient:

$$\nabla(\bar{r} \cdot \bar{A}) = \frac{\partial}{\partial x}(A_1x + A_2y + A_3z)\hat{i} + \frac{\partial}{\partial y}(A_1x + A_2y + A_3z)\hat{j} + \frac{\partial}{\partial z}(A_1x + A_2y + A_3z)\hat{k}$$

$$\nabla(\bar{r} \cdot \bar{A}) = A_1\hat{i} + A_2\hat{j} + A_3\hat{k} = \bar{A}$$

This completes the proof.

15.2 Q2 (05)

Show that

$$\nabla \times (\nabla \times \bar{A}) = -\nabla^2\bar{A} + \nabla(\nabla \cdot \bar{A}).$$

ID	PYQ-2018-4a
Appeared in	2018 Q4(a) [05 marks], 2021 Q2(c) [04 marks]
Frequency	⋆⋆ (2 times)

Answer:
Let the vector be

$$\bar{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$

We calculate the curl of $\bar{A}$:

$$\nabla \times \bar{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{i} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{j} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{k}$$

Let this vector be

$$\bar{V} = \nabla \times \bar{A}$$

. Now we find the curl of $\bar{V}$:

$$\nabla \times (\nabla \times \bar{A}) = \nabla \times \bar{V} = \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z}\right)\hat{i} + \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x}\right)\hat{j} + \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y}\right)\hat{k}$$

We evaluate the x-component:

$$(\nabla \times \bar{V})_x = \frac{\partial}{\partial y}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right) - \frac{\partial}{\partial z}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)$$

$$(\nabla \times \bar{V})_x = \frac{\partial^2 A_y}{\partial y \partial x} - \frac{\partial^2 A_x}{\partial y^2} - \frac{\partial^2 A_x}{\partial z^2} + \frac{\partial^2 A_z}{\partial z \partial x}$$

We add and subtract

$$\frac{\partial^2 A\_\_x}{\partial x^2}$$

to the expression:

$$(\nabla \times \bar{V})_x = \left(\frac{\partial^2 A_x}{\partial x^2} + \frac{\partial^2 A_y}{\partial y \partial x} + \frac{\partial^2 A_z}{\partial z \partial x}\right) - \left(\frac{\partial^2 A_x}{\partial x^2} + \frac{\partial^2 A_x}{\partial y^2} + \frac{\partial^2 A_x}{\partial z^2}\right)$$

$$(\nabla \times \bar{V})_x = \frac{\partial}{\partial x}\left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}\right) - \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) A_x$$

$$(\nabla \times \bar{V})_x = \frac{\partial}{\partial x}(\nabla \cdot \bar{A}) - \nabla^2 A_x$$

By using the same steps for the y and z-components, we get the vector identity:

$$\nabla \times (\nabla \times \bar{A}) = \nabla(\nabla \cdot \bar{A}) - \nabla^2 \bar{A}$$

15.3 Q3 (04)

Prove that

$$\nabla^2(1/r) = 0.$$

ID	PYQ-2019-3b
Source	2019 Q3(b) [04 marks]

Answer:

Let

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

Then

$$r = \sqrt{x^2 + y^2 + z^2}.$$

The partial derivatives of r with respect to coordinates are:

$$\frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}$$

We find the gradient of $\frac{1}{r}$:

$$\vec{\nabla}\left(\frac{1}{r}\right) = \hat{i}\frac{\partial}{\partial x}(r^{-1}) + \hat{j}\frac{\partial}{\partial y}(r^{-1}) + \hat{k}\frac{\partial}{\partial z}(r^{-1})$$

Using the chain rule:

$$\frac{\partial}{\partial x}(r^{-1}) = -r^{-2}\frac{\partial r}{\partial x} = -r^{-2}\left(\frac{x}{r}\right) = -\frac{x}{r^3}$$

So the gradient vector is:

$$\vec{\nabla}\left(\frac{1}{r}\right) = -\frac{x\hat{i} + y\hat{j} + z\hat{k}}{r^3} = -\frac{\vec{r}}{r^3}$$

The Laplacian

$$\nabla^2\left(\frac{1}{r}\right)$$

is the divergence of the gradient:

$$\nabla^2\left(\frac{1}{r}\right) = \vec{\nabla}\cdot\vec{\nabla}\left(\frac{1}{r}\right) = -\left[\frac{\partial}{\partial x}\left(\frac{x}{r^3}\right) + \frac{\partial}{\partial y}\left(\frac{y}{r^3}\right) + \frac{\partial}{\partial z}\left(\frac{z}{r^3}\right)\right]$$

Using the quotient rule:

$$\frac{\partial}{\partial x}\left(\frac{x}{r^3}\right) = \frac{r^3(1) - x(3r^2\frac{\partial r}{\partial x})}{r^6} = \frac{r^3 - 3x^2 r}{r^6} = \frac{1}{r^3} - \frac{3x^2}{r^5}$$

Similarly:

$$\frac{\partial}{\partial y}\left(\frac{y}{r^3}\right) = \frac{1}{r^3} - \frac{3y^2}{r^5}$$

$$\frac{\partial}{\partial z}\left(\frac{z}{r^3}\right) = \frac{1}{r^3} - \frac{3z^2}{r^5}$$

Summing these three partial derivatives:

$$\nabla^2\left(\frac{1}{r}\right) = -\left[\frac{3}{r^3} - \frac{3(x^2+y^2+z^2)}{r^5}\right] = -\left[\frac{3}{r^3} - \frac{3r^2}{r^5}\right] = -\left[\frac{3}{r^3} - \frac{3}{r^3}\right] = 0$$

This completes the proof.

15.4 Q4 (03)

Evaluate $\nabla(\vec{A}\times\vec{r})$ if

$$\nabla\times\vec{A} = 0.$$

ID	PYQ-2024-3a
Source	2024 Q3(a) [03 marks]

Answer:
We assume that the expression $\nabla(\vec{A}\times\vec{r})$ refers to the divergence:

$$\nabla\cdot(\vec{A}\times\vec{r})$$

Here,

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

is the position vector.
We use the vector identity for the divergence of a cross product:

$$\nabla\cdot(\vec{A}\times\vec{r}) = \vec{r}\cdot(\nabla\times\vec{A}) - \vec{A}\cdot(\nabla\times\vec{r})$$

We are given that

$$\nabla\times\vec{A} = 0$$

. So, the first term is zero.
Next, we calculate the curl of the position vector $\vec{r}$:

$$\nabla\times\vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = \hat{i}(0) - \hat{j}(0) + \hat{k}(0) = \vec{0}$$

So, the second term is also zero:

$$\vec{A} \cdot (\nabla \times \vec{r}) = \vec{A} \cdot \vec{0} = 0$$

Thus, we find:

$$\nabla \cdot (\vec{A} \times \vec{r}) = 0$$

If the question meant the curl

$$\nabla \times (\vec{A} \times \vec{r})$$

, we use another identity:

$$\nabla \times (\vec{A} \times \vec{r}) = (\vec{r} \cdot \nabla)\vec{A} - (\vec{A} \cdot \nabla)\vec{r} + \vec{A}(\nabla \cdot \vec{r}) - \vec{r}(\nabla \cdot \vec{A})$$

We calculate the terms: *

$$\nabla \cdot \vec{r} = 1 + 1 + 1 = 3$$

. So,

$$\vec{A}(\nabla \cdot \vec{r}) = 3\vec{A}.$$

•

$$(\vec{A} \cdot \nabla)\vec{r} = \left(A_1 \frac{\partial}{\partial x} + A_2 \frac{\partial}{\partial y} + A_3 \frac{\partial}{\partial z} \right) (x\hat{i} + y\hat{j} + z\hat{k}) = \vec{A}$$

This gives:

$$\nabla \times (\vec{A} \times \vec{r}) = 2\vec{A} + (\vec{r} \cdot \nabla)\vec{A} - \vec{r}(\nabla \cdot \vec{A})$$

If $\vec{A}$ is a constant vector field, then its derivatives are zero. In that case, the curl is:

$$\nabla \times (\vec{A} \times \vec{r}) = 2\vec{A}$$

16 Topic 20: Divergence Theorem

This file contains the organized questions and answers for **Divergence Theorem**, priority ranked as **Priority 20** based on frequency and exam weight.

16.1 Q1 (06)

Verify the divergence theorem for

$$\bar{A} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$$

taken over the region bounded by

$$x^2 + y^2 = 4, \quad z = 0, \quad z = 3$$

ID	PYQ-2018-3b
Appeared in	2018 Q3(b) [06 marks], 2021 Q4(b) [06 marks]
Frequency	⋆⋆ (2 times)

Answer:

The divergence theorem states:

$$\iiint_V \bar{\nabla} \cdot \bar{A} dV = \iint_S \bar{A} \cdot \hat{n} dS$$

16.1.0.1 1. Evaluate Volume Integral Calculate the divergence of $\bar{A}$:

$$\bar{\nabla} \cdot \bar{A} = \frac{\partial}{\partial x}(4x) + \frac{\partial}{\partial y}(-2y^2) + \frac{\partial}{\partial z}(z^2) = 4 - 4y + 2z$$

We use cylindrical coordinates:

$$x = r\cos\phi, \quad y = r\sin\phi, \quad z = z$$

And $dV = rdrd\phi dz$. The limits are:

$$r \in [0, 2], \quad \phi \in [0, 2\pi], \quad z \in [0, 3]$$

Set up the integral:

$$\iiint_V \bar{\nabla} \cdot \bar{A} dV = \int_{z=0}^{3} \int_{\phi=0}^{2\pi} \int_{r=0}^{2} (4 - 4r\sin\phi + 2z) rdrd\phi dz$$

Integrate with respect to ϕ first:

$$\int_0^{2\pi} (4r - 4r^2\sin\phi + 2rz)d\phi = \left[(4r + 2rz)\phi + 4r^2\cos\phi\right]_0^{2\pi} = 2\pi(4r + 2rz)$$

Now integrate with respect to r:

$$2\pi \int_0^2 (4r + 2rz)dr = 2\pi \left[2r^2 + r^2 z\right]_0^2 = 2\pi(8 + 4z)$$

Now integrate with respect to z:

$$2\pi \int_0^3 (8 + 4z)dz = 2\pi \left[8z + 2z^2\right]_0^3 = 2\pi(24 + 18) = 84\pi$$

16.1.0.2 2. Evaluate Surface Integral The cylinder surface S has three parts: 1. Top surface S_1 $(z=3)$: The normal vector is

$$\hat{n}=\hat{k}$$

2. Bottom surface S_2 $(z=0)$: The normal vector is

$$\hat{n}=-\hat{k}$$

3. Curved wall S_3 (

$$x^2+y^2=4$$

): The normal vector is

$$\hat{n}=\frac{x\hat{i}+y\hat{j}}{2}$$

Evaluate the top surface integral:

$$\bar{A}\cdot\hat{n}=\bar{A}\cdot\hat{k}=z^2=9$$

$$\iint_{S_1}\bar{A}\cdot\hat{n}dS=9\iint_{S_1}dS=9\times\pi(2)^2=36\pi$$

Evaluate the bottom surface integral:

$$\bar{A}\cdot\hat{n}=\bar{A}\cdot(-\hat{k})=-z^2=0$$

$$\iint_{S_2}\bar{A}\cdot\hat{n}dS=0$$

Evaluate the curved wall integral:

$$\bar{A}\cdot\hat{n}=(4x\hat{i}-2y^2\hat{j}+z^2\hat{k})\cdot\left(\frac{x\hat{i}+y\hat{j}}{2}\right)=2x^2-y^3$$

We use cylindrical coordinates:

$$x=2\cos\phi,\quad y=2\sin\phi,\quad dS=2d\phi dz$$

$$\iint_{S_3}\bar{A}\cdot\hat{n}dS=\int_0^3\int_0^{2\pi}\left[2(4\cos^2\phi)-8\sin^3\phi\right]2d\phi dz$$

We use the standard integral properties:

$$\int_0^{2\pi}\cos^2\phi d\phi=\pi$$

and

$$\int_0^{2\pi}\sin^3\phi d\phi=0$$

$$\iint_{S_3}\bar{A}\cdot\hat{n}dS=\int_0^3 16\pi dz=48\pi$$

Add the three surface integrals:

$$\iint_S\bar{A}\cdot\hat{n}dS=36\pi+0+48\pi=84\pi$$

Both the volume and surface integrals equal 84π. The divergence theorem is verified.

17 Topic 21: Stokes' Theorem

This file contains the organized questions and answers for **Stokes' Theorem**, priority ranked as **Priority 21** based on frequency and exam weight.

17.1 Q1. State and Prove Stokes' theorem. (05)

ID	PYQ-2023-3a
Source	2023 Q3(a) [05 marks]

Answer:

17.1.0.1 Statement Let S be an open, two-sided surface bounded by a closed, non-self-intersecting curve C. If $\vec{A}$ has continuous first-order partial derivatives, then:

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\vec{\nabla} \times \vec{A}) \cdot \hat{n} dS$$

Here the boundary curve C is traversed in the positive (counterclockwise) direction.

17.1.0.2 Proof Let us prove this for a surface S which can be projected uniquely onto coordinate planes. Let the surface equation be $z = f(x, y)$ over a region R in the xy-plane.

Let the vector field be:

$$\vec{A} = A_1\hat{i} + A_2\hat{j} + A_3\hat{k}$$

We prove the identity for the component A_1:

$$\oint_C A_1 dx = \iint_S \left[\vec{\nabla} \times (A_1\hat{i})\right] \cdot \hat{n} dS \quad \ldots (1)$$

Since the curve C is the boundary of the surface $z = f(x, y)$, we project the line integral onto the xy-plane:

$$\oint_C A_1(x, y, z)dx = \oint_{C'} A_1(x, y, f(x, y))dx$$

Apply Green's theorem in the plane to the projected curve C':

$$\oint_{C'} A_1(x, y, f(x, y))dx = \iint_R -\frac{\partial}{\partial y}\left[A_1(x, y, f(x, y))\right] dA$$

Using the chain rule, we compute:

$$\frac{\partial}{\partial y}\left[A_1(x, y, f(x, y))\right] = \frac{\partial A_1}{\partial y} + \frac{\partial A_1}{\partial z}\frac{\partial z}{\partial y}$$

So the integral becomes:

$$\oint_C A_1 dx = \iint_R \left(-\frac{\partial A_1}{\partial y} - \frac{\partial A_1}{\partial z}\frac{\partial z}{\partial y}\right) dxdy \quad \ldots (2)$$

Now calculate the term

$$\vec{\nabla} \times (A\_1\hat{i}) :$$

$$\vec{\nabla} \times (A_1\hat{i}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & 0 & 0 \end{vmatrix} = \frac{\partial A_1}{\partial z}\hat{j} - \frac{\partial A_1}{\partial y}\hat{k}$$

For the surface $z - f(x, y) = 0$, the normal vector gives the relation:

$$\hat{n}dS = \left(-\frac{\partial z}{\partial x}\hat{\imath} - \frac{\partial z}{\partial y}\hat{\jmath} + \hat{k}\right) dxdy$$

Calculate the dot product:

$$\left[\vec{\nabla} \times (A_1\hat{\imath})\right] \cdot \hat{n}dS = \left(\frac{\partial A_1}{\partial z}\hat{\jmath} - \frac{\partial A_1}{\partial y}\hat{k}\right) \cdot \left(-\frac{\partial z}{\partial x}\hat{\imath} - \frac{\partial z}{\partial y}\hat{\jmath} + \hat{k}\right) dxdy$$

$$\left[\vec{\nabla} \times (A_1\hat{\imath})\right] \cdot \hat{n}dS = \left(-\frac{\partial A_1}{\partial z}\frac{\partial z}{\partial y} - \frac{\partial A_1}{\partial y}\right) dxdy \quad \ldots (3)$$

Comparing equations (2) and (3) shows that they are equal.
By applying the same projection method for the components A_2 and A_3, we get:

$$\oint_C A_2 dy = \iint_S \left[\vec{\nabla} \times (A_2\hat{\jmath})\right] \cdot \hat{n}dS$$

$$\oint_C A_3 dz = \iint_S \left[\vec{\nabla} \times (A_3\hat{k})\right] \cdot \hat{n}dS$$

Adding these three relations gives:

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\vec{\nabla} \times \vec{A}) \cdot \hat{n}dS$$

The proof is complete.

17.2 Q2 (05)

Verify Stoke's theorem for

$$\vec{A} = (y - z + 2)\hat{\imath} + (yz + 4)\hat{\jmath} - xz\hat{k}$$

where S is the surface of the cube $x = y = z = 0$; $x = y = z = 2$ above xy plane.

ID	PYQ-2024-4b
Source	2024 Q4(b) [05 marks]

Answer:
Stokes' theorem states:

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\nabla \times \vec{A}) \cdot \hat{n}dS$$

The surface S consists of the five faces of the cube above the xy-plane. The bottom face $z = 0$ is open. The boundary curve C is the square in the xy-plane ($z = 0$) bounded by

$$x = 0, \quad x = 2, \quad y = 0$$

And $y = 2$.

17.2.0.1 1. Evaluate the Line Integral Along the boundary curve C in the plane $z = 0$, we have $z = 0$ and $dz = 0$. The vector field simplifies to:

$$\vec{A} = (y + 2)\hat{\imath} + 4\hat{\jmath}$$

The line integral is:

$$\oint_C \vec{A} \cdot d\vec{r} = \oint_C (y + 2)dx + 4dy$$

We evaluate the integral along the four segments of the square C: * **Path 1 (C_1):** From $(0, 0, 0)$ to $(2, 0, 0)$. Here, $y = 0$ and $dy = 0$.

$$\int_{C_1} = \int_0^2 2dx = 4$$

- **Path 2 (C_2):** From $(2,0,0)$ to $(2,2,0)$. Here, $x=2$ and $dx=0$.
$$\int_{C_2} = \int_0^2 4dy = 8$$
- **Path 3 (C_3):** From $(2,2,0)$ to $(0,2,0)$. Here, $y=2$ and $dy=0$.
$$\int_{C_3} = \int_2^0 4dx = -8$$
- **Path 4 (C_4):** From $(0,2,0)$ to $(0,0,0)$. Here, $x=0$ and $dx=0$.
$$\int_{C_4} = \int_2^0 4dy = -8$$

We sum these line integrals:
$$\oint_C \vec{A}\cdot d\vec{r} = 4+8-8-8 = -4$$

17.2.0.2 2. Evaluate the Surface Integral

First, we find the curl of $\vec{A}$:
$$\nabla\times\vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y-z+2 & yz+4 & -xz \end{vmatrix}$$
$$= \hat{i}\,[0-y] - \hat{j}\,[-z-(-1)] + \hat{k}\,[0-1] = -y\hat{i} + (z-1)\hat{j} - \hat{k}$$
The surface S has five faces: * **Face 1 (Top, $z=2$):**
$$\hat{n} = \hat{k}$$
$dS = dxdy$.
$$(\nabla\times\vec{A})\cdot\hat{n} = -1 \implies \iint_{\text{Top}} -1dxdy = -1\times\text{Area} = -4$$
- **Face 2 (Front, $x=2$):**
$$\hat{n} = \hat{i}$$
$dS = dydz$.
$$(\nabla\times\vec{A})\cdot\hat{n} = -y \implies \iint_{\text{Front}} -ydydz = \int_0^2 dz\int_0^2 -ydy = 2\left[-\frac{y^2}{2}\right]_0^2 = -4$$
- **Face 3 (Back, $x=0$):**
$$\hat{n} = -\hat{i}$$
$dS = dydz$.
$$(\nabla\times\vec{A})\cdot\hat{n} = y \implies \iint_{\text{Back}} ydydz = \int_0^2 dz\int_0^2 ydy = 2\left[\frac{y^2}{2}\right]_0^2 = 4$$
- **Face 4 (Right, $y=2$):**
$$\hat{n} = \hat{j}$$
$dS = dxdz$.
$$(\nabla\times\vec{A})\cdot\hat{n} = z-1 \implies \iint_{\text{Right}} (z-1)dxdz = \int_0^2 dx\int_0^2 (z-1)dz = 2\left[\frac{z^2}{2}-z\right]_0^2 = 2(2-2) = 0$$
- **Face 5 (Left, $y=0$):**

$$\hat{n} = -\hat{j}$$

$dS = dxdz.$

$$(\nabla \times \vec{A}) \cdot \hat{n} = -(z-1) \implies \iint_{\text{Left}} (1-z)dxdz = \int_0^2 dx \int_0^2 (1-z)dz = 2\left[z - \frac{z^2}{2}\right]_0^2 = 2(2-2) = 0$$

We sum the integrals over all five faces:

$$\iint_S (\nabla \times \vec{A}) \cdot \hat{n} dS = -4 - 4 + 4 + 0 + 0 = -4$$

Both integrals yield -4. Stokes' theorem is verified.

17.3 SECTION - B

18 Topic 22: Surface Integrals

This file contains the organized questions and answers for **Surface Integrals**, priority ranked as **Priority 22** based on frequency and exam weight.

18.1 Q1 (06)

Evaluate

$$\iint_S \vec{A} \cdot \hat{n} dS$$

Where

$$\vec{A} = z\hat{i} + x\hat{j} - 3y^2 z\hat{k}$$

and S is the surface of the cylinder

$$x^2 + y^2 = 16$$

included in the first octant between $z = 0$ and $z = 5$.

ID	PYQ-2017-2b
Appeared in	2017 Q2(b) [06 marks], 2020 Q4(a) [06 marks]
Frequency	⋆⋆ (2 times)

Answer:
Let the cylinder surface be defined by the function:

$$g(x, y, z) = x^2 + y^2 - 16 = 0$$

The gradient of g points normal to this surface:

$$\vec{\nabla} g = 2x\hat{i} + 2y\hat{j}$$

We find the unit outward normal vector $\hat{n}$:

$$\hat{n} = \frac{\vec{\nabla} g}{|\vec{\nabla} g|} = \frac{2x\hat{i} + 2y\hat{j}}{\sqrt{4x^2 + 4y^2}} = \frac{2x\hat{i} + 2y\hat{j}}{2\sqrt{16}} = \frac{x\hat{i} + y\hat{j}}{4}$$

Now calculate the dot product $\vec{A} \cdot \hat{n}$:

$$\vec{A} \cdot \hat{n} = (z\hat{i} + x\hat{j} - 3y^2 z\hat{k}) \cdot \left(\frac{x\hat{i} + y\hat{j}}{4}\right) = \frac{xz + xy}{4}$$

Next we project the cylinder surface S onto the yz-plane. The projection region R is bounded by:

$$y \in [0, 4], \quad z \in [0, 5]$$

For this projection, the surface area element dS is related to $dydz$ by:

$$dS = \frac{dydz}{|\hat{n} \cdot \hat{i}|} = \frac{dydz}{|x/4|} = \frac{4}{x} dydz$$

Now we set up the surface integral over the region R:

$$\iint_S \vec{A} \cdot \hat{n} dS = \iint_R \left(\frac{xz + xy}{4}\right) \left(\frac{4}{x} dydz\right)$$

Simplify the integrand:

$$\iint_S \vec{A} \cdot \hat{n} dS = \iint_R (z + y) dydz$$

Set up the limits of integration:

$$\iint_S \vec{A} \cdot \hat{n} dS = \int_{y=0}^{4} \int_{z=0}^{5} (y+z) dz dy$$

Integrate with respect to z:

$$\iint_S \vec{A} \cdot \hat{n} dS = \int_0^4 \left[yz + \frac{z^2}{2} \right]_0^5 dy = \int_0^4 \left(5y + \frac{25}{2} \right) dy$$

Now integrate with respect to y:

$$\iint_S \vec{A} \cdot \hat{n} dS = \left[\frac{5y^2}{2} + \frac{25}{2} y \right]_0^4 = \frac{5(16)}{2} + \frac{25(4)}{2} = 40 + 50 = 90$$

So the value of the surface integral is 90.

19 Topic 23: Volume Integrals

This file contains the organized questions and answers for **Volume Integrals**, priority ranked as **Priority 23** based on frequency and exam weight.

19.1 Q1 (06)

Let

$$\Phi = 45x^2y$$

and let V denote the closed region bounded by the planes $4x + 2y + z = 8$, $x = 0, y = 0, z = 0$. Evaluate

$$\iiint_V \Phi dV$$

ID	PYQ-2018-3a
Source	2018 Q3(a) [06 marks]

Answer:
We set up the limits of integration for the region V: * The variable z goes from the plane $z = 0$ to the plane $z = 8 - 4x - 2y$. * For $z = 0$, the boundary is $4x + 2y = 8 \implies y = 4 - 2x$. So y goes from 0 to $4 - 2x$. * For $y = z = 0$, the boundary is $4x = 8 \implies x = 2$. So x goes from 0 to 2.

We write the triple integral:

$$\iiint_V \Phi dV = \int_{x=0}^{2} \int_{y=0}^{4-2x} \int_{z=0}^{8-4x-2y} 45x^2 ydzdydx$$

Integrate with respect to z:

$$\iiint_V \Phi dV = 45 \int_0^2 x^2 \int_0^{4-2x} y[z]_0^{8-4x-2y} dydx = 45 \int_0^2 x^2 \int_0^{4-2x} y(8 - 4x - 2y)dydx$$

$$\iiint_V \Phi dV = 45 \int_0^2 x^2 \int_0^{4-2x} \left[(8 - 4x)y - 2y^2\right] dydx$$

Integrate with respect to y:

$$\iiint_V \Phi dV = 45 \int_0^2 x^2 \left[(8 - 4x)\frac{y^2}{2} - \frac{2y^3}{3}\right]_0^{4-2x} dx$$

$$\iiint_V \Phi dV = 45 \int_0^2 x^2 \left[(4 - 2x)(4 - 2x)^2 - \frac{2}{3}(4 - 2x)^3\right] dx$$

$$\iiint_V \Phi dV = 45 \int_0^2 x^2 \left[(4 - 2x)^3 - \frac{2}{3}(4 - 2x)^3\right] dx = 45 \int_0^2 x^2 \left[\frac{1}{3}(4 - 2x)^3\right] dx$$

Factor out

$$2^3 = 8$$

from $(4 - 2x)^3$:

$$\iiint_V \Phi dV = 15 \int_0^2 x^2 \left[8(2 - x)^3\right] dx = 120 \int_0^2 x^2(2 - x)^3 dx$$

Expand

$$(2 - x)^3 = 8 - 12x + 6x^2 - x^3 :$$

$$\iiint_V \Phi dV = 120 \int_0^2 (8x^2 - 12x^3 + 6x^4 - x^5) dx$$

Now integrate with respect to x:

$$\iiint_V \Phi dV = 120 \left[\frac{8}{3}x^3 - 3x^4 + \frac{6}{5}x^5 - \frac{x^6}{6} \right]_0^2$$

$$\iiint_V \Phi dV = 120 \left(\frac{64}{3} - 48 + \frac{192}{5} - \frac{64}{6} \right) = 120 \left(\frac{32}{3} - 48 + \frac{192}{5} \right)$$

$$\iiint_V \Phi dV = 120 \left(\frac{160 - 720 + 576}{15} \right) = 120 \left(\frac{16}{15} \right) = 8 \times 16 = 128$$

So the value of the triple integral is 128.

19.2 Q2 (06)

Evaluate

$$\iiint_V \nabla \cdot \bar{F} dV$$

where

$$\bar{F} = (2x^2 - 3z)\hat{\imath} - 2xy\hat{\jmath} - 4x\hat{k}$$

and V is the closed region bounded by the planes $x = 0, y = 0, z = 0$ and $2x + 2y + z = 4$.

ID	PYQ-2019-4a
Source	2019 Q4(a) [06 marks]

Answer:

First, we calculate the divergence of the vector field $\vec{F}$:

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial}{\partial x}(2x^2 - 3z) + \frac{\partial}{\partial y}(-2xy) + \frac{\partial}{\partial z}(-4x)$$

$$= 4x - 2x + 0 = 2x$$

The region V is a tetrahedron bounded by the coordinate planes and the plane $2x + 2y + z = 4$. We find the integration limits:

- The variable z goes from 0 to $4 - 2x - 2y$.
- For $z = 0$, the boundary is $2x + 2y = 4 \implies y = 2 - x$. So y goes from 0 to $2 - x$.
- For $y = z = 0$, the boundary is $2x = 4 \implies x = 2$. So x goes from 0 to 2.

We set up the triple integral:

$$I = \iiint_V 2x \, dV = \int_{x=0}^{2} \int_{y=0}^{2-x} \int_{z=0}^{4-2x-2y} 2x \, dz \, dy \, dx$$

Integrate with respect to z:

$$I = \int_{x=0}^{2} \int_{y=0}^{2-x} 2x[z]_0^{4-2x-2y} \, dy \, dx = \int_{x=0}^{2} \int_{y=0}^{2-x} 2x(4 - 2x - 2y) \, dy \, dx$$

$$= \int_0^2 2x \left[(4 - 2x)y - y^2 \right]_0^{2-x} \, dx$$

Since $4 - 2x = 2(2 - x)$, we substitute:

$$I = \int_0^2 2x \left[2(2 - x)(2 - x) - (2 - x)^2 \right] \, dx = \int_0^2 2x(2 - x)^2 \, dx$$

$$= \int_0^2 2x(4-4x+x^2)\,dx = \int_0^2 (8x-8x^2+2x^3)\,dx$$

Integrate with respect to x:

$$I = \left[4x^2 - \frac{8}{3}x^3 + \frac{1}{2}x^4\right]_0^2 = 4(4) - \frac{8}{3}(8) + \frac{1}{2}(16) = 16 - \frac{64}{3} + 8 = 24 - \frac{64}{3} = \frac{8}{3}$$

So the value of the triple integral is $\frac{8}{3}$.

19.3 Q3 (04)

Let

$$\vec{F} = 2xz\hat{\imath} - x\hat{\jmath} + y^2\hat{k}$$

Evaluate

$$\iiint_V \vec{F}dV$$

Where V is the region bounded by the surfaces

$$x = 0, y = 0, y = 6, z = x^2, z = 4.$$

ID	PYQ-2021-3c
Source	2021 Q3(c) [04 marks]

Answer:

The boundaries of the region V are x from 0 to 2, y from 0 to 6, and z from x^2 to 4.

We set up the triple integral:

$$\iiint_V \vec{F}dV = \int_{y=0}^{6}\int_{x=0}^{2}\int_{z=x^2}^{4}(2xz\hat{\imath} - x\hat{\jmath} + y^2\hat{k})dzdxdy$$

We integrate each component separately:

19.3.0.1 1. $\hat{\imath}$-component

$$\int_0^6\int_0^2\int_{x^2}^4 2xzdzdxdy = \left(\int_0^6 dy\right)\int_0^2 \left[xz^2\right]_{x^2}^4 dx = 6\int_0^2(16x - x^5)dx$$

$$= 6\left[8x^2 - \frac{x^6}{6}\right]_0^2 = 6\left(32 - \frac{64}{6}\right) = 192 - 64 = 128$$

19.3.0.2 2. $\hat{\jmath}$-component

$$\int_0^6\int_0^2\int_{x^2}^4 -xdzdxdy = 6\int_0^2 -x(4-x^2)dx = 6\int_0^2(x^3 - 4x)dx$$

$$= 6\left[\frac{x^4}{4} - 2x^2\right]_0^2 = 6(4-8) = -24$$

19.3.0.3 3. $\hat{k}$-component

$$\int_0^6\int_0^2\int_{x^2}^4 y^2dzdxdy = \left(\int_0^6 y^2dy\right)\left(\int_0^2(4-x^2)dx\right)$$

$$= \left[\frac{y^3}{3}\right]_0^6 \times \left[4x - \frac{x^3}{3}\right]_0^2 = 72 \times \left(8 - \frac{8}{3}\right) = 72 \times \frac{16}{3} = 384$$

19.3.0.4 Combine Components

$$\iiint_V \vec{F} dV = 128\hat{\imath} - 24\hat{\jmath} + 384\hat{k}$$

20 Topic 24: Curvature and Torsion

This file contains the organized questions and answers for **Curvature and Torsion**, priority ranked as **Priority 24** based on frequency and exam weight.

20.1 Q1 (05)

For the space curve

$$x = t, y = t^2, z = \frac{2}{3}t^3$$

, find the curvature K and the torsion T.

ID	PYQ-2020-2b
Appeared in	2020 Q2(b) [05 marks], 2023 Q1(c) [04 marks]
Frequency	⋆⋆ (2 times)

Answer:

We write the curve as a vector function:

$$\vec{r}(t) = t\hat{i} + t^2\hat{j} + \frac{2}{3}t^3\hat{k}$$

Calculate the first three derivatives of $\vec{r}$:

$$\vec{r}'(t) = \hat{i} + 2t\hat{j} + 2t^2\hat{k}$$

$$\vec{r}''(t) = 2\hat{j} + 4t\hat{k}$$

$$\vec{r}'''(t) = 4\hat{k}$$

Calculate the cross product of the first two derivatives:

$$\vec{r}' \times \vec{r}'' = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2t & 2t^2 \\ 0 & 2 & 4t \end{vmatrix} = \hat{i}(8t^2 - 4t^2) - \hat{j}(4t - 0) + \hat{k}(2 - 0) = 4t^2\hat{i} - 4t\hat{j} + 2\hat{k}$$

Find the magnitudes:

$$|\vec{r}'| = \sqrt{1 + 4t^2 + 4t^4} = \sqrt{(2t^2+1)^2} = 2t^2 + 1$$

$$|\vec{r}' \times \vec{r}''| = \sqrt{16t^4 + 16t^2 + 4} = 2\sqrt{4t^4 + 4t^2 + 1} = 2(2t^2 + 1)$$

20.1.0.1 1. Curvature K

$$K = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|^3} = \frac{2(2t^2+1)}{(2t^2+1)^3} = \frac{2}{(2t^2+1)^2}$$

20.1.0.2 2. Torsion T

Calculate the scalar triple product:

$$(\vec{r}' \times \vec{r}'') \cdot \vec{r}''' = (4t^2\hat{i} - 4t\hat{j} + 2\hat{k}) \cdot 4\hat{k} = 8$$

$$T = \frac{(\vec{r}' \times \vec{r}'') \cdot \vec{r}'''}{|\vec{r}' \times \vec{r}''|^2} = \frac{8}{4(2t^2+1)^2} = \frac{2}{(2t^2+1)^2}$$

So the curvature and the torsion are equal:

$$K = T = \frac{2}{(2t^2+1)^2}$$

21 Topic 11: Vector Space, Subspace, and Direct Sum

This file contains the organized questions and answers for **Vector Space, Subspace, and Direct Sum**, priority ranked as **Priority 11** based on frequency and exam weight.

21.1 Q1. Define vector space and subspace. (04)

ID	PYQ-2017-4a
Source	2017 Q4(a) [04 marks]

Answer:

21.1.0.1 Vector Space Let V be a non-empty set of objects called vectors. Let K be a field of scalars. We define two operations on this set: vector addition $(u + v)$ and scalar multiplication (ku). The set V is called a vector space over the field K if it satisfies these ten rules:

1. **Closure under Addition:** If $u, v \in V$, then $u + v \in V$.
2. **Commutative Law:** $u + v = v + u$ for all $u, v \in V$.
3. **Associative Law:** $(u + v) + w = u + (v + w)$ for all $u, v, w \in V$.
4. **Additive Identity:** There is a zero vector $0 \in V$ such that $u + 0 = u$ for all $u \in V$.
5. **Additive Inverse:** For each $u \in V$, there is a vector $-u \in V$ such that $u + (-u) = 0$.
6. **Closure under Scalar Multiplication:** If $u \in V$ and $k \in K$, then $ku \in V$.
7. **Distributive Law 1:** $k(u + v) = ku + kv$ for all $u, v \in V$ and $k \in K$.
8. **Distributive Law 2:** $(a + b)u = au + bu$ for all $u \in V$ and $a, b \in K$.
9. **Associativity of Multiplication:** $a(bu) = (ab)u$ for all $u \in V$ and $a, b \in K$.
10. **Unitary Identity:** $1u = u$ for all $u \in V$, where 1 is the multiplicative identity in K.

21.1.0.2 Subspace Let W be a non-empty subset of a vector space V over a field K. We call W a subspace of V if W is itself a vector space under the same operations defined on V.

In practice, W is a subspace if and only if it satisfies these two closure properties: 1. **Closed under Addition:** If $u, v \in W$, then $u + v \in W$. 2. **Closed under Scalar Multiplication:** If $u \in W$ and $k \in K$, then $ku \in W$.

21.2 Q2 (08)

Let V be a vector space over a field k, prove that:

ID	PYQ-2017-4b
Source	2017 Q4(b) [08 marks]

Answer:

- **i)** $k0 = 0$
- **ii)** $0u = 0$
- **iii)** If $ku = 0$, then $k = 0$ or $u = 0$
- **iv)** $(-k)u = k(-u) = -ku$

Answer:

21.2.0.1 Proof of i): $k0 = 0$ Using properties of the zero vector, we know:

$$0 + 0 = 0$$

Multiply both sides by the scalar k:

$$k(0 + 0) = k0$$

Use the distributive property of vector spaces:

$$k0 + k0 = k0$$

Now add the additive inverse vector $-k0$ to both sides:

$$(k0 + k0) + (-k0) = k0 + (-k0)$$

Apply the associative law of addition:

$$k0 + (k0 + (-k0)) = 0$$

$$k0 + 0 = 0 \implies k0 = 0$$

This completes the proof.

21.2.0.2 Proof of ii): $0u = 0$

Using properties of scalar addition in the field k, we know:

$$0 + 0 = 0$$

Multiply both sides by the vector u:

$$(0 + 0)u = 0u$$

Use the distributive property:

$$0u + 0u = 0u$$

Now add the additive inverse vector $-0u$ to both sides:

$$(0u + 0u) + (-0u) = 0u + (-0u)$$

Apply the associative law of addition:

$$0u + (0u + (-0u)) = 0$$

$$0u + 0 = 0 \implies 0u = 0$$

This completes the proof.

21.2.0.3 Proof of iii): If $ku = 0$, then $k = 0$ or $u = 0$

Assume $ku = 0$. If the scalar $k = 0$, then the statement is already true.
But if $k \neq 0$, then its multiplicative inverse k^{-1} exists in the field k. We multiply both sides of the equation by k^{-1}:

$$k^{-1}(ku) = k^{-1}0$$

Use the associative property of scalar multiplication and the fact that

$$k^{-1}0 = 0$$

(from property i):

$$(k^{-1}k)u = 0$$

$$1u = 0$$

Since $1u = u$ (unitary property):

$$u = 0$$

So either $k = 0$ or $u = 0$. This completes the proof.

21.2.0.4 Proof of iv): $(-k)u = k(-u) = -ku$

The vector $-ku$ is the unique additive inverse of ku, which means $ku + (-ku) = 0$.

To show $(-k)u = -ku$, we add them together and show the result is the zero vector:

$$ku + (-k)u = (k + (-k))u = 0u = 0 \quad \text{(using property ii)}$$

Since the sum is zero, $(-k)u$ is indeed the additive inverse of ku:

$$(-k)u = -ku$$

To show $k(-u) = -ku$, we do the same:

$$ku + k(-u) = k(u + (-u)) = k0 = 0 \quad \text{(using property i)}$$

So $k(-u)$ is also the additive inverse of ku:

$$k(-u) = -ku$$

So we have shown that $(-k)u = k(-u) = -ku$.

21.3 SECTION - B

21.4 Q3. Define vector space and subspace with examples. (03)

ID	PYQ-2018-8a
Source	2018 Q8(a) [03 marks]

Answer:

21.4.0.1 Vector Space

Let V be a non-empty set of vectors and K be a field of scalars. V is a vector space if it satisfies closure, commutative, associative, identity, inverse, and distributive properties under vector addition and scalar multiplication.

Example: The space $\mathbb{R}^3$ of all real triples under standard addition and multiplication.

21.4.0.2 Subspace

A non-empty subset W of a vector space V is a subspace if W is closed under addition ($u, v \in W \implies u + v \in W$) and closed under scalar multiplication ($u \in W, k \in K \implies ku \in W$).

Example: The plane $z = 0$ in $\mathbb{R}^3$, defined by $W = \{(x, y, 0) \mid x, y \in \mathbb{R}\}$.

21.5 Q4 (06)

Prove that the vector space V is the direct sum of its subspaces U and W if and only if (i) $V = U + W$, (ii) $U \cap W = \{0\}$.

ID	PYQ-2019-8a
Source	2019 Q8(a) [06 marks]

Answer:

21.5.0.1 1. Forward Direction ($\implies$)

Assume V is the direct sum of U and W ($V = U \oplus W$). By definition of direct sum, every vector $v \in V$ can be uniquely written as:

$$v = u + w \quad \text{where } u \in U \text{ and } w \in W$$

- Since every $v \in V$ can be written as $u + w$, we have:
$$V = U + W$$
- Now let us prove $U \cap W = \{0\}$. Let $v \in U \cap W$.
  - Since $v \in U$, we can write $v = v + 0$ (where $v \in U$ and $0 \in W$).
  - Since $v \in W$, we can write $v = 0 + v$ (where $0 \in U$ and $v \in W$).
  - But by uniqueness of the decomposition of v, these two expressions must be identical:
$$v = 0 \quad \text{and} \quad 0 = v \implies v = 0$$

Thus, $U \cap W = \{0\}$.

21.5.0.2 2. Reverse Direction ($\impliedby$)

Assume (i) $V = U + W$ and (ii) $U \cap W = \{0\}$. Since $V = U + W$, every vector $v \in V$ can be written as:

$$v = u + w \quad \text{where } u \in U \text{ and } w \in W$$

We must show that this representation is unique.
Suppose v can be represented in two ways:

$$v = u_1 + w_1 \quad \text{and} \quad v = u_2 + w_2$$

where $u\_1, u\_2 \in U$ and $w\_1, w\_2 \in W$.
Equating the two expressions:

$$u_1 + w_1 = u_2 + w_2 \implies u_1 - u_2 = w_2 - w_1$$

- Since U is a subspace, $u\_1 - u\_2 \in U$.
- Since W is a subspace, $w\_2 - w\_1 \in W$.

Since
$$u\_1 - u\_2 = w\_2 - w\_1$$
, this vector belongs to both U and W:

$$u_1 - u_2 \in U \cap W \quad \text{and} \quad w_2 - w_1 \in U \cap W$$

But we are given $U \cap W = \{0\}$. Therefore:

$$u_1 - u_2 = 0 \implies u_1 = u_2$$

$$w_2 - w_1 = 0 \implies w_1 = w_2$$

Since
$$u\_1 = u\_2$$
and
$$w\_1 = w\_2$$
, the representation is unique. Thus, V is the direct sum of U and W ($V = U \oplus W$).

21.6 Q5. Define sum and direct sum with examples. (02)

ID	PYQ-2023-8a
Source	2023 Q8(a) [02 marks]

Answer:

- **Sum of Subspaces:** Let U and W be subspaces of a vector space V. The sum $U + W$ is the set of all vectors $u + w$ where $u \in U$ and $w \in W$. *Example:* In $\mathbb{R}^2$, let U be the x-axis and W be the y-axis. Then
$$U + W = \mathbb{R}^2.$$
- **Direct Sum of Subspaces:** Let U and W be subspaces of V. The sum $U + W$ is a direct sum (written $U \oplus W$) if every element in the sum can be written uniquely as $u + w$. This occurs if and only if $U \cap W = \{0\}$. *Example:* In $\mathbb{R}^2$, let U be the x-axis and W be the y-axis. Their intersection is only the origin $\{(0, 0)\}$. So
$$U \oplus W = \mathbb{R}^2.$$

21.7 Q6. Define direct sum. Give an example of direct sum. (02)

ID	PYQ-2024-7b
Source	2024 Q7(b) [02 marks]

Answer:

21.7.0.1 1. Definition Let V be a vector space. Let U and W be subspaces of V. We say V is the direct sum of U and W, written as $V = U \oplus W$, if: * $V = U + W$. This means every vector $v \in V$ can be written as $v = u + w$ for some $u \in U$ and $w \in W$. * $U \cap W = \{0\}$. This means the intersection of U and W contains only the zero vector.

Equivalently, every vector $v \in V$ can be written uniquely as $v = u + w$, where $u \in U$ and $w \in W$.

21.7.0.2 2. Example Let
$$V = \mathbb{R}^2$$
be the two-dimensional real plane. Let U be the x-axis, which is the subspace of all vectors of the form $(x, 0)$:
$$U = \{(x, 0) \mid x \in \mathbb{R}\}$$
Let W be the y-axis, which is the subspace of all vectors of the form $(0, y)$:
$$W = \{(0, y) \mid y \in \mathbb{R}\}$$
We show that
$$\mathbb{R}^2 = U \oplus W :$$

- Any vector
$$v = (a, b) \in \mathbb{R}^2$$

can be written as $(a, b) = (a, 0) + (0, b)$, where $(a, 0) \in U$ and $(0, b) \in W$. * The only vector that lies on both the x-axis and the y-axis is the origin $(0, 0)$. So, $U \cap W = \{(0, 0)\}$.

Thus, V is the direct sum of U and W.

21.8 Q7. What is vector space? Also define the basis and dimension of a vector space. (02)

ID	PYQ-2024-8a
Source	2024 Q8(a) [02 marks]

Answer:

21.8.0.1 1. Vector Space A vector space V over a field F is a set of elements (called vectors) together with two operations: vector addition and scalar multiplication. These operations must satisfy the following eight axioms for all vectors $u, v, w \in V$ and scalars $a, b \in F$: * **Associativity of addition:** $u+(v+w) = (u+v)+w$ * **Commutativity of addition:** $u + v = v + u$ * **Identity element of addition:** There exists a zero vector $0 \in V$ such that $u + 0 = u$. * **Inverse elements of addition:** For every $u \in V$, there exists an element $-u \in V$ such that $u + (-u) = 0$. * **Compatibility of scalar multiplication:** $a(bu) = (ab)u$ * **Identity element of scalar multiplication:** $1u = u$, where 1 is the multiplicative identity in F. * **Distributivity of scalar multiplication over vector addition:** $a(u + v) = au + av$ * **Distributivity of scalar multiplication over field addition:** $(a + b)u = au + bu$

21.8.0.2 2. Basis of a Vector Space A subset

$$B = \{v\_1, v\_2, \dots, v\_n\}$$

of a vector space V is a basis of V if: * The set B is linearly independent. * The set B spans V. This means any vector in V can be written as a linear combination of vectors in B.

21.8.0.3 3. Dimension of a Vector Space The dimension of a vector space V is the number of vectors in any basis of V.

21.9 Q8 (04)

Let U consists of all vectors in $\mathbb{R}^3$ whose entries are equal that is $u = \{(a, b, c) \mid a = b = c\}$, prove that U is a vector space over $\mathbb{R}$.

ID	PYQ-2024-8b
Source	2024 Q8(b) [04 marks]

Answer:
Since U is a subset of the vector space $\mathbb{R}^3$, we can show it is a subspace. A subspace of $\mathbb{R}^3$ is itself a vector space. We check the three subspace criteria:

21.9.0.1 1. Contains the Zero Vector The zero vector of $\mathbb{R}^3$ is $(0, 0, 0)$. Since $0 = 0 = 0$, we have:

$$(0, 0, 0) \in U$$

21.9.0.2 2. Closed Under Vector Addition Let

$$u\_1 = (a\_1, a\_1, a\_1) \in U$$

and

$$u\_2 = (a\_2, a\_2, a\_2) \in U$$

, where $a\_1, a\_2 \in \mathbb{R}$. We add these vectors:

$$u_1 + u_2 = (a_1 + a_2, a_1 + a_2, a_1 + a_2)$$

All three components of the sum vector are equal to $a\_1 + a\_2$. So, the sum vector is in U.

21.9.0.3 3. Closed Under Scalar Multiplication Let $u = (a, a, a) \in U$ and $k \in \mathbb{R}$ be a scalar. We multiply the vector by the scalar:

$$ku = (ka, ka, ka)$$

All three components of the resulting vector are equal to ka. So, this vector is in U.

Since U satisfies all three conditions, it is a subspace of $\mathbb{R}^3$. Therefore, U is a vector space over $\mathbb{R}$.

22 Topic 12: Basis and Dimension

This file contains the organized questions and answers for **Basis and Dimension**, priority ranked as **Priority 12** based on frequency and exam weight.

22.1 Q1 (05)

Define basis and dimension. Let w be the subspace of $\mathbb{R}^4$ generated by the vectors $(1, -2, 5, -3)$, $(2, 3, 1, -4)$ and $(3, 8, -3, -5)$. Find a basis and the dimension of w.

ID	PYQ-2018-8c
Source	2018 Q8(c) [05 marks]

Answer:

22.1.0.1 1. Definitions

- **Basis:** A set of vectors in a space is a basis if they are linearly independent and span the space.
- **Dimension:** The dimension of a space is the number of vectors in its basis.

22.1.0.2 2. Find Basis and Dimension of w We write the generating vectors as rows of a matrix:

$$\begin{bmatrix} 1 & -2 & 5 & -3 \\ 2 & 3 & 1 & -4 \\ 3 & 8 & -3 & -5 \end{bmatrix}$$

Apply row operations: * $R\_2 \to R\_2 - 2R\_1$ * $R\_3 \to R\_3 - 3R\_1$
This gives:

$$\begin{bmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 14 & -18 & 4 \end{bmatrix}$$

Now eliminate row 3 using row 2 ($R\_3 \to R\_3 - 2R\_2$):

$$\begin{bmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The non-zero rows of this echelon matrix form a basis for w:

$$\text{Basis} = \{(1, -2, 5, -3), (0, 7, -9, 2)\}$$

The number of vectors in the basis is 2, so the dimension is:

$$\text{Dimension} = 2$$

22.2 Q2 (04)

Let w be the space generated by the polynomials:

ID	PYQ-2020-8c
Appeared in	2020 Q8(c) [04 marks], 2023 Q8(b) [04 marks]
Frequency	⋆⋆ (2 times)

Answer:

$$v_1 = t^3 - 2t^2 + 4t + 1$$

$$v_2 = 2t^3 - 3t^2 + 9t - 1$$
$$v_3 = t^3 + 6t - 5$$
$$v_4 = 2t^3 - 5t^2 + 7t + 5$$

Find a basis and the dimension of w.

Answer:

We represent each polynomial as a coordinate vector in the standard basis $\{t^3, t^2, t, 1\}$:

$$v_1 = (1, -2, 4, 1), \quad v_2 = (2, -3, 9, -1), \quad v_3 = (1, 0, 6, -5), \quad v_4 = (2, -5, 7, 5)$$

We write these vectors as the rows of a matrix:

$$\begin{bmatrix} 1 & -2 & 4 & 1 \\ 2 & -3 & 9 & -1 \\ 1 & 0 & 6 & -5 \\ 2 & -5 & 7 & 5 \end{bmatrix}$$

Apply row operations: * $R\_2 \to R\_2 - 2R\_1$ * $R\_3 \to R\_3 - R\_1$ * $R\_4 \to R\_4 - 2R\_1$

This gives:

$$\begin{bmatrix} 1 & -2 & 4 & 1 \\ 0 & 1 & 1 & -3 \\ 0 & 2 & 2 & -6 \\ 0 & -1 & -1 & 3 \end{bmatrix}$$

Now perform operations on row 3 and row 4: * $R\_3 \to R\_3 - 2R\_2$ * $R\_4 \to R\_4 + R\_2$

This gives the echelon form:

$$\begin{bmatrix} 1 & -2 & 4 & 1 \\ 0 & 1 & 1 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The non-zero rows form the basis vectors for the subspace:

$$\text{Basis} = \{t^3 - 2t^2 + 4t + 1, \quad t^2 + t - 3\}$$

The number of polynomials in the basis is 2, so the dimension is:

$$\text{Dimension} = 2$$

22.3 Q3 (04)

Determine whether $(1, 1, 1, 1)$, $(1, 2, 3, 2)$, $(2, 5, 6, 4)$, $(2, 6, 8, 5)$ form a basis of $\mathbb{R}^4$. If not, find the dimension of the subspace they span.

ID	PYQ-2021-8b
Appeared in	2021 Q8(b) [04 marks], 2024 Q8(c) [04 marks]
Frequency	⋆⋆ (2 times)

Answer:

We write the vectors as rows of a matrix:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 2 \\ 2 & 5 & 6 & 4 \\ 2 & 6 & 8 & 5 \end{bmatrix}$$

Apply row operations: * $R\_2 \to R\_2 - R\_1$ * $R\_3 \to R\_3 - 2R\_1$ * $R\_4 \to R\_4 - 2R\_1$

This gives:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 3 & 4 & 2 \\ 0 & 4 & 6 & 3 \end{bmatrix}$$

Apply operations: * $R\_3 \to R\_3 - 3R\_2$ * $R\_4 \to R\_4 - 4R\_2$
This gives:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -2 & -1 \\ 0 & 0 & -2 & -1 \end{bmatrix}$$

Subtract row 3 from row 4 ($R\_4 \to R\_4 - R\_3$):

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Since the last row is zero, the vectors are linearly dependent. Therefore, they do not form a basis of $\mathbb{R}^4$.
The number of non-zero rows is 3. So the dimension of the subspace they span is:

$$\text{Dimension} = 3$$

22.4 Q4 (05)

Let U be the subspace of $\mathbb{R}^3$ spanned by the vectors $(1, 2, 1), (0, -1, 0)$ and $(2, 0, 2)$. Find a basis and the dimension of U.

ID	PYQ-2023-7b
Source	2023 Q7(b) [05 marks]

Answer:
We write the spanning vectors as the rows of a matrix:

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 0 \\ 2 & 0 & 2 \end{bmatrix}$$

Perform the operation $R\_3 \to R\_3 - 2R\_1$:

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 0 \\ 0 & -4 & 0 \end{bmatrix}$$

Perform the operation $R\_3 \to R\_3 - 4R\_2$:

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The non-zero rows of the echelon matrix form a basis:

$$\text{Basis} = \{(1, 2, 1), \quad (0, 1, 0)\}$$

The number of vectors in the basis is 2, so the dimension is:

$$\text{Dimension} = 2$$

23 Topic 13: Linear Dependence and Independence

This file contains the organized questions and answers for **Linear Dependence and Independence**, priority ranked as **Priority 13** based on frequency and exam weight.

23.1 Q1 (06)

Show that the vectors

$$\bar{A} = 2\hat{i} + \hat{j} - 3\hat{k}, \quad \bar{B} = \hat{i} - 4\hat{k}, \quad \bar{C} = 4\hat{i} + 3\hat{j} - \hat{k}$$

are linearly dependent and determine a relation between them and hence show that the terminal points are collinear.

ID	PYQ-2018-1a
Source	2018 Q1(a) [06 marks]

Answer:

23.1.0.1 1. Show Linear Dependence We set up a matrix with the given vectors as its rows. The vectors are linearly dependent if the determinant of this matrix is zero:

$$D = \begin{vmatrix} 2 & 1 & -3 \\ 1 & 0 & -4 \\ 4 & 3 & -1 \end{vmatrix}$$

We expand the determinant along the first row:

$$D = 2(0(-1) - (-4)(3)) - 1(1(-1) - (-4)(4)) - 3(1(3) - 0(4))$$

$$D = 2(12) - 1(-1 + 16) - 3(3) = 24 - 15 - 9 = 0$$

Since the determinant is zero, the vectors are coplanar. So they are linearly dependent.

23.1.0.2 2. Determine the Relation We write the linear combination equal to the zero vector:

$$x\bar{A} + y\bar{B} + z\bar{C} = 0$$

Substitute the vectors:

$$x(2\hat{i} + \hat{j} - 3\hat{k}) + y(\hat{i} - 4\hat{k}) + z(4\hat{i} + 3\hat{j} - \hat{k}) = 0$$

This gives a system of three equations:

$$2x + y + 4z = 0 \quad \ldots(1)$$

$$x + 3z = 0 \implies x = -3z \quad \ldots(2)$$

$$-3x - 4y - z = 0 \quad \ldots(3)$$

We substitute $x = -3z$ into equation (1):

$$2(-3z) + y + 4z = 0 \implies -6z + y + 4z = 0 \implies y = 2z$$

Let $z = 1$. This gives:

$$x = -3, \quad y = 2, \quad z = 1$$

Substitute these values into the linear combination:

$$-3\bar{A} + 2\bar{B} + \bar{C} = 0 \implies \bar{C} = 3\bar{A} - 2\bar{B}$$

This is the required relation.

23.1.0.3 3. Show Terminal Points are Collinear

Let P, Q, R be the terminal points of the position vectors $\bar{A}, \bar{B}, \bar{C}$ from the origin. We find the vectors representing the line segments:

$$\vec{PQ} = \bar{B} - \bar{A} = (\hat{\imath} - 4\hat{k}) - (2\hat{\imath} + \hat{\jmath} - 3\hat{k}) = -\hat{\imath} - \hat{\jmath} - \hat{k}$$

$$\vec{PR} = \bar{C} - \bar{A} = (4\hat{\imath} + 3\hat{\jmath} - \hat{k}) - (2\hat{\imath} + \hat{\jmath} - 3\hat{k}) = 2\hat{\imath} + 2\hat{\jmath} + 2\hat{k}$$

We see that:

$$\vec{PR} = -2\vec{PQ}$$

So the vectors $\vec{PR}$ and $\vec{PQ}$ are parallel. Since they share the common point P, the terminal points P, Q, R are collinear.

23.2 Q2 (04)

Define linearly dependent and independent vectors. Determine whether or not the vectors in $\mathbb{R}^3$ are linearly dependent: $(1, 2, -3)$, $(1, -3, 2)$, $(2, -1, 5)$.

ID	PYQ-2018-8b
Source	2018 Q8(b) [04 marks]

Answer:

23.2.0.1 1. Definitions

- **Linearly Dependent:** A set of vectors is linearly dependent if a non-trivial linear combination of them equals the zero vector.
- **Linearly Independent:** A set of vectors is linearly independent if the only linear combination that equals the zero vector is the trivial one (where all coefficients are zero).

23.2.0.2 2. Check the given vectors

We set up a matrix with these vectors as its columns and calculate its determinant:

$$D = \begin{vmatrix} 1 & 1 & 2 \\ 2 & -3 & -1 \\ -3 & 2 & 5 \end{vmatrix}$$

Expand along the first row:

$$D = 1((-3)(5) - (-1)(2)) - 1(2(5) - (-1)(-3)) + 2(2(2) - (-3)(-3))$$

$$D = 1(-15 + 2) - 1(10 - 3) + 2(4 - 9) = -13 - 7 - 10 = -30$$

Since the determinant is not zero ($D = -30 \neq 0$), the vectors are linearly independent. So they are not linearly dependent.

23.3 Q3 (04)

Suppose the vectors u, v, w are linearly independent. Show that the vectors $u + v, u - v, u - 2v + w$ are also linearly independent.

ID	PYQ-2020-8b
Appeared in	2020 Q8(b) [04 marks], 2021 Q8(c) [04 marks]
Frequency	⋆⋆ (2 times)

Answer:

We set a linear combination of these vectors equal to the zero vector:

$$c_1(u+v)+c_2(u-v)+c_3(u-2v+w)=0$$

Group the terms by vectors u, v, and w:

$$(c_1+c_2+c_3)u+(c_1-c_2-2c_3)v+c_3w=0$$

Since the vectors u, v, w are linearly independent, their coefficients must equal zero:

$$c_1+c_2+c_3=0 \quad \dots(1)$$

$$c_1-c_2-2c_3=0 \quad \dots(2)$$

$$c_3=0 \quad \dots(3)$$

Substitute

$$c\_3=0$$

from equation (3) into equations (1) and (2):

$$c_1+c_2=0$$

$$c_1-c_2=0$$

Adding these equations gives

$$2c\_1=0 \implies c\_1=0$$

. This then gives

$$c\_2=0.$$

Since the only solution is

$$c\_1=c\_2=c\_3=0$$

, the vectors $u+v$, $u-v$, and $u-2v+w$ are linearly independent.

23.4 Q4 (05)

Show that a necessary and sufficient condition that the vectors

$$\vec{A}=A_1\hat{\imath}+A_2\hat{\jmath}+A_3\hat{k}, \quad \vec{B}=B_1\hat{\imath}+B_2\hat{\jmath}+B_3\hat{k}, \quad \vec{C}=C_1\hat{\imath}+C_2\hat{\jmath}+C_3\hat{k}$$

be linearly independent is that the determinant

ID	PYQ-2024-1b
Source	2024 Q1(b) [05 marks]

Answer:

$$\begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}$$

be different from zero.

Answer:

Let a, b, and c be scalars. We set their linear combination to zero:

$$a\vec{A}+b\vec{B}+c\vec{C}=\vec{0}$$

We substitute the vector components:

$$a(A_1\hat{\imath}+A_2\hat{\jmath}+A_3\hat{k})+b(B_1\hat{\imath}+B_2\hat{\jmath}+B_3\hat{k})+c(C_1\hat{\imath}+C_2\hat{\jmath}+C_3\hat{k})=\vec{0}$$

We group the components for

$$\hat{i}, \quad \hat{j}, \quad \hat{k}:$$

$$(aA_1 + bB_1 + cC_1)\hat{i} + (aA_2 + bB_2 + cC_2)\hat{j} + (aA_3 + bB_3 + cC_3)\hat{k} = \vec{0}$$

This gives a homogeneous system of three linear equations:

$$\begin{aligned} A_1 a + B_1 b + C_1 c &= 0 \\ A_2 a + B_2 b + C_2 c &= 0 \\ A_3 a + B_3 b + C_3 c &= 0 \end{aligned}$$

The vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$ are linearly independent if and only if the system has only the trivial solution:

$$a = b = c = 0$$

A homogeneous system has only the trivial solution if and only if the determinant of its coefficient matrix is not zero. We write this determinant:

$$D = \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$$

We know that transposing a matrix does not change its determinant. So we can swap rows and columns:

$$\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}$$

Therefore, the vectors are linearly independent if and only if:

$$\begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} \neq 0$$

This completes the proof.

23.5 Q5. What is a linearly independent vector? Determine whether the vectors are linearly independent or not: (05)

ID	PYQ-2024-7c
Source	2024 Q7(c) [05 marks]

Answer:

- (i) $u = (1, 1, 0), v = (1, 3, 2), w = (4, 9, 5)$
- (ii) $u = (1, 2, 3), v = (2, 5, 7), w = (1, 3, 5)$

Answer:

23.5.0.1 1. Definition of Linearly Independent Vectors

A set of vectors

$$\{v\_1, v\_2, \ldots, v\_n\}$$

is linearly independent if the equation:

$$c_1 v_1 + c_2 v_2 + \cdots + c_n v_n = 0$$

has only the trivial solution

$$c\_1 = c\_2 = \cdots = c\_n = 0$$

. If there is a solution with some non-zero coefficients, the vectors are linearly dependent.

23.5.0.2 2. Determine Linear Independence

23.5.0.2.1 (i) $u = (1, 1, 0), v = (1, 3, 2), w = (4, 9, 5)$ We set up a matrix with these vectors as rows and check its determinant:

$$D = \begin{vmatrix} 1 & 1 & 0 \\ 1 & 3 & 2 \\ 4 & 9 & 5 \end{vmatrix}$$

We calculate the determinant:

$$D = 1(15 - 18) - 1(5 - 8) + 0 = -3 - (-3) = 0$$

The determinant is zero. So, the vectors are **linearly dependent**. We can find a non-trivial linear combination:

$$3u + 5v - 2w = 3(1, 1, 0) + 5(1, 3, 2) - 2(4, 9, 5) = (3 + 5 - 8, 3 + 15 - 18, 0 + 10 - 10) = (0, 0, 0)$$

23.5.0.2.2 (ii) $u = (1, 2, 3), v = (2, 5, 7), w = (1, 3, 5)$ We set up a matrix with these vectors as rows and check its determinant:

$$D = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 1 & 3 & 5 \end{vmatrix}$$

We calculate the determinant:

$$D = 1(25 - 21) - 2(10 - 7) + 3(6 - 5) = 4 - 6 + 3 = 1$$

The determinant is 1, which is not zero. So, the vectors are **linearly independent**.

24 Topic 14: Linear Combination

This file contains the organized questions and answers for **Linear Combination**, priority ranked as **Priority 14** based on frequency and exam weight.

24.1 Q1 (04)

Write the vector $v = (1, -2, 5)$ as a linear combination of the vectors

$$e\_1 = (1, 1, 1),$$

$$e\_2 = (1, 2, 3)$$

and

$$e\_3 = (2, -1, 1).$$

ID	PYQ-2020-8a
Appeared in	2020 Q8(a) [04 marks], 2021 Q8(a) [04 marks]
Frequency	⋆⋆ (2 times)

Answer:
We write the relation:

$$v = c_1e_1 + c_2e_2 + c_3e_3 \implies (1, -2, 5) = c_1(1, 1, 1) + c_2(1, 2, 3) + c_3(2, -1, 1)$$

This gives the system of equations:

$$c_1 + c_2 + 2c_3 = 1$$

$$c_1 + 2c_2 - c_3 = -2$$

$$c_1 + 3c_2 + c_3 = 5$$

We write the augmented matrix of this system:

$$\begin{bmatrix} 1 & 1 & 2 & | & 1 \\ 1 & 2 & -1 & | & -2 \\ 1 & 3 & 1 & | & 5 \end{bmatrix}$$

Apply row operations: * $R\_2 \to R\_2 - R\_1$ * $R\_3 \to R\_3 - R\_1$
This gives:

$$\begin{bmatrix} 1 & 1 & 2 & | & 1 \\ 0 & 1 & -3 & | & -3 \\ 0 & 2 & -1 & | & 4 \end{bmatrix}$$

Now perform the operation $R\_3 \to R\_3 - 2R\_2$:

$$\begin{bmatrix} 1 & 1 & 2 & | & 1 \\ 0 & 1 & -3 & | & -3 \\ 0 & 0 & 5 & | & 10 \end{bmatrix}$$

By back substitution, we solve the coefficients: * From the third row:

$$5c\_3 = 10 \implies c\_3 = 2.$$

- From the second row:

$$c\_2 - 3(2) = -3 \implies c\_2 = 3.$$

- From the first row:

$$c\_1 + 3 + 2(2) = 1 \implies c\_1 = -6.$$

So the linear combination is:

$$v = -6e_1 + 3e_2 + 2e_3$$

24.2 Q2 (05)

Consider the vectors

$$v\_1 = (2, 1, 4), v\_2 = (1, -1, 3)$$

and

$$v\_3 = (3, 2, 5)$$

in $\mathbb{R}^3$. Show that $v = (5, 9, 5)$ is a linear combination of v_1, v_2, and v_3.

ID	PYQ-2023-7a
Source	2023 Q7(a) [05 marks]

Answer:

We set up the relation:

$$v = c_1 v_1 + c_2 v_2 + c_3 v_3 \implies (5, 9, 5) = c_1(2, 1, 4) + c_2(1, -1, 3) + c_3(3, 2, 5)$$

This gives the system:

$$2c_1 + c_2 + 3c_3 = 5$$

$$c_1 - c_2 + 2c_3 = 9$$

$$4c_1 + 3c_2 + 5c_3 = 5$$

We write the augmented matrix and swap row 1 and row 2 ($R\_1 \leftrightarrow R\_2$):

$$\begin{bmatrix} 1 & -1 & 2 & | & 9 \\ 2 & 1 & 3 & | & 5 \\ 4 & 3 & 5 & | & 5 \end{bmatrix}$$

Apply row operations: * $R\_2 \to R\_2 - 2R\_1$ * $R\_3 \to R\_3 - 4R\_1$

This gives:

$$\begin{bmatrix} 1 & -1 & 2 & | & 9 \\ 0 & 3 & -1 & | & -13 \\ 0 & 7 & -3 & | & -31 \end{bmatrix}$$

Perform the operation $R\_3 \to 3R\_3 - 7R\_2$:

$$\begin{bmatrix} 1 & -1 & 2 & | & 9 \\ 0 & 3 & -1 & | & -13 \\ 0 & 0 & -2 & | & -2 \end{bmatrix}$$

Now solve for the coefficients by back substitution: * From row 3:

$$-2c\_3 = -2 \implies c\_3 = 1.$$

- From row 2:

$$3c\_2 - 1 = -13 \implies 3c\_2 = -12 \implies c\_2 = -4.$$

- From row 1:

$$c\_1 - (-4) + 2(1) = 9 \implies c\_1 + 6 = 9 \implies c\_1 = 3.$$

So we can write v as:

$$v = 3v_1 - 4v_2 + v_3$$

This shows that the vector is a linear combination of the given vectors.

24.3 Q3 (03)

Express the polynomial

$$v = 3t^2 + 5t - 5$$

as a linear combination of the polynomials

$$P\_1 = t^2 + 2t + 1,$$

$$P\_2 = 2t^2 + 5t + 4$$

and

$$P\_3 = t^2 + 3t + 6.$$

ID	PYQ-2024-7a
Source	2024 Q7(a) [03 marks]

Answer:

We write v as a linear combination:

$$v = c_1 P_1 + c_2 P_2 + c_3 P_3$$

We substitute the polynomials:

$$3t^2 + 5t - 5 = c_1(t^2 + 2t + 1) + c_2(2t^2 + 5t + 4) + c_3(t^2 + 3t + 6)$$

We expand the terms and group them by powers of t:

$$3t^2 + 5t - 5 = (c_1 + 2c_2 + c_3)t^2 + (2c_1 + 5c_2 + 3c_3)t + (c_1 + 4c_2 + 6c_3)$$

We match the coefficients of t^2, t, and the constant terms: 1.

$$c\_1 + 2c\_2 + c\_3 = 3$$

2.

$$2c\_1 + 5c\_2 + 3c\_3 = 5$$

3.

$$c\_1 + 4c\_2 + 6c\_3 = -5$$

We solve this system. From equation (1):

$$c_1 = 3 - 2c_2 - c_3$$

We substitute this expression for c_1 into equations (2) and (3):

In equation (2):

$$2(3 - 2c_2 - c_3) + 5c_2 + 3c_3 = 5$$

$$6 - 4c_2 - 2c_3 + 5c_2 + 3c_3 = 5 \implies c_2 + c_3 = -1 \implies c_2 = -1 - c_3 \quad \ldots(4)$$

In equation (3):

$$(3 - 2c_2 - c_3) + 4c_2 + 6c_3 = -5$$

$$3 + 2c_2 + 5c_3 = -5 \implies 2c_2 + 5c_3 = -8 \quad \ldots(5)$$

We substitute equation (4) into equation (5):

$$2(-1 - c_3) + 5c_3 = -8$$

$$-2 - 2c_3 + 5c_3 = -8$$

$$3c_3 = -6 \implies c_3 = -2$$

We use

$$c\_3 = -2$$

in equation (4):

$$c_2 = -1 - (-2) = 1$$

We use

$$c\_2 = 1$$

and

$$c\_3 = -2$$

to find c_1:

$$c_1 = 3 - 2(1) - (-2) = 3$$

Thus, the linear combination is:

$$v = 3P_1 + P_2 - 2P_3$$

25 Topic 25: Linear Transformations

This file contains the organized questions and answers for **Linear Transformations**, priority ranked as **Priority 25** based on frequency and exam weight.

25.1 Q1 (06)

Define Kernel and image of a linear mapping. Let $F : V \to U$ be a linear mapping. Show that the Kernel of F is a subspace of V and the image of F is a subspace of U.

ID	PYQ-2019-8b
Source	2019 Q8(b) [06 marks]

Answer:

25.1.0.1 1. Definitions

- **Kernel of F:** The set of all vectors in V that map to the zero vector in U. It is denoted by $\mathrm{Ker}(F)$:
$$\mathrm{Ker}(F) = \{v \in V \mid F(v) = 0_U\}$$
- **Image of F:** The set of all vectors in U that are mapped to by at least one vector in V. It is denoted by $\mathrm{Im}(F)$:
$$\mathrm{Im}(F) = \{u \in U \mid \exists v \in V \text{ such that } F(v) = u\}$$

25.1.0.2 2. Prove $\mathrm{Ker}(F)$ is a subspace of V

To show that $\mathrm{Ker}(F)$ is a subspace of V, we must verify three properties:

1. **Non-emptiness / Zero Vector:** Since F is linear:
$$F(0_V) = 0_U \implies 0_V \in \mathrm{Ker}(F)$$
2. **Closure under Addition:** Let $v\_1, v\_2 \in \mathrm{Ker}(F)$. Then
$$F(v\_1) = 0\_U$$

and

$$F(v\_2) = 0\_U.$$

$$F(v_1 + v_2) = F(v_1) + F(v_2) = 0_U + 0_U = 0_U \implies v_1 + v_2 \in \mathrm{Ker}(F)$$

3. **Closure under Scalar Multiplication:** Let $v \in \mathrm{Ker}(F)$ and let k be a scalar in the field. Then
$$F(v) = 0\_U.$$
$$F(kv) = kF(v) = k(0_U) = 0_U \implies kv \in \mathrm{Ker}(F)$$

Since all three properties are satisfied, $\mathrm{Ker}(F)$ is a subspace of V.

25.1.0.3 3. Prove $\mathrm{Im}(F)$ is a subspace of U

To show that $\mathrm{Im}(F)$ is a subspace of U, we verify the three properties:

1. **Non-emptiness / Zero Vector:** Since
$$F(0\_V) = 0\_U$$

, the zero vector 0_U has a pre-image in V:

$$0_U \in \mathrm{Im}(F)$$

2. **Closure under Addition:** Let $u\_1, u\_2 \in \mathrm{Im}(F)$. Then there exist $v\_1, v\_2 \in V$ such that

$$F(v\_1) = u\_1$$

and

$$F(v\_2) = u\_2.$$

$$u_1 + u_2 = F(v_1) + F(v_2) = F(v_1 + v_2)$$

Since $v\_1 + v\_2 \in V$ (as V is a vector space), $u\_1 + u\_2$ is the image of $v\_1 + v\_2$, so:

$$u_1 + u_2 \in \mathrm{Im}(F)$$

3. **Closure under Scalar Multiplication:** Let $u \in \mathrm{Im}(F)$ and let k be a scalar. Then there exists $v \in V$ such that $F(v) = u$.

$$ku = kF(v) = F(kv)$$

Since $kv \in V$, ku is the image of kv, so:

$$ku \in \mathrm{Im}(F)$$

Since all three properties are satisfied, $\mathrm{Im}(F)$ is a subspace of U.

25.2 Q2 (04)

Identify that the following mappings F are linear or not linear:

ID	PYQ-2023-8c
Source	2023 Q8(c) [04 marks]

Answer:

- **(i)**

$$F : \mathbb{R}^3 \to \mathbb{R}$$

defined by $F(x, y, z) = 2x - 3y + 4z$ * **(ii)**

$$F : \mathbb{R}^2 \to \mathbb{R}^3$$

defined by $F(x, y) = (x + 1, 2y, x + y)$

Answer:

25.2.0.1 (i) Mapping $F(x, y, z) = 2x - 3y + 4z$

Let

$$u = (x\_1, y\_1, z\_1)$$

and

$$v = (x\_2, y\_2, z\_2)$$

be vectors in $\mathbb{R}^3$. * **Check Addition:**

$$F(u + v) = F(x_1 + x_2, y_1 + y_2, z_1 + z_2) = 2(x_1 + x_2) - 3(y_1 + y_2) + 4(z_1 + z_2)$$

$$F(u + v) = (2x_1 - 3y_1 + 4z_1) + (2x_2 - 3y_2 + 4z_2) = F(u) + F(v)$$

- **Check Scalar Multiplication:**

$$F(ku) = F(kx_1, ky_1, kz_1) = 2(kx_1) - 3(ky_1) + 4(kz_1) = k(2x_1 - 3y_1 + 4z_1) = kF(u)$$

So the mapping is **linear**.

25.2.0.2 (ii) Mapping $F(x, y) = (x + 1, 2y, x + y)$

A linear mapping must map the zero vector to the zero vector. We check the image of $(0, 0)$:

$$F(0, 0) = (0 + 1, 2(0), 0 + 0) = (1, 0, 0) \neq (0, 0, 0)$$

Since the image of the zero vector is not the zero vector, this mapping is **not linear**.